

Operations Project Authorization Form

Approved on February 5, 2026 EPAC

[Link to Meeting Minutes](#)

[Non-redacted detailed PAF link](#)

Project Title: Hyde Park Battery Energy Storage System – Rev 2	Project Number: 21334 (DSS) & 24211 (D-Line)
Date Prepared: 3/25/2024 (Rev 0), 01/12/2024 (Rev 1) 01/27/2026 (Rev 2)	Company: Eversource EMA
Organization: System Planning	Class(es) of Plant: D SS D Line
Project Initiator: Amsa Mangga	Project Category: Stations - Batteries
Managing Organization: Project Mgt and Construction	Schedule ID#: 80089926
Project Manager: Adam Kelley	Project Type: Specific
Project Sponsor: Juan Martinez	Capital Investment part of original Op Plan: Yes
Estimated in service date: December 2028	Emergency Related Request? No
Eng./Constr. Resources Budgeted? Yes	Facility Type (check all that apply): ☐ PTF ☐ non-PTF ☐ local PTF ☒ Distribution ☐ N/A (General Plant)
EJ Community: Yes	Authorization Type: Partial Funding
Total Capital Request: \$5,767K	

[1] Executive Summary

Rev 2 - This revised partial funding Project Authorization Form (PAF) is a request for an additional \$1,758K and a new total request of \$5,767K for completion of preliminary engineering, a Limited Notice to Proceed (LNTF) and 10% down for procurement of engineering drawings for the Battery System, project management, siting, environmental and project services support and preparation of a full funding PAF package. Once the items included in this PAF request are completed, the next partial funding phase will commence to cover the procurement of the remaining long lead materials which include the switchgear and the Generator Step-Up (GSU) Transformer. The next PAF request would be similar to this request which will incorporate a 10% LNTF for the procurement of engineering drawings of the switchgear and the GSU Transformer, project management, siting, environmental and project services support.

The requirements for the Article 80 Siting filing includes the battery system design, site survey, tree survey for areas affected by project, site plan (site layout, grading & drainage, infrastructure, stormwater management, landscaping and screening), vehicular/pedestrian access, traffic impacts in the vicinity of the site, location, dimensions, and architectural elevations of structures and buildings, visual renderings to show views of project from

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Form Template Revised 03/13/2024

APS 1 – Project Authorization Policy Issued 12/31/21 Rev. 6

Policy Sponsor: EVP and CFO

JA-AM-2001-A Rev. 7

abutting properties, have been better defined with Subject Matter Expert (SME) input on deliverables and durations. Additional partial funding will be requested in June 2026 for the remaining long lead materials. The full funding PAF is anticipated to be presented in January 2028 and the conceptual level estimate for the full funding request is approximately \$43,926K.

Partial funding of \$4,009K was approved in PowerPlan on July 10, 2024 (Reference 9). To date, \$2,579K has been spent primarily on internal completion of Conceptual Engineering, project support activities, external project management support and related overheads and AFUDC. Engineering has been involved in the initial and rebid of the BESS System Procurement process. The initial Request for Proposal (RFP) process resulted in no qualified bidders which then required a rebid process to secure a technically qualified bidder. Procurement for a Battery System Vendor required a re-bid process to secure a Technically Qualified bidder. This was a four (4)-month delay to the schedule, pushing the in service date (ISD) from August 2028 to December 2028.

Schedule development with SMEs found that the approved shop drawings for several long lead materials are required for both Civil and Substation Electrical to complete Preliminary engineering.

Current long lead material lead times have increased since the last funding request. These lead times for Manufacturer shop drawing needs, fabrication lead time, and the logic changes based on SME needs for Engineering deliverables are still under review by the project team to maintain the targeted December 2028 ISD.

This project is to construct a 10MW/20MWh Battery Energy Storage System (BESS) at 401-433 Cummins Highway (future Hyde Park BESS Station #359) in the Hyde Park section of the City of Boston to help relieve the heavy loading condition of Hyde Park Substation #496 in Hyde Park, MA.

The key drivers for this revised funding request are as follows.

- **Procurement Package #1 (Battery System Packager)**
Includes Batteries, Racks, Containers, Disconnects, Inverters, All container auxiliaries, Ground Banks, Fire Protection, Local Controller 1-4, UPS, HVAC, BMS 1-4, PCS 1-4, Gas and Fire Safety Panel.
 - Procurement Package estimated total - \$12,106,850.00 (This partial funding request includes 10% payment (\$1,211K).

The Hyde Park Substation #496 peak load forecast for year 2025 exceeded the station firm capacity and keeps increasing incrementally. A mid-term solution to bring additional capacity to the area is needed before a long-term solution (21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives project will be in service. Partial Funding (Rev 2) was approved at the Eversource Project Approval Committee (EPAC) meeting on February 27, 2025 for preliminary engineering (Reference 1). The expected in service

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date for the long-term project is March 2033. This proposed BESS project is the mid-term solution.

The project 22345 - 23375 - Hyde Park-Dorchester Area Plan Land Acquisition (Reference 2) was approved by EPAC on August 3, 2023 for the amount of \$6.275M and the property was purchased in October 2023. The property at 161-191 New England Avenue, Boston has been purchased to locate the transmission substation in the project 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 – Hyde Park-Dorchester Area Supply Initiatives project. A separate project 25172 - Hyde Park - Dorchester Area Plan: 200 New England Ave Acquisition (Reference 3) was approved at EPAC on April 11, 2025 for the purchase of an adjacent property to locate the distribution substation in project 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives project.

This Hyde Park BESS project had initial funding of \$200K (Reference 4) approved in PowerPlan on June 16, 2021, and a revised 21334 - Hyde Park BESS – Battery Energy Storage System Solution, Rev 1 IFR (Reference 5) for \$300K approved in PowerPlan on May 25, 2023. The initial funding was used to complete conceptual level engineering and estimates and development of a partial funding PAF for \$4,009K and Solution Selection Form (SSF) (Reference 6). The partial funding PAF and SSF were presented to and conditionally approved at the Solution Design Committee (SDC) on February 15, 2024. The partial funding was to fund the preliminary engineering, siting, and permitting for the BESS.

In the meantime, another Partial Funding PAF (Rev 0) for \$500K was approved in PowerPlan on April 15, 2024 (Reference 7) to continue engineering and siting until approval of the partial funding PAF presented at SDC in February 2024. The SSF and partial funding PAF presented to the SDC on February 15, 2024 received final approval in May and July 2024, respectively.

Partial Funding of \$2,085K for the D Line portion of the project under project number 24211 was approved by the Eversource Massachusetts Project Approval Committee (MPAC) on August 5, 2024 (Reference 8) for completion of D Line preliminary engineering and permitting. This project was approved by SDC on February 15, 2024.

This revised partial funding of \$5,767K will be used for the following:

- **10% LNTF for the Battery System (Procurement Package #1) – \$1,211K**
- **Preliminary Engineering and Design of Procurement Package #1 – \$500K**
- **Continuation of Environmental, Siting and Outreach evaluations – \$200K**
- **Indirects and AFUDC – \$994K**
- **Project Management Support – \$283K**
- **Actuals to Date – \$2,579K**

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[2] Project Costs Summary

Note: Dollar values are in thousands and rounded to nearest thousand

Line item Category	Prior Authorized	Actuals to Date	2026	2027	2028	Total Request
1. ROW / Easements / Land Acquisition	\$	\$16	\$		\$	\$16
2. Environmental Approvals / Permits	\$323	\$15	\$50		\$	\$65
3. Outreach	\$39	\$60	\$50		\$	\$110
4. Siting Approvals / Permits	\$244	\$35	\$100		\$	\$135
5. Engineering / Design	\$374	\$343	\$500		\$	\$843
6. Materials (Eversource purchased)	\$	\$2	\$1,211		\$	\$1,213
7. Construction (incl materials by contractors)	\$36	\$	\$		\$	\$
8. Testing / Commissioning	\$	\$25	\$		\$	\$25
9. Project Mgmt Team	\$382	\$577	\$283		\$	\$860
10. Removals	\$	\$	\$		\$	\$
11. Other	\$4	\$24	\$			\$24
12. Risks	\$	\$	\$		\$	\$
SUBTOTAL DIRECTS W/ RISKS	\$1,401	\$1,096	\$2,194		\$	\$3,2909
13. Indirects/Overhead	\$2,282	\$1,280	\$819		\$	\$2,099
14. AFUDC	\$326	\$202	\$175		\$	\$378
PROJECT TOTAL – BASELINE BUDGET	\$4,009	\$2,579	\$3,188		\$	\$5,767
15. Contingency	\$	\$	\$		\$	\$
TOTAL CAPITAL REQUEST	\$4,009	\$2,579	\$3,188		\$	\$5,767
16. Reimbursables/Customer Contribution	\$	\$	\$		\$	\$
PROJECT TOTAL (LESS REIMBURSABLES)	\$4,009	\$2,579	\$3,188		\$	\$5,767
O&M	\$	\$	\$		\$	\$
TOTAL REQUEST	\$4,009	\$2,579	\$3,188		\$	\$5,767

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Detail other costs, project risk items and their allocations:

Other (\$24K): Taxes

Risk	Allocation
1. Risks will be accounted for in Full Funding Request	\$
2.	\$
3.	\$
4.	\$
5.	\$
Total	\$0

Explain unique payment provisions, if applicable:

The following material will need to be ordered using the Material Order Exception Request (MOER) (Reference 10) process prior to full funding to meet the current in-service date due to material lead times, which are 15 months after receipt of order, and drawing lead times, which are 12-16 weeks after receipt of order. Drawings are required to complete preliminary design deliverables for EPAC.

- Battery Energy Storage System
 - DC Power Blocks and Tier 1 controls with required HVAC and Fire protection

The following is a list for the Hyde Park BESS System payment comment for pre and post full funding.

- Milestone Payment Commitments Prior to Full funding: Battery Energy Storage System
 - 10% Payment (\$1,210,685) to support drawing development and reserve manufacturing slot.

Identify and detail any unique O&M impacts resulting from this project:

The Project will incorporate new equipment not normally maintained by Eversource, including battery modules, inverters, and other energy storage system specific equipment. Contractual language and requirements have been placed in the Battery Energy Storage System Vendor Request for Proposal to require training of Eversource personnel, material warranty and spares, and O&M manuals for all non-standard equipment.

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[3] Technical Justification

Project Need, Objective and Justification

Project Need

This project is triggered because of the increasing load growth at the Hyde Park #496 Substation as a result of step loads and electrification. Hyde Park Substation #496 2021 peak load was 157 MVA, which exceeded the 2021 Load Forecast by 7 MVA; it is 108% or 11 MVA above Substation firm capacity of 146 MVA. The substation firm capacity is the remaining capacity with the loss of a Hyde Park Substation 115/13.8kV 75/100/140 MVA transformer. In 2021, 7.7 MVA was transferred permanently off of Hyde Park Substation after the Substation peaked for that year. The 2022 post load transfer peak load was still 152 MVA, 104% of the Substation firm capacity of 146 MVA. About an additional 5 MVA of load was transferred in 2022 after the Substation peaked for that year. About 2.5 MVA was transferred from Hyde Park Substation #496 to Colburn St Substation #350 in 2024. The substation was loaded to about 97% of its firm capacity in 2024. Even with these load transfers, the Hyde Park Substation #496 peak load forecast for year 2025 exceeded the station firm capacity and will keep increasing incrementally.

Load transfers are not a feasible option because they reduce the existing circuit transfer capability, increase the existing distribution lines' exposure, and reduce the spare capacity at the receiving Substation. Building a new Substation to address this thermal constraint is expected to take up to March 2033 (at least) and is a long-term solution to address the capacity constraint in the area. Therefore, a near-term mitigation plan is urgently needed to relieve this thermal Substation constraint until the long-term plan is in service.

Further transfers can only be done temporarily to relieve the Hyde Park Substation #496 load if the station is loaded above its firm capacity under loss of one (1) of the station 115/13.8kV transformers. As noted above, these transfers are not favorable.

Project Objective

The objective of the project is to alleviate Hyde Park Substation #496 heavily loaded condition, to prevent the Substation from being overloaded before the long-term solution is in place. The long-term solution is the new Hyde Park Substation #454 (New England Ave Substation).

Four (4) 2.5MW / 5 MWh battery units will be installed at 401-433 Cummins Highway, each connected to a Hyde Park 13.8kV distribution feeder. The batteries would be dispatched with a peak load shave functionality. They will charge during the light load and discharge during the peak load in order to lower the Substation peak. The battery can operate through multi-day peak load events.

Hyde Park substation load forecast continues to gradually increase beyond its firm capacity. The forecast will continue to be monitored to evaluate whether the capacity required would need to be delivered by other options until the BESS is in service. Those options may include deployment of spot generation, voltage reduction, and load shedding during peak load hours typically from 6 PM to 8 PM. Spot generation would be required to supply load during peak load hours and can remain on standby for the remainder of the time.

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Project Scope

The Hyde Park BESS project consists of a 10MW/20MWh energy storage system. The Project will be located at 401-433 Cummins Highway in Boston on Eversource-owned property. The Hyde Park BESS will be operated in parallel with Eversource's (the Owner's) electrical grid. The BESS shall be comprised of a system of battery racks, inverters, control system(s), pad mounted transformers, and switchgear. The BESS will be located and interconnected to Eversource's existing 13.8kV distribution infrastructure adjacent to the BESS facility. Constructability reviews were performed on 7/20/2023 and 10/31/2023 as part of the partial funding Rev 1 PAF (Attachment C).

- D Line (Street) work: The BESS facility is proposed to interconnect to the following 13.8kV circuits supplied from Hyde Park substation:
 - 441-1401H
 - 362-1207H
 - 362-1209H
 - 496-H7
- BESS Preliminary Scope of Distribution Line Work as referenced in Attachment D include:
 - Eversource Property: Manhole Installation: (3) concrete manholes with inside dimension of 6'x10'x8'
 - Eversource Property: Conduit Installation (1,270') 6-5" PVC type EB encased in concrete from MH12362 to BESS via three (3) new manholes
 - Street work: Primary Underground Cable Removal (3,646') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (10 sections)
 - Street work: Primary Underground Cable Installation (9,940') of 3-700 FS / 3-1000 FS 15kV primary cable (29 sections)
- Street work: Interconnection to Circuit 496-H7
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.
- Street work: Interconnection to Circuit 441-1401H
 - Cut and remove 186 feet of 800/3 cable from MH15729 to MH12365 (1 section).
 - Install new cable from MH15729 to MH12362 with 1,400 feet of 3-700 FS (4 sections) and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.
 - Note: 1 single 3-way joint in MH15729 and 3 single multiple joints in MH12365.
- Street work: Interconnection to Circuit 362-1207H
 - Cut and remove cable from MH15729 to MH12361 (5 sections).
 - Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

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- Street work: Interconnection to Circuit 362-1209H
 - Cut and remove cable from MH12365 to MH12361 (4 sections).
 - Install 1,650 feet of 3-700 FS (4 sections) from MH12365 to MH12361 and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.
 - Note: 1 single 3-way joint in MH12361 and 3 single multiple joints in MH12362.

The partial funding (Rev 2) being requested will be used for material only for 10% down on the Battery System procurement package, as well as for the following:

Distribution System Planning

- Coordinate with the various stakeholders to carry the project up until full funding.
- Conduct PSCAD dynamic study using the specific battery inverter that will be procured for the project.

Transmission System Planning

- Perform transmission system impact study and obtain approvals of Proposed Plan Applications (PPA) from ISO-NE.
 - Develop study scope and report for the study
 - Completed
 - Conduct steady state, stability, short circuit and EMT analysis for the System Impact Study.
 - Completed
 - Conduct project inverter model validation in PSCAD.
 - Have weekly technical discussion with ISO-NE on the study

Capital Project Engineering

- Coordinate the overall engineering effort for the external and internal disciplines below.
 - Battery Energy Storage System Vendor
 - Substation Engineering
 - Protection & Control Engineering
 - Telecom Engineering
 - Distribution Engineering
- Finalize technical specifications for the Battery Energy Storage System Vendor Request for Proposal. Engineering has decided to bring on an engineering vendor to support this tech spec development.
 - Completed
- Procurement of Battery Energy Storage System vendor to develop conceptual and preliminary packages required for the installation and interconnection of the battery system as outlined above.

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Project Management

- Overall project planning and scheduling
- Project reporting, document control and management, and financial projections
- Battery Energy Storage System Vendor procurement coordination

Siting

• Coordinate and lead siting effort which includes pre-filing meetings with the City of Boston for the Article 80 Small Project siting path. Coordination with Eversource legal and outside legal counsel to assist with the submittal for Small Project Review, along with assistance throughout the duration of the permitting process and securing approval at the Boston Zoning Board of Appeals.

Project Engagement

- Continue the community outreach and engagement strategy required for the project through the Project Engagement and Community Relations teams. This outreach strategy has been reviewed and communicated with the City of Boston and Boston Planning and Development Authority and has been aligned and coordinated with the outreach strategy for project 21192 – Hyde Park - Dorchester Area Supply Initiatives.
- Project Collateral, i.e., posters and printing services – for community outreach, mailings and public meetings, door to door outreach
- Cost of tabling events – can vary from free to \$500
- Translation services – Translations into target languages of community for all collateral, live translations, up to five (5) languages.
- Branded collateral – totes, coloring books, etc.
- Food and refreshments for community events

Equity and Environmental Justice (EJ)

• Identify and engage EJ communities, populations, Community Based Organizations (CBOs), influencers, advocates, etc. impacted by or who are relevant for outreach regarding this Project. Identify languages and procure services for language translation/interpretation in EJ stakeholder communities to enhance strategic communication with multilingual customers. Identify and communicate with EJ communities and stakeholders in a broad geographic area to understand historical engagement, determine potential Project, Program, or policy impact, and assist in resolution efforts to open doors, build trust and enable fair outcomes.

Environmental

- Internal project support from the Environmental Affairs Department for Licensing & Permitting and Remediation groups to carry the project up until full funding.
- External project support by the environmental licensing & permitting vendor for:
 - Development of a Soil and Groundwater Management Plan for implementation during project construction that dictates regulatory requirements for construction phase handling, management,

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and disposal of contaminated soils and groundwater regulated under the Massachusetts Contingency Plan.

- Development/acquisition of necessary environmental-related permits from Massachusetts Department of Environmental Protection (MassDEP) Solid Waste Bureau for any geotechnical investigations proposed on the capped landfill upon which the BESS facility is being proposed to be built.
- Coordination with MassDEP on any pre-application meetings associated with disturbance/alteration of the capped landfill upon which the BESS facility is being proposed to be built.
- Development/acquisition of necessary environmental-related permits from applicable environmental regulatory agencies for:
 - SW36 Landfill Post-Closure Use Permit from MassDEP Bureau of Solid Waste
 - National Pollution Discharge Elimination System (NPDES) Construction General Permit (CGP) authorization and Stormwater Pollution Prevention Plan (SWPPP) from U.S. Environmental Protection Agency (USEPA)
 - NPDES Dewatering and Remediation General Permit (DRGP) authorization and Best Management Practices Plan (BMPP) from USEPA (if necessary)

Substation Equipment

Station	Device Position / Nomenclature	Device Removals <i>include manufacturer</i>	Device Additions <i>include manufacturer</i>
Hyde Park BESS Station #359	N/A	N/A	Battery System Packager
Hyde Park BESS Station #359	N/A	N/A	Transformer
Hyde Park BESS Station #359	N/A	N/A	Shelter Aisle Gear (Switchgear)
Hyde Park BESS Station #359	N/A	N/A	Control Enclosure

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Background

The 13.8kV Hyde Park Substation is forecasted to have a steady increase in electrification load. The system needs for the northern area can be classified into several broad categories:

Capacity Deficiency: Under an N-1 contingency of a Hyde Park 115/13.8kV transformer, the remaining Hyde Park Substation #496 transformer is overloaded. Load is typically transferred during a Substation transformer contingency via distribution circuit ties utilizing automatic field switches. However, with this load transfer, the Hyde Park Substation #496 load is still forecasted to contribute to overloads under an N-1 contingency. 2025 summer peak load was over the Station Firm capacity and future years forecast show an increase in the load. Hyde Park Substation #496 2021 peak load exceeded the Substation firm capacity and almost exceeded the Substation Load Carrying Capability (LCC). The firm capacity is the capacity based on the loss of a Hyde Park Substation #496 transformer and the LCC is the firm capacity in addition to the amount of load that can be transferred to nearby Substations utilizing distribution circuit ties. In 2021 and 2022, load was transferred permanently off of Hyde Park Substation #496 after the Substation peaked in those years. However, even with these load transfers, the Hyde Park Substation #496 is forecasted to exceed the firm capacity of the Substation and reach 95% of the Hyde Park Substation #496 LCC by 2024. Circuit 496-H1XY runs 80-90% of its summer normal ampacity and circuit 362-1209H runs 60-90% of its summer normal ampacity in the summer months. Circuit 496-H1XY supplies Milton and Dorchester while circuit 362-1209 supplies Milton.

By July of 2027, the capacity to be delivered by the project 21334 will need to be delivered by other means. Those options may include deployment of spot generation, voltage reduction, and load shedding during peak load hours during the summer timeframe typically from 18:00 PM to 20:00 PM. Based on the 2023 forecast, two (2) generators of one (1) MW capacity would be needed in 2027 and 2028, and two (2) generators of two (2) MW capacity would be needed in 2029 until the BESS is installed. Eight (8) MW of generation would be needed in the worst-case scenario where the BESS does not get installed until the station is at 100% of LCC. Spot generation would be required to supply load during peak load hours and can remain on standby for the remainder of the time. System Planning will monitor the annual forecast that may shift the plan laid out above.

Transmission Reliability: Various N-1 transmission contingencies may result in loss of the 115kV line supplying the Hyde Park 115/13.8kV transformer including: faults on the 115kV Baker St - Hyde Park lines; or 115kV bus faults at Hyde Park or Baker St; or 115kV breaker failures at Hyde Park or Baker St. These would cause the loss of a Hyde Park 115/13.8kV transformer, which would leave Hyde Park Substation #496 supplied by only one remaining transformer. This results in capacity issues explained in the section above.

Distribution Reliability: Circuits 496-H1XY and 362-1209H run up to 90% of their summer normal ampacity. This thermal constraint will be explained further in sections below.

Asset Condition: Both the 110A and 110B transformers at Hyde Park Substation #496 are over 50 years old and are already in a five (5) year long term plan for replacement.

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Project Background

The Substations surrounding Hyde Park Substation #496 are also heavily loaded including Dewar Street and Andrew Square Substations. Dewar St substation was forecasted to be above 100% of the firm capacity from 2024 onwards and Andrew Square Substation is forecasted to be above 100% of firm capacity by 2027. Additional capacity is needed in the area to resolve the Hyde Park Substation #496 overload and nearby Substations' heavily loaded conditions. A new Substation is considered as the long-term solution to address the area capacity issues. The project 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives project, which addresses the ten (10)-year long-term solution had a partial funding PAF approved; the PAF was then revised and approved to include the 15-year plan.

Below is the Hyde Park ten (10)-year load forecast and percentage of Firm Capacity.

	Station		2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
496	Hyde Park	Forecast Load	156.8	157.1	158.1	158.9	160.4	161.4	164.4	166.6	169	172
		%age of Firm	107.4%	107.6%	108.3%	108.8%	109.9%	110.5%	112.6%	114.1%	115.8%	117.8%

Table 1. Hyde Park Load Forecast

Load transfers are not a feasible option because they reduce the existing circuit transfer capability, increase the existing distribution lines' exposure, and reduce the spare capacity at the receiving Substation. Building a new Substation to address this thermal constraint is expected to take up to March 2033 (at least) and is a long-term solution to address the capacity constraint in the area. Therefore, a near term mitigation plan is urgently needed to relieve this thermal Substation constraint until the long-term plan is in service.

Further transfers can only be done temporarily to relieve the Hyde Park substation load if the station is loaded above its firm capacity under loss of one of the stations' 115/15kV transformers. As noted above, these transfers are not favorable.

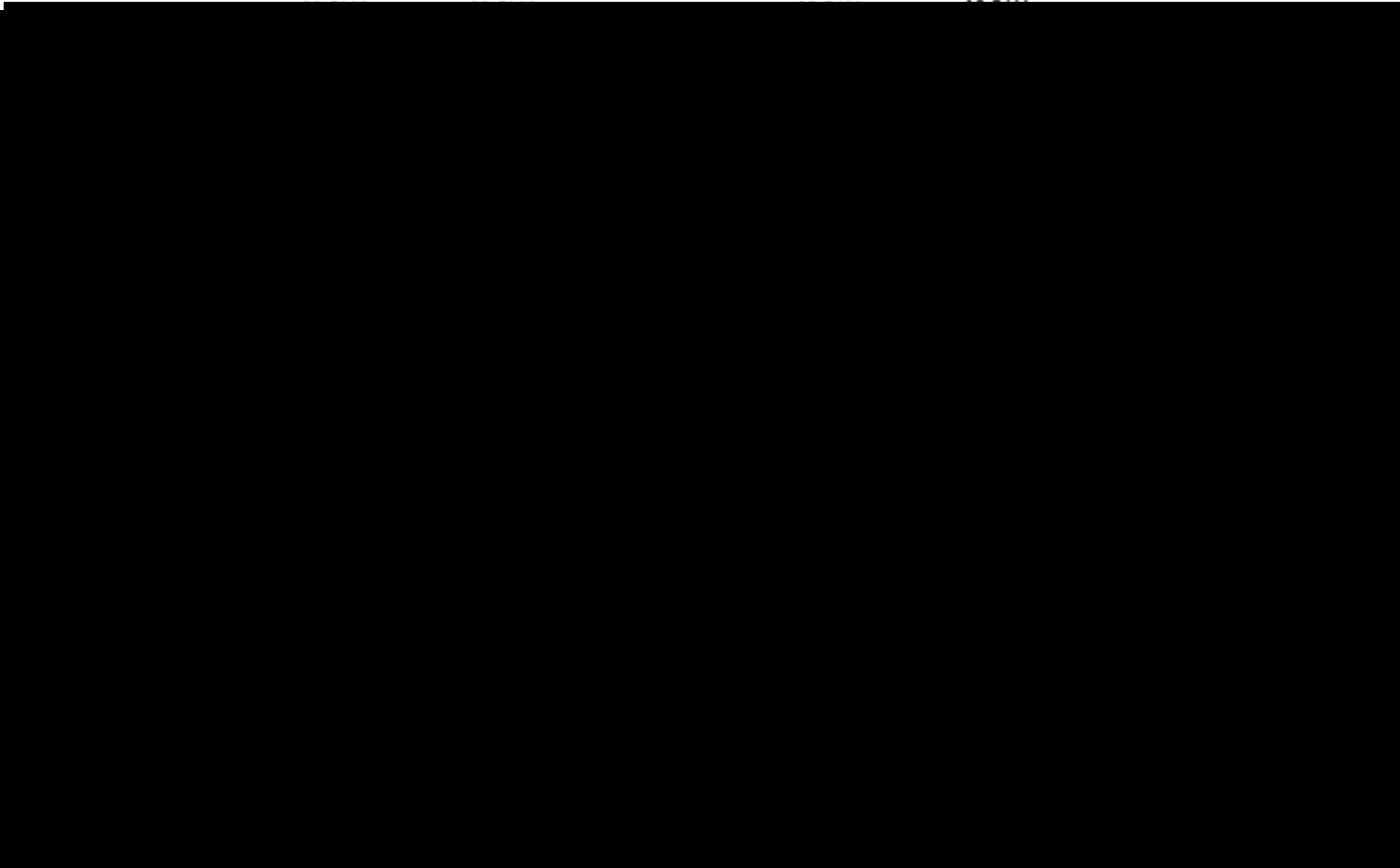
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[4] Alternatives Considered

Alternative 1 (Preferred Solution): 10MW/20MWh Battery Energy Storage System

The proposed BESS system is expected to perform peak load shave functionality to relieve the Hyde Park Substation #496 heavy loading condition. The solution was evaluated based on the Dynamic & Steady State Analysis conducted (see Attachment F).

See Figure 1 for the proposed BESS interconnection one line diagram.



Advantages:

- BESS will help relieve heavily loaded Hyde Park Substation.
- Construction is confined to a single site where the batteries can each interconnect to four (4) different 13.8kV Hyde Park distribution feeders and provide substation bus diversity.
- Can be completed in a minimum of three (3) years.
- This option has the least overall construction impact on the communities.

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- Shorter timeline for Outreach as compared to other alternatives since discussions with the City of Boston is already underway. As part of the overall 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives project and the BESS, outreach and community engagement have already begun with active engagement and participation from the City of Boston and Hyde Park neighborhoods. There has been early involvement with the city for this alternative and the site selection, which helps to smooth the path for siting the BESS with city support and reduces opposition.
- After the five (5)-year window between the BESS installation and the long-term project (new substation installation), the BESS can be redeployed to other locations where it can be useful.

Disadvantages:

- This alternative will require a City of Boston Article 80E Small Project Review and a conditional use permit from the Boston Zoning Board of Appeals. The estimated timeframe for approval is 12 months.
- If the Article 80E and Zoning Relief process is not successful, a zoning exemption would be required from the Energy Facilities Siting Board (EFSB) which would add significant time to the project.
- Local and neighborhood opposition and concerns have been expected for the BESS development. Project Services and Community Relations have been engaging with the City on this issue and is performing targeted community outreach for this project and the Hyde Park solutions as a whole.

Alternative 2: Addition of Two Distribution Feeders

Bring additional capacity to Hyde Park area by constructing two 13.8kV distribution lines from Needham Substation. See Attachment E.

Advantages:

- This alternative would relieve the heavily loaded Hyde Park Substation.
- A filing with the EFSB or Department of Public Utilities (DPU) is not anticipated since this is a distribution related solution.

Disadvantages:

- Construction will be crossing multiple towns, busy highways and bridges, having an adverse impact on the community.

This alternative will either take the same amount of time as Alternative 1 or more, due to the extensive in-street work required for this Alternative. Project Services and Community Relations would have to undertake substantial community outreach and dialogue with City officials, which has not commenced. Expected timeline for outreach is approximately 18 – 24 months and continue ongoing thereafter. Extensive permitting would also be needed in order to cross multiple towns.

- Opposition in Needham is expected since a project was just completed through Needham. If residents object or create organized opposition, significant delays to approvals are expected.
- Routing options would need to be discussed with cities and other stakeholders.

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- Rights would need to be obtained from the towns/cities to install the conduits. Grant of location would be needed for Needham, Dedham and Boston.
- Permits would be needed to cross Route 128 and the bridges, and other entities such as Massachusetts Water Resources Authority (MWRA) or Department of Conservation and Resources (DCR); this can take up to a year.
- The following caveats pose as disadvantages as well and can affect the cost estimates for this alternative:
 - South St in Needham has been recently paved, so it is uncertain whether curb to curb paving would be required. Costs to mill and pave this section are included in the estimate.
 - Two (2) overhead lines will be installed to cross Route 128. This may result in increased labor costs and also decreases reliability of these circuits.
 - Conduit/pipes need to be installed over a bridge crossing at the Dedham/Needham line.
- Expect significant mitigation requests for roadwork restoration, such as curb-to-curb/gutter-to-gutter paving, the Americans with Disabilities Act (ADA) ramp restoration, new sidewalks, and/or traffic calming measures like speed humps, and other community benefits, which could include monetary payments for trees or other types of added green space.
- Depending on the extent of work required at the Needham and Hyde Park substations (not yet known), zoning relief may be required from the Town of Needham and the City of Boston.
- Based on the current forecast, this alternative would not remain as useful after the five (5)-year window between the construction of this alternative and the long-term project; new 115kV/13.8kV substation construction.

Alternative Summary

Although the cost for the Alternative 2 is lower than the preferred alternative, it is critical to note the hurdles and timing delay that would be faced if this alternative were to be implemented as explained above. Opposition in Needham is expected since a project was just completed through Needham. If residents object or create organized opposition, significant delays to approvals are expected. There are also potential hurdles involved in gaining rights and permits for alternative 2. Alternative 2 may take significantly longer than the BESS to implement as explained above. Outreach for the BESS Alternative 1 is already ongoing which helps to smooth the path for siting the BESS with City support and reduces opposition whilst there has been no outreach for Alternative 2.

Two (2) of the feeders that the BESS would connect to would also benefit from the BESS, Circuit 496-H7 supplied from Bus 1 historically runs 80-95% of its summer normal ampacity in the summer. The circuit 362-1209 supplied from Bus 4 historically runs 60-90% of its summer normal in the summer. The BESS would help to reduce the loading on these circuits.

The BESS can be redeployed to other stations where it can be useful after the five (5)-year window between the BESS installation and the long-term project construction. However, based on the current forecast, Alternative 2 would not be as useful after the five (5)-year window when the long-term project or new 115/13.8kV substation is in service to relieve load from Hyde Park and surrounding substations.

There was a River Street BESS solution considered but it was rejected as it did not include as much flexibility in connecting to different distribution circuits and would entail further System Planning studies and

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construction work, which would delay the project significantly. Such a delay would defeat the purpose of having a short-term solution to the capacity issue at this substation. Both alternatives would involve supplying a minority Environmental Justice zone.

Non-Wires Solution Alternatives

A pre-screening assessment was performed based on the scope identified. Projects that answer 'No' to one or more item of the questionnaire do not need to have a non-wires alternative (NWA) solution screening performed. As shown in Table 2 below, all responses are 'Yes' and an NWA screening has been performed.

NWA Pre-Screening Criteria	Response
Preferred Solution Capital Cost > \$3M	Yes
Capacity Need > 3+ Years	Yes
Main Project Driver is Capacity	Yes

Table 2. Non-Wires Alternative pre-screening assessment

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[5] Project Schedule

Milestone	Scheduled Completion Date
IFR Approval	June 2021 May 2023 (Rev 1)
SDC Approval (for applicable projects)	February 2024
Conversion of Tx development costs, 183 to 107 (applicable to projects that require SSF approval and/or ISO preferred alternative selection)	N/A
EPAC Partial Funding Approval (for applicable projects)	February 2024 March 2024 February 2026 June 2026
Preliminary Design Completion (or latest pre-full funding design phase)	December 2027
EPAC Full Funding Approval	January 2028
Siting Approval (for applicable projects)	July 2027
Start of Construction	February 2028
Project In-Service Date (ISD)	December 2028
Project Need Date (for applicable projects)	Present
ISO Related Milestones: (for applicable projects)	
• ISO Preferred Alternative Selected	N/A
• PAC Presentation	N/A
• TCA Submittal	N/A

See Attachment B for the P6 schedule.

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[6] Regulatory Approvals

Regulatory Approvals will be included in the Full Funding PAF.

- The installation of the BESS is expected to require Article 80 Small Project Review from the Boston Planning Department and a Conditional Use Permit from the Boston Zoning Board of Appeal. Siting, with the assistance of outside legal counsel, would file for and obtain the required zoning relief.
- If the City is unsupportive of providing zoning relief, then Siting would file for a zoning exemption request to the EFSB.
- The expected timeframe for a zoning exemption would be about 18 months to 24 months from date of filing to a decision.
- Necessary environmental permit approvals include:
 - o Environmental Protection Agency (EPA) National Pollution Discharge Elimination System (NPDES) Construction General Permit (CGP) and Stormwater Pollution Prevention Plan (SWPPP)
 - ☐ Requires development/engineering of structural stormwater management system for facility and development of Operations and Maintenance (O&M) Plan for long term operation of stormwater management system, including semi-annual inspection, maintenance, and record keeping requirements
 - ☐ Requires regular construction phase erosion and sediment control inspections and reporting
 - o Massachusetts Department of Environmental Protection (MassDEP) Landfill Post-Closure Use Permit (PCUP)
 - ☐ Requires both construction phase and long-term operation inspection, monitoring, and reporting requirements
 - ☐ May require installation of sophisticated gas monitoring system inside facility building to send real-time alert to Operations group for any buildup of combustible and/or toxic gasses within enclosed spaces
 - o Soil and Groundwater Management Plan
 - ☐ Dictates regulatory requirements for construction phase handling, management, and disposal of contaminated soils and groundwater regulated under Massachusetts Contingency Plan
 - o Spill Prevention, Control, and Countermeasure Plan (SPCC) and Tier II Reporting to EPA, MassDEP, Massachusetts Emergency Management Agency (MEMA), Local Fire Dept / Emergency Response Agency
 - ☐ Required for any new oil filled equipment with capacity >1,320 gal.
 - ☐ Required for onsite storage/use of large quantities of hazardous materials (batteries)

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EJ Community Considerations:

EJ Mapping

Identify EJ Communities and populations relevant to Project/Program, through creation of an EJ Map. EJ Map will identify, by Census Block:

- Neighborhoods with English isolation
- Low-income areas
- Areas with concentrations of racial/ethnic minorities
- Areas with vulnerable populations (including those with high existing environmental burdens)
- Areas with combinations of any of the above
- Distressed municipalities and any sub-areas with concentrations of EJ communities
- Areas containing sensitive infrastructure or critical facilities (e.g., public housing, nursing/elder care homes or facilities, parks, hospitals, schools)
- Areas containing high residency of low to moderate income renters

Stakeholders Mapping & Engagement

Identify and communicate with communities and stakeholders in a broad geographic area to determine historical engagement and determine potential Project, Program, or policy impact, including local community-based organizations and leaders, state-wide Non-Governmental Organizations (NGOs) and indigenous tribes.

Language Translation Services

Identify languages spoken in stakeholder communities to enhance strategic communication with multilingual customers. Language identification data shall be used to inform and secure interpreters, generate appropriate translations of materials, conversations, dialogue, and in-community events.

Public Engagement

Host public engagement processes, including through non-traditional and community-based communication channels, to solicit public input. This may include, as applicable:

- Determining the level of community interest
- Conducting analysis of stakeholders who will be immediately impacted by the delivery or implementation of the program/project/services
- Creating and implementing engagement & communication strategies
- Hosting in person and or virtual public feedback sessions to clearly define program/project/services scopes of and timelines
- Identify additional stakeholders to attend future public engagement activities and follow-up on feedback received

Feedback

Establish a database to track all outreach events, interactions, materials, communications received from communities, individuals, and organizations; record all responses, all changes/modifications to projects or programs, and all remediation, mitigation, and community benefit arrangements made to or with respect to relevant EJ communities.

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[7] Risk and Risk Mitigation Plans

Contingency:

Contingency will be included in the full funding PAF.

Risk Mitigation:

Risk	Mitigation
1. Risks and associated mitigation plans will be included in the full funding PAF.	
2.	
3.	
4.	
5.	

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[8] References

References
1. 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives Partial Funding - PAF Rev 2, approved 2/7/2025
2. 22345 - 23375 - Hyde Park-Dorchester Area Plan Land Acquisition Full Funding – PAF Rev 2, approved 8/14/2025
3. 25172 - Hyde Park - Dorchester Area Plan: 200 New England Ave Acquisition Full Funding – PAF, approved 4/11/2025
4. 21334 – Hyde Park BESS – Battery Energy Storage System Solution - IFR, approved 6/11/2021
5. 21334 - Hyde Park BESS – Battery Energy Storage System Solution IFR Rev 1, approved 5/9/2023
6. 21334 – Hyde Park BESS – Battery Energy Storage System Solution, Solution Selection Form, approved 2/15/2024
7. 21334 - Hyde Park Battery Eng Storage Sys - Partial Funding – PAF, approved 3/28/2024
8. 21334 - 24211 - Hyde Park Batter Energy Storage System Partial Funding – PAF, MPAC approved 08/05/2024
9. 21334 - Hyde Park BESS - Battery Eng Stgr Sys - Partial Funding – PAF (Rev 1), approved 2/15/2024
10. 21334-Hyde Park BESS_Material Order Exception Request (MOER) - Package #1 only_Final_pjm (002) _tpr

[9] Attachments (One-Line Diagrams, Images, etc.)

Attachment	Included in Submittal?
A1. 21334 E-24-149 Hyde Park D Station Overhead_AFUDC Calculation REV 25.xlsm R2 (Rev 0)	YES
A2. 21334 E-24-197 Hyde Park BESS (D Station) EST Rev EXT (Rev 1)	YES
A3. ESF - Hyde Park BESS (PF) r0-2 (Rev 2)	YES
B1. Hyde Park Battery Storage Project - Full Remaining Activities Schedule 03-06-24 (Rev 0)	YES
B2. Hyde Park Battery Storage Project - All Activities Schedule 06-05-24 (Rev 1)	YES
B3. 21334 - Hyde Park - All Remaining Activities (December 2028 ISD) 01-26-26 (Rev 2)	YES
C. 21334 Constructability Review Form (Rev 1)	YES
D1. 21334 Hyde Park BESS Scope of Work Dist (002) (Rev 1)	YES
D2. Hyde Park BESS Scope of work (Rev 2)	YES
E. New Needham 13.8kV Feeder One Lines (Rev 1)	YES
F. Hyde Park BESS Dynamic & Steady State Study Report (Rev 1)	YES

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EVERSOURCE

Project Authorization Form

INSTRUCTIONS:

It is the responsibility of the initiator to contact the subject matter experts in the listed area disciplines to determine if the project considerations contained in this list are applicable to their project. They should fill out the checklist and determine a transition plan for the purpose of project execution. See the PAF Guide for additional instruction.

Project Checklist - Transmission & Substation Capital Project	
Project Name: Hyde Park Battery Energy Storage System – Rev 2	Project Number: 21334 (D SS), 24211 (D Line)
Facility Type: ☐ BPS ☐ BES ☐ PTF ☐ local PTF ☐ non-PTF ☐ CIP ☒ Distribution ☐ General Plant ☐ CONVEX ☐ SCADA	
PLANNING	
Is a NX-9 required?	No
Is an ISO-NE PAC presentation required?	No
ISO-NE Presentation Date (if completed):	N/A
Cost in ISO-NE Presentation (\$M) (if completed):	N/A
Is a PPA required?	Yes
ISO-NE Approval Date (if completed):	TBD
Is a TCA Application Required?	No
TCA Submittal Date (if submitted):	N/A
Cost in TCA Submittal (\$M) (if submitted):	N/A
Enter ISO-NE RSP / Asset Condition ID, if assigned:	ES-23-LSP-032
PLANNING/PROTECTION & CONTROLS	
Are RAS/SPS/UVLs affected?	No
OPERATIONS	
Outage Required?	☐ Primary Equipment (Power Transfer) ☒ Secondary Equipment (P&C only) ☐ Outage Not Required
Do SCLL Conditions Exist?	No
Has an outage schedule been approved?	No
Are Operations & Maintenance procedures/training required?	Yes
STANDARDS	
Does the project include standard equipment and designs?	Yes
SUBSTATION ENGINEERING	
Does this impact Revenue Metering	No
Is preliminary short circuit/ breaker duty analysis required?	No
Are there any changes to the baseline audible noise?	Yes
Is there an impact to the existing ground grid?	No
Is a Transient Over Voltage (TOV) analysis required?	Yes

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EVERSOURCE

Project Authorization Form

Project Checklist - Transmission & Substation Capital Project	
Project Name: Hyde Park Battery Energy Storage System – Rev 2	Project Number: 21334 (D SS), 24211 (D Line)
P&C ENGINEERING	
OP-22 - Are PMUs and DDR required?	No
If BPS, is an NPCC Directory #4 presentation required?	No
Are ISO-NE OP-24 Appendix B updates necessary?	No
TRANSMISSION LINE ENGINEERING	
Are there any changes that affect the baseline EMF?	No
Are there any changes that affect the baseline EMI?	No
SITING	
Has Siting reviewed this project?	Yes
Name of Siting contact:	Eamon McGilligan
Is a Siting filing required? <i>(If yes, list in regulatory approvals section)</i>	Yes
PERMITTING	
Is there any permitting required? <i>(If yes, list in regulatory approvals section)</i>	Yes
PROJECT SERVICES (OUTREACH) and ROW WORK COORDINATION	
Has Project Services reviewed this project?	Yes
Name of Project Services contact:	Meredith Boericke
What is the level of outreach expected?	High
Has ROW Coordination reviewed this project?	Yes
Is this project coordinating with other existing scheduled work?	No
ENVIRONMENTAL JUSTICE COMMUNITIES	
Has the Equity and Environmental Justice team reviewed this project?	Yes
Name of Equity and Environmental Justice contact:	Alicine Francois
Is any part of this project in an Environmental Justice Community?	Yes
Was the Equity Framework Checklist completed?	No
Has enhanced outreach required per State Regulatory requirements been completed?	No
INITIATOR	
Has a field constructability review been completed?	Yes

Project Checklist - Transmission & Substation Capital Project

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EVERSOURCE

Project Authorization Form

Project Name: Hyde Park Battery Energy Storage System – Rev 2	Project Number: 21334 (D SS), 24211 (D Line)
INVESTMENT RECOVERY	
Does the project require development of an Investment Recovery plan?	No _____
COST ESTIMATING	
What team/firm prepared the cost estimate?	Estimate was prepared by the Eversource Estimating Team _____
Name of the person who prepared the estimate:	Muthana Al-Quraishi _____
Was the estimate reviewed by Eversource Estimating?	Yes - Full Review _____

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Project Number:

Project Title:

D- Station

PAF (dollars in thousands)

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Line item Category	Prior Authorized	Actuals to Date	2024 to Go	2025	2026	2027	2028	2029	Total
1. ROW / Easements / Land purchases									
2. Environmental Approvals / Permits		\$6	\$3						\$8
3. Outreach		\$12	\$3						\$15
4. Siting Approvals / Permits		\$12	\$9						\$21
5. Engineering / Design		\$17	\$7						\$24
6. Materials (Eversource purchased)		\$2							\$2
7. Construction (incl matl's by contractors)		\$11							\$11
8. Testing / Commissioning									
9. PMT Project Mgmt Team		\$88	\$3						\$91
10. Removals									
11. Other		-\$1							-\$1
12. Risks									
Project Directs Total w/Risks		\$148	\$23						\$170
13. Indirects/Overhead		\$222	\$47						\$269
14. AFUDC		\$30	\$30						\$60
Project Total - Baseline Budget		\$400	\$100						\$500
15. Contingency									
#N/A									
Project Total - Funding Approval Level		\$400	\$100						\$500
16. Reimbursables/Customer Contribution									
#N/A									
Project Total (less reimbursables)		\$400	\$100						\$500
Total O&M									
TOTAL REQUEST	\$300	\$400	\$100						\$500

Breakout costs incl above

ST Labor		\$108	\$17						\$125
OT Labor									
Outside services - Labor		\$38	\$6						\$44
Materials (All materials incl. outside service purchased which is shown in constr. above)		\$2							\$2
Removals									
Other		-\$1							-\$1
Indirects		\$222	\$47						\$269
AFUDC		\$30	\$30						\$60
TOTAL CAPITAL REQUEST - with out Reimbursables	\$300	\$400	\$100						\$500

- Instructions:**
1. Input the project type info into the green cells on the Overall summary tab / answer escalation question
 2. Input direct costs (DOLLARS AND CENTS) and customer contribution (if applicable) in the green highlighted cells for the appropriate company which are separate tabs (green tabs)
 3. Input customer contribution as a negative number
 4. For D and Live work and station work have different tender rates so make sure you fill out the correct template section of each tab
 5. Input zero to green cells on overall summary tab. They then flow into the other tabs.

Template version: rev22

Template version: rev22

D - Station Work

Direct Costs									
WBS Code	WBS Description	Prior Authorized	Actuals to Date	2024	2025	2026	2027	2028	Total
Costs									
Eversource Internal									
1	ROW / Eversource / Land purchases								
2	Environmental Approvals / Permits			\$ 5,522	\$ 3,000				
3	Outreach			\$ 8,686	\$ 3,000				
4	Sting Approvals / Permits			\$ 12,649	\$ 6,000				
5	Engineering / Design			\$ 5,509	\$ 2,000				
6	Materials			\$ 2,358					
7	Construction			\$ 10,963					
8	Testing / Commissioning								
9	PM/ Project Mgmt Team			\$ 63,811	\$ 1,000				
10	Removals								
11	Other			\$ (1,037)					
12	Risk								
15	Contingency								
Outside vendor or contractor									
1	ROW / Eversource / Land purchases								
2	Environmental Approvals / Permits								
3	Outreach			\$ 2,633	\$ -				
4	Sting Approvals / Permits								
5	Engineering / Design			\$ 11,155	\$ 4,000				
6	Materials								
7	Construction								
8	Testing / Commissioning								
9	PM/ Project Mgmt Team			\$ 24,633	\$ 1,000				
10	Removals								
11	Other								
12	Risk								
15	Contingency								
		\$ -	\$ 147,344	\$ 22,000	\$ -	\$ -	\$ -	\$ -	\$ 176,444

Escalation (if not included above)									
Escalation rates - Eversource Labor				3.0%		6.1%		9.3%	
Escalation rates - External Labor				3.0%		6.1%		9.3%	
Escalation rates - Materials				1.5%		3.0%		4.6%	
Eversource Internal									
1	ROW / Eversource / Land purchases	\$	-	\$	-	\$	-	\$	-
2	Environmental Approvals / Permits	\$	-	\$	-	\$	-	\$	-
3	Outreach	\$	-	\$	-	\$	-	\$	-
4	Sting Approvals / Permits	\$	-	\$	-	\$	-	\$	-
5	Engineering / Design	\$	-	\$	-	\$	-	\$	-
6	Materials	\$	-	\$	-	\$	-	\$	-
7	Construction	\$	-	\$	-	\$	-	\$	-
8	Testing / Commissioning	\$	-	\$	-	\$	-	\$	-
9	PM/ Project Mgmt Team	\$	-	\$	-	\$	-	\$	-
10	Removals	\$	-	\$	-	\$	-	\$	-
11	Other	\$	-	\$	-	\$	-	\$	-
12	Risk	\$	-	\$	-	\$	-	\$	-
15	Contingency								
Outside vendor or contractor									
1	ROW / Eversource / Land purchases	\$	-	\$	-	\$	-	\$	-
2	Environmental Approvals / Permits	\$	-	\$	-	\$	-	\$	-
3	Outreach	\$	-	\$	-	\$	-	\$	-
4	Sting Approvals / Permits	\$	-	\$	-	\$	-	\$	-
5	Engineering / Design	\$	-	\$	-	\$	-	\$	-
6	Materials	\$	-	\$	-	\$	-	\$	-
7	Construction	\$	-	\$	-	\$	-	\$	-
8	Testing / Commissioning	\$	-	\$	-	\$	-	\$	-
9	PM/ Project Mgmt Team	\$	-	\$	-	\$	-	\$	-
10	Removals	\$	-	\$	-	\$	-	\$	-
11	Other	\$	-	\$	-	\$	-	\$	-
12	Risk								
15	Contingency								
		\$	-	\$	-	\$	-	\$	-

Escalated costs									
Eversource Internal									
1	ROW / Eversource / Land purchases	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
2	Environmental Approvals / Permits	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
3	Outreach	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
4	Sting Approvals / Permits	\$ -	\$ 5,522	\$ 3,000	\$ -	\$ -	\$ -	\$ -	\$ -
5	Engineering / Design	\$ -	\$ 8,686	\$ 3,000	\$ -	\$ -	\$ -	\$ -	\$ -
6	Materials	\$ -	\$ 12,649	\$ 6,000	\$ -	\$ -	\$ -	\$ -	\$ -
7	Construction	\$ -	\$ 3,509	\$ 2,000	\$ -	\$ -	\$ -	\$ -	\$ -
8	Testing / Commissioning	\$ -	\$ 2,633	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
9	PM/ Project Mgmt Team	\$ -	\$ 10,963	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
10	Removals	\$ -	\$ 63,811	\$ 1,000	\$ -	\$ -	\$ -	\$ -	\$ -
11	Other	\$ -	\$ (1,037)	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
12	Risk	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
15	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Outside vendor or contractor									
1	ROW / Eversource / Land purchases	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
2	Environmental Approvals / Permits	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
3	Outreach	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
4	Sting Approvals / Permits	\$ -	\$ 2,633	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
5	Engineering / Design	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
6	Materials	\$ -	\$ 11,155	\$ 4,000	\$ -	\$ -	\$ -	\$ -	\$ -
7	Construction	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
8	Testing / Commissioning	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
9	PM/ Project Mgmt Team	\$ -	\$ 24,633	\$ 1,000	\$ -	\$ -	\$ -	\$ -	\$ -
10	Removals	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
11	Other	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
12	Risk	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
15	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
		\$ -	\$ 147,344	\$ 22,000	\$ -	\$ -	\$ -	\$ -	\$ 176,644

PM/ Vendors														
1	Laber (PM/ROW/Env/Outreach/Sting)	\$ -	\$ 91,300	\$ 14,500	\$ -	\$ -	\$ -	\$ -	\$ -					
2	Laber (Eng/Const/Test)	\$ -	\$ 16,002	\$ 3,000	\$ -	\$ -	\$ -	\$ -	\$ -					
3	Materials	\$ -	\$ 2,503	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -					
4	Outside Services	\$ -	\$ 26,421	\$ 6,000	\$ -	\$ -	\$ -	\$ -	\$ -					
5	Other	\$ -	\$ (1,037)	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -					
6	Risk	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -					
7	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -					
Total Direct Costs		\$ -	\$ 147,844	\$ 22,500	\$ -	\$ -	\$ -	\$ -	\$ 176,644					
Customer Contribution		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -					
Indirect Costs														
Indirect Costs														
Non-Productive(BNP)	Non-Productive(BNP)	10.0%	\$ -	\$ 17,960	10.0%	\$ 1,150	10.0%	\$ -	10.0%	\$ -	10.0%	\$ -	10.0%	\$ 21,095
Project Development(PDP)	Project Development(PDP)	8.0%	\$ -	\$ 8,184	8.0%	\$ 1,000	8.0%	\$ -	8.0%	\$ -	8.0%	\$ -	8.0%	\$ 1,496
General Services(GS)	General Services(GS)	70.0%	\$ -	\$ 84,714	70.0%	\$ 12,270	70.0%	\$ -	70.0%	\$ -	70.0%	\$ -	70.0%	\$ 96,794
MISC(GM)	MISC(GM)	1.0%	\$ -	\$ 1,517	1.0%	\$ 1,400	1.0%	\$ -	1.0%	\$ -	1.0%	\$ -	1.0%	\$ 1,617
EM/OPS (BIL 2B)	EM/OPS (BIL 2B)	61.0%	\$ -	\$ 89,202	61.0%	\$ 24,438	61.0%	\$ -	61.0%	\$ -	61.0%	\$ -	61.0%	\$ 124,849
AS&M(GM)	AS&M(GM)	6.0%	\$ -	\$ 8,503	6.0%	\$ 1,000	6.0%	\$ -	6.0%	\$ -	6.0%	\$ -	6.0%	\$ 1,567
Lobby Staff	Lobby Staff	\$ -	\$ -	\$ 1,500	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Stations and STS	Stations and STS	\$ -	\$ -	\$ 1,500	\$ -	\$ 1,500	\$ -	\$ 1,500	\$ -	\$ 1,500	\$ -	\$ 1,500	\$ -	\$ 4,500
Vehicles(DV)	Vehicles(DV)	21.2%	\$ -	\$ 855	21.2%	\$ 3,495	21.2%	\$ -	21.2%	\$ -	21.2%	\$ -	21.2%	\$ 4,350
Total Indirect Costs			\$ -	\$ 140,000		\$ 30,150		\$ -		\$ -		\$ -		\$ 66,750
All Costs (incl. Indirect)				\$ 300,000		\$ 176,150		\$ -		\$ -		\$ -		\$ 243,394
AT&M(GM)		\$ -	\$ 20,000		\$ 20,000									\$ 20,000
TOTALS				\$ 300,000		\$ 196,150		\$ -		\$ -		\$ -		\$ 263,394

Project Number:

Project Title:

D- Station

PAF (dollars in thousands)

And on years e version; rev25

Line item Category	Prior Authorized	Actuals to Date	2024 to Go	2025	2026	2027	2028	2029	Total
1. ROW / Easements / Land purchases									
2. Environmental Approvals / Permits		\$6	\$156	\$161					\$323
3. Outreach		\$14	\$13	\$13					\$39
4. Siting Approvals / Permits		\$13	\$114	\$117					\$244
5. Engineering / Design		\$30	\$169	\$174					\$374
6. Materials (Eversource purchased)									
7. Construction (incl matl's by contractors)		\$14	\$11	\$11					\$36
8. Testing / Commissioning									
9. PMT Project Mgmt Team		\$70	\$154	\$158					\$382
10. Removals									
11. Other		\$4							\$4
12. Risks									
Project Directs Total w/Risks		\$151	\$616	\$635					\$1,401
13. Indirects/Overhead		\$235	\$1,008	\$1,038					\$2,282
14. AFUDC		\$33	\$86	\$208					\$326
Project Total - Baseline Budget		\$418	\$1,710	\$1,880					\$4,009
15. Contingency									
#N/A									
Project Total - Funding Approval Level	\$500	\$418	\$1,710	\$1,880					\$4,009
16. Reimbursables/Customer Contribution									
#N/A									
Project Total (less reimbursables)	\$500	\$418	\$1,710	\$1,880					\$4,009
Total O&M									
TOTAL REQUEST	\$500	\$418	\$1,710	\$1,880					\$4,009

Breakout costs incl above

ST Labor		\$114	\$301	\$310					\$726
OT Labor									
Outside services - Labor		\$30	\$315	\$324					\$670
Materials (All materials incl. outside service purchased which is shown in constr. above)		\$2							\$2
Removals									
Other		\$4							\$4
Indirects		\$235	\$1,008	\$1,038					\$2,282
AFUDC		\$33	\$86	\$208					\$326
TOTAL CAPITAL REQUEST - with out Reimbursables	\$500	\$418	\$1,710	\$1,880					\$4,009

[illegible]

Project name **Hyde Park BESS (PF)**
Project no **21334**
Project manager **ADAM KELLEY**
Project estimator **Muthana Al-Quraishi**

[Return to TOC](#)

PAF (dollars in th And on years

Line item Category	Prior Authorized	Actuals to Date	2026 to Go	2030	Total
1. ROW / Easements / Land purchases		16	-	-	16
2. Environmental Approvals / Permits		15	50	-	65
3. Project Services		60	50	-	110
4. Siting Approvals / Permits		35	100	-	135
5. Engineering / Design		343	500	-	843
6. Materials (Eversource purchased)		2	1,211	-	1,213
7. Construction (incl contractor sourced material)		-	-	-	-
8. Testing / Commissioning		25	-	-	25
9. PMT Project Mgmt Team		577	283	-	860
10. Removals		-	-	-	-
11. Other (including CWIP & Taxes)		24	-	-	24
12. Risks		-	-	-	-
Project Directs Total w/Risks	-	1,096	2,194	-	3,290
13. Indirects/Overhead		1,280	819	-	2,099
14. AFUDC		202	175	-	378
Project Total - Baseline Budget	-	2,579	3,188	-	5,767
15. Contingency		-	-	-	-
Project Total - Approval Level	4,009	2,579	3,188	-	5,767
16. Reimbursables/Customer Contribution		-	-	-	-
Project Total (less reimbursables)	4,009	2,579	3,188	-	5,767
Total O&M					-
TOTAL REQUEST	4,009	2,579	3,188	-	5,767

Hyde Park BESS (PF) - REV01					
1. ROW / Easements / Land purchases		16	-	-	16
2. Environmental Approvals / Permits		15	50	-	65
3. Project Services		60	50	-	110
4. Siting Approvals / Permits		35	100	-	135
5. Engineering / Design		343	500	-	843
6. Materials (Eversource purchased)		2	1,211	-	1,213
7. Construction (incl contractor sourced material)		-	-	-	-
8. Testing / Commissioning		25	-	-	25
9. PMT Project Mgmt Team		577	283	-	860
10. Removals		-	-	-	-
11. Other (including CWIP & Taxes)		24	-	-	24
12. Risks		-	-	-	-
Project Directs Total w/Risks	-	1,096	2,194	-	3,290
13. Indirects/Overhead		1,280	819	-	2,099
14. AFUDC		202	175	-	378
Project Total - Baseline Budget	-	2,579	3,188	-	5,767
15. Contingency		-	-	-	-
Project Total - Approval Level	4,009	2,579	3,188	-	5,767
16. Reimbursables/Customer Contribution		-	-	-	-
Project Total (less reimbursables)	4,009	2,579	3,188	-	5,767
Total O&M					-
TOTAL REQUEST	4,009	2,579	3,188	-	5,767

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24

BESS - Remaining TJP

06-Mar-24 15:55

Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028
Project Lead: Adam Kelley Adam S. Kelley		1037d	15-Jun-21 A	09-Mar-28											
80089926 Hyde Park BESS - Battery Energy Storage System Solution 03-06-24		1037d	15-Jun-21 A	09-Mar-28											
80089926.01 PROJECT PLANNING & FUNDING		686d	15-Jun-21 A	05-Nov-26											
PPF1000	Project Planning & Funding	686d	15-Jun-21 A	05-Nov-26	In Progress										
80089926.01.12 T-Minus Milestones		661d	29-Mar-24	05-Nov-26											
AT2605	Preliminary Design / Engineering Completed	0d		04-Jun-25	Not Started	ENG-PR	ENG.ST.SS		DE2120, DE2100, DE2140, DE2060, DE2040, DE2160						
AT1811	Detail Design/Engineering Completed	0d		18-Dec-25	Not Started	ENG-DT	ENG.ST.SS		COM1000, PC1110, UG1280, OH1430, EE1210, EE1230						
AT0805	Final Engineering Issued for Construction W/BOM	0d		09-Jan-26	Not Started	ENG-IFC	ENG.ST.SS		COM1030, PC1330, UG1370, OH1690, EE1210, EE1250	ISD					
80089926.01.12.01 Civil T-Minus Milestones		930d	29-Mar-24	15-Oct-26											
AT3001	Scope Document Approved	0d		29-Mar-24	Not Started	ENG-SC	ENG.ST.SS		PE1720, ATCGE1010	AT00					
BT2615	Civil Preliminary Design Complete	0d		08-Apr-25	Not Started	ENG-PR	ENG.ST.SS		DE2160	BD1010, PP1000, PHZ1010					
AT1030	ISO I.3.9 Approval	0d		30-May-25	Not Started		ENG.SYS.I		ISO1190	AT00					
BT2625	Civil Detail Design Complete	0d		18-Dec-25	Not Started	ENG-DT	ENG.ST.SS		EE1330	AT00					
AT0805C	IFC - Civil (BG)	0d		09-Jan-26	Not Started	AG-IFC	ENG.ST.SS		EE1350	AT00					
AT0401	Civil Long Lead Items Delivered	0d		19-Feb-26	Not Started		PURC		CMAT1010	AT00					
AT0814	Civil Construction Contracts Awarded - Civil BG	0d		17-Mar-26	Not Started		PURC		BCN2150	AT00					
AT0203	Civil Constr Auth Notice (CAN) completed [go/ no go]	0d		25-Jun-26	Not Started		PM			AT00, BCN2090					
AT00	T-00 Construction Start Civil	0d	26-Jun-26*		Not Started		PM		AT0401, AT0203, AT0805C, BT2625, AT0814, AT0801, AT0203	BCN2090					
AT0801	Environmental & Local Permits Received	0d		15-Oct-26	Not Started		PM		PP1000, BD1000, BD1010	AT00					
80089926.01.12.02 Electrical T-Minus Milestones		533d	20-May-25	05-Nov-26											
BT2605	Electrical Preliminary Design Complete	0d		20-May-25	Not Started	ENG-PR	ENG.ST.SS		DE2040	BT00, BD1010, PP1000					
BT2404	Assign Sta Tech (Comissioning) Engineer	0d		20-May-25	Not Started		ENG.ST.SL		DE2040	CT00					
BT1811	Electrical Detail Design Complete	0d		17-Oct-25	Not Started	ENG-DT	ENG.ST.SS		EE1210	BT00					
BT0805E	IFC - Electrical (AG)	0d		07-Nov-25	Not Started	AG-IFC	ENG.ST.SS		EE1310	BT00					
BT0814	Electrical Construction Contracts Awarded - Electrical AG	0d		26-Dec-25	Not Started		PURC		BCN2050	BT00					
BT0801	Environmental & Local Permits Received	0d		15-Oct-26	Not Started		PM		PRE1010, PP1000, BD1000, BD1010	BT00					
BT0203	Electrical Constr Auth Notice (CAN) completed [go/ no go]	0d		04-Nov-26	Not Started		PM			BT00					
BT0401	Electrical Long Lead Items Delivered	0d		04-Nov-26	Not Started		PURC		EMAT1230	BT00					
BT00	T-00 Construction Start Electrical	0d	05-Nov-26		Not Started		PM		BT0401, BT0203, BT0805E, BT1811, BU171000, BT2605	BCN1050, BCN1030, ECN1040					
80089926.01.12.06 P&C T-Minus Milestones		390d	22-Apr-25	29-Oct-26											

Actual Work

Remaining Work

Critical Remaining Work

Milestone

Summary

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TASK filter: EMA REPORT - Remaining TJP.

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Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP								06-Mar-24 15:55						
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028
	DE2160	ES Engineering Review Civil Substation Preliminary Design		10d	26-Mar-25	08-Apr-25	Not Started	ES-PRE	ENG.ST.SS	Shodhan Patel	DE2150	AT2605, BT2615, EE1320, DE2120, DE1000					
	80089926.01.2.3.00.3 Electrical Conceptual & Preliminary Engineering Design			60d	26-Feb-25	20-May-25											
	80089926.01.2.3.00.3.02 Electrical Preliminary Design			60d	26-Feb-25	20-May-25											
	DE2030	Engineer Develop Electrical Substation Preliminary Design		50d	26-Feb-25	06-May-25	Not Started		ENG.ST.SS	Shodhan Patel	PE1100, EOC1040	DE2040					
	DE2040	ES Engineering Review Electrical Substation Preliminary Design		10d	07-May-25	20-May-25	Not Started	ES-PRE	ENG.ST.SS	Shodhan Patel	DE2030	AT2605, BT2404, BT2605, CMPL1000, CMPL1010, EE1320					
	80089926.01.2.3.00.9 P&C Conceptual & Preliminary Engineering Design			40d	26-Feb-25	22-Apr-25											
	80089926.01.2.3.00.9.06 P&C Preliminary Design			40d	26-Feb-25	22-Apr-25											
	DE2090	Engineer Develop P&C Preliminary Design		30d	26-Feb-25	08-Apr-25	Not Started		ENG.ST.P8	P&C Engineer	PE1400, EOC1040	DE2100					
	DE2100	ES Engineering Review P&C Preliminary Design		10d	09-Apr-25	22-Apr-25	Not Started	ES-PRE	ENG.ST.P8	P&C Engineer Generic	DE2090	AT2605, ET2605, ET2404, SST1000, FC1100, CMPL1000					
	80089926.01.2.3.00.07 Telecom Conceptual & Preliminary Engineering Design			70d	26-Feb-25	04-Jun-25											
	DE2110	Develop Communications Preliminary Scope Engineering Design in Accordance with Deliverables Checklist		30d	26-Feb-25	08-Apr-25	Not Started		ENG.ST.SL	Bechir El Khoury Hanna	PE1500, EOC1040	DE2120					
	DE2120	ES Engineering Review Communications Preliminary Design		10d	21-May-25	04-Jun-25	Not Started	ES-PRE	ENG.ST.SL	Bechir El Khoury Hanna	DE2110, PE1500, PE1100, PE1300, PE1720, DE2160, DE2040	FT2605, COM1040, AT2605, COM1010, DE1000					
	80089926.02 ROW & EASEMENTS			44d	21-May-25	23-Jul-25											
	ROW1000	ROW & Easements		44d	21-May-25	23-Jul-25	Not Started										
	80089926.02.01 Right of Way			44d	21-May-25	23-Jul-25											
	80089926.02.01.02 Easements			44d	21-May-25	23-Jul-25											
	PRE1000	Identify Required Takes & Easements		22d	21-May-25	20-Jun-25	Not Started		RE	Jennifer Valente	PF1010, DE2040	PRE1010					
	PRE1010	Obtain Required Takes & Easements		22d	23-Jun-25	23-Jul-25	Not Started		RE	Jennifer Valente	PRE1000, PF1010	BT0801, FINCO1000					
	80089926.03 ENVIRONMENTAL APPROVALS & PERMITS			434d	05-Feb-25	15-Oct-26											
	PHZ1010	Construction Environmental Permits		254d	05-Feb-25*	05-Feb-26	Not Started		ENG.ENV.I	Christopher Newhall	BT2615	AT00					
	PR1000	C A N [go / no go] Submitted & Approved		1d	15-Oct-26	15-Oct-26	Not Started		PM	Adam S. Kelley		FINCO1000, BCN1030, FCN1010, FCN1020					
	PERM1000	Environmental Approvals & Permits DEP		434d	05-Feb-25	15-Oct-26	Not Started										
	80089926.03.06 Environmental Services / Affairs			35d	25-Feb-26	14-Apr-26											
	PHZ1000	Plan(s) for PCB's; Asbestos; Contaminated Soil		35d	25-Feb-26	14-Apr-26	Not Started		ENG.ENV.I	Christopher Newhall	BD1000, BCN2120	BT00, BCN2090					
	80089926.04 SITING APPROVALS / PERMITS / PROJECT SERVICES			672d	25-Aug-22 A	15-Oct-26											
	SIT1000	Siting Approvals / Permits / Project Services		672d	25-Aug-22 A	15-Oct-26	In Progress										
	80089926.04.02 Regional Approvals			343d	16-Feb-24 A	03-Jul-25											
	80089926.04.02.01 ISO Technical Approval (I.3.9)			343d	16-Feb-24 A	03-Jul-25											
	80089926.04.02.01.01 Solutions Study			184d	16-Feb-24 A	14-Nov-24											
	ISO1000	Define PPA Level (0.4)(2.4 req'r Soln study)		0d	26-Feb-24		Not Started		ENG.SYS.I		PI1200, PI1000, SDC1000	ISO1010					
	ISO1010	Create Rpt Pkg of Needs/ Solutions, -25/ +50 Estimate		54d	16-Feb-24 A	09-May-24	In Progress		ENG.SYS.I	Mariam A. Chatha	ISO1000, PI1000, SDC1000	ISO1020					
	ISO1020	Soln Pkg, -25/ +50 Est Complete		0d		09-May-24	Not Started		ENG.SYS.I		ISO1010	ISO1030					
	ISO1040	ISO supports Solution proposal		0d		05-Aug-24	Not Started		ENG.SYS.I		ISO1030	ISO1050					
	ISO1030	ISO review/ comment, re-Submit Pkg, ISO review		60d	10-May-24	05-Aug-24	Not Started		ENG.SYS.I	Mariam A. Chatha	ISO1020	ISO1040					

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24				BESS - Remaining TJP							06-Mar-24 15:55													
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	20242025202620272028											
ISO1060		TO supports Solution proposal		0d		03-Sep-24	Not Started		ENG.SYS.:		ISO1050	ISO1070	◆ TO supports Solution proposal											
ISO1050		Trans Owner (TO) Review/ Comment		20d	06-Aug-24	03-Sep-24	Not Started		ENG.SYS.:	Mariam A. Chatha	ISO1040	ISO1060	■ Trans Owner (TO) Review/ Comment											
ISO1070		Plan'g Advisory Committee (PAC) Review/ Comment		30d	04-Sep-24	16-Oct-24	Not Started		ENG.SYS.:	Mariam A. Chatha	ISO1060	ISO1080	■ Plan'g Advisory Committee (PAC) Review/ Comment											
ISO1080		Incorporate PAC comments		20d	17-Oct-24	14-Nov-24	Not Started		ENG.SYS.:	Mariam A. Chatha	ISO1070	ISO1090	■ Incorporate PAC comments											
ISO1090		Issue Final Solution Study Report Accepted (ISO Ltr Rec'd)		0d		14-Nov-24	Not Started		ENG.SYS.:		ISO1080	DE1600, ISO1100	◆ Issue Final Solution Study Report Accepted (ISO L											
80089926.04.02.01.02 ISO 1.3.9				159d	15-Nov-24	03-Jul-25							▼ 03-Jul-25, 80089926.04.02.01.02 ISO 1											
80089926.04.02.01.02.1 System Impact Study				63d	15-Nov-24	18-Feb-25							▼ 18-Feb-25, 80089926.04.02.01.02.1 System I											
ISO1100		Develop/ Present Scope of Work for SIS (System Impact Study)		21d	15-Nov-24	16-Dec-24	Not Started		ENG.SYS.:	Mariam A. Chatha	ISO1090	ISO1110	■ Develop/ Present Scope of Work for SIS (System											
ISO1110		Perform System Impact Study (SIS)		42d	17-Dec-24	18-Feb-25	Not Started		ENG.SYS.:	Mariam A. Chatha	ISO1100	ISO1140, ISO1120	■ Perform System Impact Study (SIS)											
80089926.04.02.01.02.2 Soln Pkg & Estimate				63d	17-Dec-24	19-Mar-25							▼ 19-Mar-25, 80089926.04.02.01.02.2 Soln Pk											
ISO1130		Soln Pkg revised, -25/ +25 Estimate Complete		0d		19-Mar-25	Not Started		ENG.SYS.:		ISO1120	ISO1150	◆ Soln Pkg revised, -25/ +25 Estimate Complet											
ISO1120		Revise Soln Pkg, -25/ +25 Estimate		63d	17-Dec-24	19-Mar-25	Not Started		ENG.SYS.:	Mariam A. Chatha	ISO1110	ISO1130	■ Revise Soln Pkg, -25/ +25 Estimate											
80089926.04.02.01.02.3 Task Forces				21d	19-Feb-25	19-Mar-25							▼ 19-Mar-25, 80089926.04.02.01.02.3 Task Fc											
ISO1140		Trans Task Force & Stability Task Force Review/ Comment		21d	19-Feb-25	19-Mar-25	Not Started		ENG.SYS.:	Mariam A. Chatha	ISO1110	ISO1150	■ Trans Task Force & Stability Task Force Review											
ISO1150		TTF & STF support Solution proposal		0d		19-Mar-25	Not Started		ENG.SYS.:		ISO1140, ISO1130	ISO1160	◆ TTF & STF support Solution proposal											
80089926.04.02.01.02.4 Reliability Committee				51d	20-Mar-25	30-May-25							▼ 30-May-25, 80089926.04.02.01.02.4 Reli											
ISO1160		Draft the PPA Form		15d	20-Mar-25	09-Apr-25	Not Started		ENG.SYS.:	Mariam A. Chatha	ISO1150	ISO1170	■ Draft the PPA Form											
ISO1170		Submit PPA Form to ISO		0d		09-Apr-25	Not Started		ENG.SYS.:		ISO1160	ISO1180	◆ Submit PPA Form to ISO											
ISO1180		Reliabilty Committee (RC) Review/ Recommendation		21d	10-Apr-25	08-May-25	Not Started		ENG.SYS.:	Mariam A. Chatha	ISO1170	ISO1190	■ Reliability Committee (RC) Review/ Recom											
ISO1190		Final Report/ Presentation		15d	09-May-25	30-May-25	Not Started		ENG.SYS.:	Mariam A. Chatha	ISO1180	AT1030, FINCO1000	■ Final Report/ Presentation											
80089926.04.02.01.02.5 P&C NPCC Directory 4				51d	23-Apr-25	03-Jul-25							▼ 03-Jul-25, 80089926.04.02.01.02.5 P&C											
NPCC1090		Tie from Tasks -Approve Application Diagram, One Lines, BOM, Floor Plan (Preliminary P&C IFC)		1d	23-Apr-25	23-Apr-25	Not Started		ENG.ST.P&	P&C Engineer	DE2100	NPCC1000	■ Tie from Tasks -Approve Application Diagram											
NPCC1000		Compile Support Drawings Package - NPCC Directory 4 Submittal		5d	24-Apr-25	30-Apr-25	Not Started		ENG.ST.P&	P&C Engineer	NPCC1090	NPCC1010	■ Compile Support Drawings Package - NPCC											
NPCC1010		Complete D4 Form (PSRF) - NPCC Directory 4 Submittal		2d	01-May-25	02-May-25	Not Started		ENG.ST.P&	P&C Engineer	NPCC1000	NPCC1020	■ Complete D4 Form (PSRF) - NPCC Directo											
NPCC1020		Complete Cover Letter- NPCC Directory 4 Submittal		3d	05-May-25	07-May-25	Not Started		ENG.ST.P&	P&C Engineer	NPCC1010	NPCC1030	■ Complete Cover Letter- NPCC Directory 4											
NPCC1030		Develop + Provide Power Point Presentation - NPCC Directory 4 Submittal		5d	08-May-25	14-May-25	Not Started		ENG.ST.P&	P&C Engineer	NPCC1020	NPCC1040	■ Develop + Provide Power Point Presentatio											
NPCC1040		Reivew & Comment - NPCC Directory 4 Submittal		3d	15-May-25	19-May-25	Not Started		ENG.ST.P&	P&C Engineer	NPCC1030	NPCC1050	■ Reivew & Comment - NPCC Directory 4 S											
NPCC1050		Submit NPCC Directory 4 Submittal		1d	20-May-25	20-May-25	Not Started		ENG.ST.P&	P&C Engineer	NPCC1040	NPCC1060, ET0814	■ Submit NPCC Directory 4 Submittal											
NPCC1060		Receive Approval (30 Day Lead Time) - NPCC Directory 4 Submittal		30d	21-May-25	02-Jul-25	Not Started		ENG.ST.P&	P&C Engineer	NPCC1050	NPCC1080	■ Receive Approval (30 Day Lead Time)-N											
NPCC1080		Tie to Milestone - Begin P/C Construction		1d	03-Jul-25	03-Jul-25	Not Started		ENG.ST.P&	P&C Engineer	NPCC1060	ET00	■ Tie to Milestone - Begin P/C Construction											
80089926.04.03 State Approvals and Permits				200d	01-Sep-23 A	09-Dec-24							▼ 09-Dec-24, 80089926.04.03 State Approvals and											
Actual Work Critical Remaining Work Summary				Page 5 of 16							TASK filter: EMA REPORT - Remaining TJP.													
Remaining Work ◆ Milestone											© Oracle Corporation													

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Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP								06-Mar-24 15:56											
Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028							
</																						

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24				BESS - Remaining TJP								06-Mar-24 15:56												
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	20242025202620272028											
</																								

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24				BESS - Remaining TJP								06-Mar-24 15:56											
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources		Predecessors	Successors	2024		2025		2026		2027		2028	

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24				BESS - Remaining TJP								06-Mar-24 15:56																			
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors																			
													2024	2025	2026	2027	2028														
	FCN1070	Constr Compl	0d		28-Sep-27	Not Started		ENG.ENGF			FCN1060	COMM1000, COMPL1000						◆ Constr													
	80089926.07.07.02 Telecom Wiring / Terminating		30d	18-Aug-27	28-Sep-27														▼ 28-Sep												
	BCN2190	Install Communication Wiring and Terminate	30d	18-Aug-27	28-Sep-27	Not Started		CONTR	O&M Southwest		BCN1040, BCN1030, BCN2080, ECN2080, FCN1000	COMPL1020, COMPL1000						■ Install C													
	80089926.08.01 ISD		0d	28-Sep-27	28-Sep-27														▼ 28-Sep												
	COMPL1000	In Service Date	0d		28-Sep-27	Not Started	ISD	PM			ECN1030, BCN1020, DCN1020, FCN1070, BCN2100, ECN1010	COMPL1010						◆ In Serv													
	ISD	ISD - In Service Date	0d	28-Sep-27	28-Sep-27	Not Started					AT0805								ISD --Ir												
	80089926.09 TEST & COMMISSIONING		504d	20-Oct-25	28-Sep-27														▼ 28-Sep												
	TC1000	Test & Commissioning	504d	20-Oct-25	28-Sep-27	Not Started													Test &												
	80089926.09.1 Isolation Plan		293d	20-Oct-25	07-Dec-26														07-Dec-26, 800899												
	A3360	PM to Request Lead Commissioning Engineer GO/NO-GO {Preliminary or Detail design is often the predecessor}	2d	20-Oct-25	21-Oct-25	Not Started	PM	PM	Adam S. Kelley		EE1210	A2930							PM to Request Lead Commissioning												
	A2930	Assign Lead Commissioning Engineer	2d	22-Oct-25	23-Oct-25	Not Started		ENG.ST.SL	LEAD COMMISSIONII ENGINEER		A3360	A3300, A2950, A3350, A1730, A2750, A3260							Assign Lead Commissioning Engine												
	A3350	Evaluate Drawings and Plan for Construction Outages	20d	24-Oct-25	21-Nov-25	Not Started		ENG.ST.SL	LEAD COMMISSIONII ENGINEER		A2930	A1730							■ Evaluate Drawings and Plan for Co												
	A2950	Develop Test and Commissioning Plan	20d	05-Nov-26	02-Dec-26	Not Started	V-COMP	TEST	LEAD COMMISSIONII ENGINEER		A3500, A3540, A3260, A2930	A3300							■ Develop Test and C												
	A3300	LCE Review and Approve Vendor's Test and Commissioning Plan	3d	03-Dec-26	07-Dec-26	Not Started	ES-COV	ENG.ST.SL	LEAD COMMISSIONII ENGINEER		A2930, A2950	A3270							LCE Review and A												
	80089926.09.04 Testing Activities		236d	03-Nov-26	28-Sep-27														▼ 28-Sep												
	A3260	Mobilize Commissioning Engineer and Testing Vendor Prior to Construction	2d	03-Nov-26	05-Nov-26	Not Started		ENG.ST.SL	LEAD COMMISSIONII ENGINEER		CT0401, A2930, BCN1050	A2950, A3270, A3280							Mobilize Commissio												
	A3280	Support Project/Testing & Execute test plan	234d	05-Nov-26	28-Sep-27	Not Started		ENG.ST.SL	LEAD COMMISSIONII ENGINEER		A3260	LCE1000, EV1000							Support												
	BCN2200	System Commissioning	81d	08-Jun-27*	28-Sep-27	Not Started		CONTR	O&M Southwest		BCN1040, BCN1030, BCN2080, ECN2080, FCN1000	COMPL1020, COMPL1000							■ System												
	80089926.09.03 LCE AUTHORIZATION (Remove)		21d	24-Oct-25	24-Nov-25														▼ 24-Nov-25, 80089926.09.03 LCE												
	A2750	Review and approve Outage Sequence Document (Sign-off)	1d	24-Oct-25	24-Oct-25	Not Started		ENG.ST.SL	LEAD COMMISSIONII ENGINEER		A2930	A3270							Review and approve Outage Seque												
	A1730	LCE Review TOA / Applications (Pred=TOA#, Dwgs and Const Plan, has T18.01 successor	1d	24-Nov-25	24-Nov-25	Not Started		ENG.ST.SL	LEAD COMMISSIONII ENGINEER		A2930, A3350	BT00							LCE Review TOA / Applications (Pr												
	80089926.10 PROJECT CLOSE OUT		365d	16-Oct-26	09-Mar-28														PROJECT CLOSE OUT												
	PCO1000	Project Close Out	365d	16-Oct-26	09-Mar-28	Not Started													PROJECT CLOSE OUT												
	80089926.10.02 Electrical Close Out		117d	29-Sep-27	09-Mar-28														Electrical Close Out												
	EV1000	Prepare Electrical Field Mark Ups	21d	06-Oct-27	03-Nov-27	Not Started	V-FLDAS	ENG.ENGF	LEAD COMMISSIONII ENGINEER		LCE1000, A3280	EV1100							■ Prepa												
	EV1100	Eversource Engineering Prepares & Sends Electrical Field Mark Up Package to Vendor	5d	04-Nov-27	10-Nov-27	Not Started	ES-ASB	ENG.ST.SS	Shodhan Patel		EV1000	EV1200							Evers												
	EV1200	Vendor Engineering Verifies Electrical Mark Ups and Updates As-Built Drawing Package. Submits Final Package to ES	30d	11-Nov-27	22-Dec-27	Not Started	V-ASB	ENG.ST.SS	Shodhan Patel		EV1100	EV1300							■ Ver												
	EV1300	Eversource Engineering Verifies Quality of Submitted Electrical Drawings	20d	23-Dec-27	19-Jan-28	Not Started	ES-ASB	ENG.ST.SS	Shodhan Patel		EV1200	EV1400							■ Ev												
	EV1400	Electrical Transmittal, Approve Final As-Built Package, and EDC Files and Issue Drawings to Field	5d	20-Jan-28	26-Jan-28	Not Started	ES-ASB	ENG.ST.SS	Shodhan Patel		COMM1300, EV1300	PM1100							El												
PM1100	PM to Document Electrical As-Built Drawings Complete (Electrical, P&C, Communications)	1d	27-Jan-28	27-Jan-28	Not Started	PM	PM	Adam S. Kelley		COMM1400, PC1310, EV1400	PC1500, COMM1500							PM													
COMPL1010	Final Inspection and Punchlist Work	90d	29-Sep-27	01-Feb-28	Not Started		PM	Adam S. Kelley		COMPL1000	COMPL1020							■ Fi													
Actual Work Remaining Work				Critical Remaining Work ◆ Milestone				Summary				Page 12 of 16								TASK filter: EMA REPORT - Remaining TJP.								© Oracle Corporation			

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP							06-Mar-24 15:56											
Activity ID		Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028					

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP								06-Mar-24 15:56															
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028									
	TMP1220	N/R 12-7-23 MR Engineering of Choice (EOC) -DR & PO Issued to Vendor (OE/CE) -Prelimi Design Effort		0d	27-Feb-24 A	18-Mar-24 A	Comple		PURC	Purchasing Department	TMP1180, PF1000	DE2130														
	DMAT1000	N/R 12-7-23 MR Issue Spec. - Long Lead Item #1		0d	19-Mar-24 A	25-Mar-24 A	Comple		ENG.D.EN	DISTRIBUTION Michael Hickey	DE2140, PF1010	DMAT1010														
	A3540	N/R 12-7-23 MR TEST SERVICES - BID - Eversource awards contract - PO Issued		0d	04-Mar-24 A	29-Mar-24 A	Comple		PURC	Purchasing Department	A3530	A2950														
	DE2140	N/R 12-7-23 MR ES Engineering Review Distribution Preliminary Design		0d	25-Jul-24 A	07-Aug-24 A	Comple	ES-PRE	ENG.D.EN	Michael Hickey	DE2130	AT2605, UG1270, DMAT1000, DT2605, DT2404, CMBL1000														
	PP1000	N/R 12-7-23 MR Permit(s) for Wetlands; ArmyCorp; Dewtr; Strmwtr; Other		0d	08-Aug-24 A	02-Jan-25 A	Comple		ENG.ENV.I	Christopher Newhall	BT2615, BT2605	BD1000, BT0801, FINCO1000, PCN2000, AT0804														
	UG1270	N/R 12-7-23 MR Vendor Engineer Develop Distribution Detail Design		0d	03-Dec-24 A	08-Jan-25 A	Comple	V-DTL	ENG.D.EN	Michael Hickey	DE2140, PF1010, FF1000	UG1280														
	UG1280	N/R 12-7-23 MR ES Engineering Review Distribution Detail Design		0d	09-Jan-25 A	23-Jan-25 A	Comple	ES-DTL	ENG.D.EN	Michael Hickey	UG1270	NX9A-NC118, AT1811, DCN2000, UG1280, CSD1050														
	UG1360	N/R 12-7-23 MR Vendor Engineer Develop Distribution IFC		0d	24-Jan-25 A	06-Feb-25 A	Comple	V-IFC	ENG.D.EN	Michael Hickey	OH1680, OH1680, UG1280	UG1370														
	CSD1020	N/R 12-7-23 MR PE Initiate CASCADE Forms (Appendix A) for Construction Package		0d	07-Feb-25 A	13-Feb-25 A	Comple	CASCADE	ENG.D.EN	Michael Hickey	UG1280, UG1370	CSD1050														
	UG1370	N/R 12-7-23 MR ES Engineering Review & Issue Distribution IFC		0d	07-Feb-25 A	13-Feb-25 A	Comple	ES-IFC	ENG.D.EN	Michael Hickey	UG1360	NX9A-NC108, AT0805, UG1380, DT0805UG														
	UG1380	N/R 12-7-23 MR Finalize Construction Specification		0d	14-Feb-25 A	28-Feb-25 A	Comple		ENG.D.EN	Michael Hickey	UG1370	DT0805UG														
	SST1000	N/R 12-7-23 MR Sta Tech Outage & Testing Procedures		0d	14-Feb-25 A	07-Mar-25 A	Comple		ENG.ST.SU	LEAD COMMISSIONII ENGINEER	DE2100, PC1330	OPERM1010														
	A-DFR1060	DFR Pts Identified N/R		0d		26-Sep-25	Not Started		ENG.ST.P8		PC1330															
	A-DFR1010	P&C Prepare Configuration for DFRs N/R		5d	10-Nov-25	17-Nov-25	Not Started		ENG.ST.P8	P&C Engineer	A-DFR1000, PC1330	A-DFR1100, A-DFR1020														
	A-DFR1100	Prepare & Issue DRs if Material is Required N/R - DFR		3d	18-Nov-25	20-Nov-25	Not Started	DR-M	ENG.ST.P8	P&C Engineer	A-DFR1010, PF1010	A-DFR1040, A-DFR1120														
	A-DFR1020	Peer Review and Comment DFR ConfigurationN/R		5d	18-Nov-25	24-Nov-25	Not Started		ENG.ST.P8	P&C Engineer	A-DFR1010	A-DFR1030														
	A-DFR1040	Verify CASCADE Forms Exists for S/S DFR, if not then Initiate FormN/R		3d	21-Nov-25	25-Nov-25	Not Started	CASCADE	ENG.ST.P8	P&C Engineer	A-DFR1100	A-DFR1050														
	A-DFR1030	P&C Issue Configuration For DFRN/R		3d	25-Nov-25	28-Nov-25	Not Started		ENG.ST.P8	P&C Engineer	A-DFR1020	A-DFR1050														
	A-DFR1050	DFR Config CompleteN/R		1d	01-Dec-25	01-Dec-25	Not Started		ENG.ST.P8	P&C Engineer	A-DFR1040, A-DFR1030															
	EMAT1110	N/R 12-7-23 MR CAT ID#/ DR#/ DR Prepare & Enter - Long Lead Item #1		0d	18-Nov-25 A	02-Dec-25 A	Comple	DR-M	ENG.ST.SS	Shodhan Patel	EMAT1100, PF1010	EMAT1120														
	PCMAT1010	N/R 12-7-23 MR CAT ID#/ DR#/ DR Prepare & Enter - Long Lead Item #1		0d	18-Nov-25 A	02-Dec-25 A	Comple		ENG.ST.P8	P&C Engineer	PCMAT1000, PF1010	EMAT1120, PCMAT1040, DMAT1020														
	COMMAT1010	N/R 12-7-23 MR CAT ID#/ DR#/ DR Prepare & Enter - GE JMUX		0d	18-Nov-25 A	02-Dec-25 A	Comple	DR-M	ENG.ST.SU	Bechir El Khoury Hanna	COMMAT1000, PF1010	COMMAT1030														
	DMAT1010	N/R 12-7-23 MR CAT ID#/ DR#/ DR Prepare & Enter - Long Lead Item #1		0d	18-Nov-25 A	02-Dec-25 A	Comple	DR-M	ENG.D.EN	Michael Hickey	DMAT1000, PF1010	DMAT1020														
	EMAT1120	N/R 12-7-23 MR CAT ID#/ DR#/ DR Approved - Long Lead Item #1		0d	03-Dec-25 A	16-Dec-25 A	Comple		PURC	Michael P. Rosenberg, Purchasing	EMAT1110, PCMAT1010, PF1010	EMAT1130, APP1000														
	PCMAT1040	N/R 12-7-23 MR CAT ID#/ DR#/ DR Approved - Long Lead Item #1		0d	03-Dec-25 A	16-Dec-25 A	Comple		ENG.ST.P8	P&C Engineer	PCMAT1010, PF1010	APP1000, PCMAT1050														
COMMAT1030	N/R 12-7-23 MR CAT ID#/ DR#/ DR Approved - GE JMUX		0d	03-Dec-25 A	16-Dec-25 A	Comple		ENG.ST.SU	Bechir El Khoury Hanna	COMMAT1010, PF1010	COMMAT1040															
DMAT1020	N/R 12-7-23 MR CAT ID#/ DR#/ DR Approved - Long Lead Item #1		0d	03-Dec-25 A	16-Dec-25 A	Comple		ENG.D.EN	Michael Hickey	DMAT1010, PCMAT1010, PF1010	DMAT1030, APP1000															
A-DFR1120	Issue PO DFR Mat N/R		20d	21-Nov-25	19-Dec-25	Not Started		PURC	Purchasing Department	A-DFR1100	A-DFR1130															
Actual Work Remaining Work		Critical Remaining Work Milestone		Summary								Page 14 of 16					TASK filter: EMA REPORT - Remaining TJP.					© Oracle Corporation				

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP								06-Mar-24 15:56											
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024		2025		2026		2027		2028	
	DMAT1110	N/R 12-7-23 MR ES Approves Shop Drawings - Long Lead Item #1		0d	09-Apr-26 A	22-Apr-26 A	Comple		ENG.ST.LII	Michael Hickey	DMAT1100	DMAT1120	N/R 12-7-23 MR ES Approve									
	DMAT1120	N/R 12-7-23 MR Material Vendor Manufactures Long Lead Item #1		0d	23-Apr-26 A	06-May-26 A	Comple		PURC	Purchasing Department	DMAT1110	DMAT1130	N/R 12-7-23 MR Material Ve									
	DMAT1130	N/R 12-7-23 MR Material Vendor Delivers Long Lead Item #1		0d	07-May-26 A	20-May-26 A	Comple	M-DELI	PURC	Purchasing Department	DMAT1120	DT0401	N/R 12-7-23 MR Material V									

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 Actual Work
 Critical Remaining Work
 Summary
 Remaining Work
 Milestone

Activity ID	Activity Name	Original Duration	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	2024																													
										May			June				July				August				September				October										
										12	19	26	02	09	16	23	30	07	14	21	28	04	11	18	25	01	08	15	22	29	06	30							
	A3280	Support Project/Testing & Execute test plan	217	217	30-Nov-2026	28-Sep-2027	Not Starte		SUBTEC	LEAD CC																													
	BCN2200	System Commissioning	81	81	01-Jul-2027*	21-Oct-2027	Not Starte		CONTR	O&M Sou																													
	80089926.09.03 LCE AUTHORIZATION (Remove)		21	21	19-Nov-2025	18-Dec-2025																																	
	A2750	Review and approve Outage Sequence Document (Sign-off)	1	1	19-Nov-2025	19-Nov-2025	Not Starte		SUBTEC	LEAD CC																													
	A1730	LCE Review TOA / Applications (Pred=TOA#,Dwgs and Const Plan, has T18.01 successor	1	1	18-Dec-2025	18-Dec-2025	Not Starte		SUBTEC	LEAD CC																													
	80089926.10 PROJECT CLOSE OUT		369	369	28-Oct-2026	27-Mar-2028																																	
	PCO1000	Project Close Out	369	369	28-Oct-2026	27-Mar-2028	Not Starte																																
	80089926.10.02 Electrical Close Out		124	124	06-Oct-2027	27-Mar-2028																																	
	EV1000	Prepare Electrical Field Mark Ups	21	21	06-Oct-2027	03-Nov-2027	Not Starte	V-FLDAS	ENG.EN	LEAD CC																													
	EV1100	Eversource Engineering Prepares & Sends Electrical Field Mark Up Packge to Vendor	5	5	04-Nov-2027	10-Nov-2027	Not Starte	ES-ASB	SS.ELEC	Shodhan																													
	EV1200	Vendor Engineering Verifies Electrical Mark Ups and Updates As-Built Drawing Package. Sub	30	30	11-Nov-2027	22-Dec-2027	Not Starte	V-ASB	SS.ELEC	Shodhan																													
	EV1300	Eversource Engineering Verifies Quality of Submitted Electrical Drawings	20	20	23-Dec-2027	19-Jan-2028	Not Starte	ES-ASB	SS.ELEC	Shodhan																													
	EV1400	Electrical Transmittal, Approve Final As-Built Package, and EDC Files and Issue Drawings to f	5	5	07-Feb-2028	11-Feb-2028	Not Starte	ES-ASB	SS.ELEC	Shodhan																													
	PM1100	PM to Document Electrical As-Built Drawings Complete (Electrical, P&C, Communications)	1	1	14-Feb-2028	14-Feb-2028	Not Starte	PM	PM	Adam S.																													
	COMPL1010	Final Inspection and Punchlist Work	90	90	22-Oct-2027	24-Feb-2028	Not Starte		PM	Adam S.																													
	COMPL1020	Project Complete	0	0		27-Mar-2028	Not Starte		PM																														
	80089926.10.03 P&C Close Out		124	124	06-Oct-2027	27-Mar-2028																																	
	PC1000	Prepare P&C Field Mark Ups	21	21	06-Oct-2027	03-Nov-2027	Not Starte	V-FLDAS	ENG.EN	LEAD CC																													
	PC1510	Eversource Engineering Prepares & Sends Electrical Field Mark Up Packge to Vendor	5	5	04-Nov-2027	10-Nov-2027	Not Starte	ES-ASB	P&C.P&C	P&C Eng																													
	PC1200	Vendor Engineering Verifies P&C Mark Ups and Updates As-Built Drawing Package. Submits	30	30	11-Nov-2027	22-Dec-2027	Not Starte	V-ASB	P&C.P&C	P&C Eng																													
	PC1300	Eversource Engineering Verifies Quality of Submitted P&C Drawings	20	20	23-Dec-2027	19-Jan-2028	Not Starte	ES-ASB	P&C.P&C	P&C Eng																													
	PC1310	P&C Transmittal, Approve Final As-Built Package, and EDC Files and Issue Drawings to Field	5	5	20-Jan-2028	26-Jan-2028	Not Starte	ES-ASB	P&C.P&C	P&C Eng																													
	PC1500	FE Files P&C Design Drawings at Station	30	30	15-Feb-2028	27-Mar-2028	Not Starte	ES-FLSA	P&C.P&C	P&C Eng																													
	80089926.10.04 Telecomm Close Out		112	112	22-Oct-2027	27-Mar-2028																																	
	COMM1000	Prepare Communication Field Mark Ups	21	21	22-Oct-2027	19-Nov-2027	Not Starte	V-FLDAS	ENG.EN	Field Con																													
	COMM1100	Eversource Engineering Prepares & Sends Electrical Field Mark Up Packge to Vendor	5	5	22-Nov-2027	26-Nov-2027	Not Starte	ES-ASB	SUBTEC	Bechir El																													
	COMM1200	Vendor Engineering Verifies Communication Mark Ups and Updates As-Built Drawing Packag	30	30	29-Nov-2027	07-Jan-2028	Not Starte	V-ASB	SUBTEC	Bechir El																													
	COMM1300	Eversource Engineering Verifies Communication Mark Ups and Updates As-Built Drawing Pa	20	20	10-Jan-2028	04-Feb-2028	Not Starte	ES-ASB	SUBTEC	Bechir El																													
	COMM1400	Communications Transmittal, Approve Final As-Built Package, and EDC Files and Issue Draw	5	5	07-Feb-2028	11-Feb-2028	Not Starte	ES-ASB	SUBTEC	Bechir El																													
	COMM1500	Operations Files Communication Design Drawings at Station	30	30	15-Feb-2028	27-Mar-2028	Not Starte	ES-FLSA	ENG.EN	Field Con																													
	80089926.10.11 LCE Closeout		5	5	29-Sep-2027	05-Oct-2027																																	
	LCE1000	LCE Review Test Data & Issue Commissioning Report (Preds=LCE Oversight of Test Contrct	5	5	29-Sep-2027	05-Oct-2027	Not Starte		SUBTEC	LEAD CC																													
	80089926.10.09 Work Order / Financial Close Out		10	10	28-Oct-2026	10-Nov-2026																																	
	FINCO1000	PM/ Engr/ Plant Acct Develop Prelim UOP & UOR List(s)	5	5	28-Oct-2026	03-Nov-2026	Not Starte	PM	PM	Adam S.																													
	FINCO1010	Review UOP, compl forms, fwd to Plant Acctng, enter in wrkbk	5	5	04-Nov-2026	10-Nov-2026	Not Starte		PM	Michael F																													
	80089926.10.10 Project Management Close Out		40	40	09-Sep-2027	03-Nov-2027																																	
	CSD1050	Construction Rep Finalize CASCADE Forms (within 14 days following ISD)	10	10	09-Sep-2027	22-Sep-2027	Not Starte		PM	Adam S.																													
CSD1060	Asset Management Complete Asset Update(s) in CASCADE	30	30	23-Sep-2027	03-Nov-2027	Not Starte		PM	Adam S.																														

Hyde Park BESS - Battery Energy Storage System Solution - Traditional				00: EMA - Remaining (MLJ)					1/27/26 14:38											
Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMENT	Resources												
									2026	2027	2028	2029								
	DE1700	Conduct Initial Risk Management & SCLL Review (Project Initiator & PM)	4	12/16/27	12/21/27	Not Started		ENG.SYS.SYSP	Mariam A. Chatha, A											Conduct Initial Risk Management & SCLL
	DE2000	Project Constructability Review	5	12/22/27	12/28/27	Not Started		CONS	Joseph F. Didomenic											Project Constructability Review
	DE1600	Create Planning Grade Estimate (+/-25% Full Funding)	10	12/29/27	1/11/28	Not Started	EST-FUL	ENG.BQA.EST	Justin R. Nidemo											Create Planning Grade Estimate (+/-25%
	PAFF2000	Submit Full Funding PAF for Internal Review	10	1/12/28	1/26/28	Not Started		ENG.SYS.SYSP	Mariam A. Chatha, A											Submit Full Funding PAF for Internal Re
	80089926.01.02.05.02 EPAC - FF		5	1/26/28	2/2/28															2/2/28, 80089926.01.02.05.02 EPAC
	PAFF2200	EPAC Full Funding Meeting	0		1/26/28	Not Started	EPAC-MD	ENG.SYS.SYSP												EPAC Full Funding Meeting
	PAFF2210	Incorporate EPAC Full Funding PAF Review Comments	5	1/27/28	2/2/28	Not Started		ENG.SYS.SYSP	Mariam A. Chatha, A											Incorporate EPAC Full Funding PAF Re
	FF1010	EPAC Full Funding Approval	0		2/2/28	Not Started	EPAC-F	ENG.SYS.SYSP												EPAC Full Funding Approval
	80089926.01.02.05.03 Power Plan - FF		15	2/3/28	2/24/28															2/24/28, 80089926.01.02.05.03 Pow
	FF1000	Full Funding - PowerPlan Approval	15	2/3/28	2/24/28	Not Started	PPA-FF	PM	Adam S. Kelley, Mari											Full Funding - PowerPlan Approval
	80089926.01.03 Civil Conceptual & Preliminary Engineering Design		75	2/8/27	5/24/27															5/24/27, 80089926.01.03 Civil Conceptual & Preliminary
	80089926.01.03.02 Civil Preliminary Design		75	2/8/27	5/24/27															5/24/27, 80089926.01.03.02 Civil Preliminary Design
	80089926.01.03.02.01 Civil Preliminary Design -Foundations & Structures		75	2/8/27	5/24/27															5/24/27, 80089926.01.03.02.01 Civil Preliminary Design
	DE2150	Engineer Develop Civil Substation Preliminary Design–Foundations & Structures	60	2/8/27	5/3/27	Not Started		ENG.ST.SS.CIVL	James D. Curtis											Engineer Develop Civil Substation Preliminary Design–Fou
	DE2160	ES Engineering Review Civil Substation Preliminary Design - Foundation & Structures	15	5/4/27	5/24/27	Not Started	ES-PREL	ENG.ST.SS.CIVL	James D. Curtis											ES Engineering Review Civil Substation Preliminary Desi
	80089926.01.03.02.1 Site Design - Preliminary Design - Stormwater, Site Plan, Topo		75	2/8/27	5/24/27															5/24/27, 80089926.01.03.02.1 Site Design - Preliminary
	DE2220	Engineer Develop Civil Substation Preliminary Design - Stormwater, Site Plan, Topo	60	2/8/27	5/3/27	Not Started		ENG.ST.SS.CIVL	James D. Curtis											Engineer Develop Civil Substation Preliminary Design- Stc
	DE2230	ES Engineering Review Civil Substation Preliminary Design - Stormwater, Site Plan, Topo	15	5/4/27	5/24/27	Not Started	ES-PREL	ENG.ST.SS.CIVL	James D. Curtis											ES Engineering Review Civil Substation Preliminary Desi
	80089926.01.04 Electrical Conceptual & Preliminary Engineering Design		75	2/8/27	5/24/27															5/24/27, 80089926.01.04 Electrical Conceptual & Prelin
	80089926.01.04.02 Electrical Preliminary Design		75	2/8/27	5/24/27															5/24/27, 80089926.01.04.02 Electrical Preliminary Design
	DE2030	Engineer Develop Electrical Substation Preliminary Design (includes Site Plan)	60	2/8/27	5/3/27	Not Started		ENG.ST.SS.ELEC	Patrick J. Fitzgerald, .											Engineer Develop Electrical Substation Preliminary Design
	DE2040	ES Engineering Review Electrical Substation Preliminary Design(includes Site Plan)	15	5/4/27	5/24/27	Not Started	ES-PREL	ENG.ST.SS.ELEC	Patrick J. Fitzgerald, .											ES Engineering Review Electrical Substation Preliminary
	80089926.01.08 P&C Conceptual & Preliminary Engineering Design		468	2/5/26	12/13/27															12/13/27, 80089926.01.08 P&C Concept
	80089926.01.08.02 P&C Preliminary Design		468	2/5/26	12/13/27															12/13/27, 80089926.01.08.02 P&C Prelin
	DE2170	Engineer Develop P&C Technical Documents	50	2/5/26	4/16/26	Not Started		ENG.ST.P&C.P&I	Benjamin A. Franco											Engineer Develop P&C Technical Documents
	DE2090	Engineer Develop P&C Preliminary Design	60	9/1/27	11/29/27	Not Started		ENG.ST.P&C.P&I	Benjamin A. Franco											Engineer Develop P&C Preliminary Design
	DE2100	ES Engineering Review P&C Preliminary Design	10	11/30/27	12/13/27	Not Started	ES-PREL	ENG.ST.P&C.P&I	Benjamin A. Franco											ES Engineering Review P&C Preliminary D
	80089926.01.09 Telecom Conceptual & Preliminary Engineering Design		70	9/1/27	12/13/27															12/13/27, 80089926.01.09 Telecom Cond
	80089926.01.09.02 Telecomm Preliminary Design		70	9/1/27	12/13/27															12/13/27, 80089926.01.09.02 Telecomm
	DE2110	Develop Communications Preliminary Scope Engineering Design in Accordance with Deliverables Check	60	9/1/27	11/29/27	Not Started		ENG.ST.SUBTEC	Syed Zohair Iqbal											Develop Communications Preliminary Scop
	DE2120	ES Engineering Review Communications Preliminary Design	10	11/30/27	12/13/27	Not Started	ES-PREL	ENG.ST.SUBTEC	Syed Zohair Iqbal											ES Engineering Review Communications f
	80089926.01.12 Distribution Conceptual & Preliminary Engineering Design		270	3/16/27	4/5/28															4/5/28, 80089926.01.12 Distributi
	80089926.01.12.2 Distribution Engineering		270	3/16/27	4/5/28															4/5/28, 80089926.01.12.2 Distrib
	DE2161	Issue WO - Resources	50	3/16/27	5/24/27	Not Started		ENG.D.ENG.ENG	Jenna M. Cloutier											Issue WO - Resources
	DE2180	ES Engineering - Distribution - Conduit Sketch Engineering & Design – 496-H7 (Bus 1)	70	5/25/27	8/31/27	Not Started	ES-PREL	ENG.D.ENG.ENG	Jenna M. Cloutier											ES Engineering - Distribution - Conduit Sketch En
	DE2190	ES Engineering - Distribution - Conduit Sketch Engineering & Design – 441-1401H (Bus 2)	70	5/25/27	8/31/27	Not Started	ES-PREL	ENG.D.ENG.ENG	Jenna M. Cloutier											ES Engineering - Distribution - Conduit Sketch En
	DE2200	ES Engineering - Distribution - Conduit Sketch Engineering & Design - 362-1207H (Bus 3)	70	5/25/27	8/31/27	Not Started	ES-PREL	ENG.D.ENG.ENG	Jenna M. Cloutier											ES Engineering - Distribution - Conduit Sketch En
	DE2210	ES Engineering - Distribution - Conduit Sketch Engineering & Design– 362-1209H (Bus 4)	70	5/25/27	8/31/27	Not Started	ES-PREL	ENG.D.ENG.ENG	Jenna M. Cloutier											ES Engineering - Distribution - Conduit Sketch En
	DE2280	ES Engineering - Distribution - Survey and Rights	150	9/1/27	4/5/28	Not Started		ENG.D.ENG.ENG	Jenna M. Cloutier											ES Engineering - Distribution - Sur
	80089926.01.10 Project Management		110	12/21/27	5/24/28															5/24/28, 80089926.01.10 Proj
	80089926.01.10.2 Compliance Authorization Notification		110	12/21/27	5/24/28															5/24/28, 80089926.01.10.2 C
	1170	Prepare CAN (Construction Authorization Notice) - Civil	10	12/21/27	1/3/28	Not Started		PM	Adam S. Kelley											Prepare CAN (Construction Authorization
	3700	Prepare CAN (Construction Authorization Notice) - Distribution	10	12/21/27	1/3/28	Not Started		PM	Adam S. Kelley											Prepare CAN (Construction Authorization
	3680	Prepare CAN (Construction Authorization Notice) - Electrical	10	2/16/28	3/1/28	Not Started		PM	Adam S. Kelley											Prepare CAN (Construction Authoriza
	3690	Prepare CAN (Construction Authorization Notice) - P&C, Telecomm	10	5/11/28	5/24/28	Not Started		PM	Adam S. Kelley											Prepare CAN (Construction Au
80089926.02 ROW & EASEMENTS		44	5/25/27	7/26/27																7/26/27, 80089926.02 ROW & EASEMENTS
ROW1000	ROW & Easements - WBS Summary	44	5/25/27	7/26/27	Not Started		ROW													ROW & Easements -WBS Summary
80089926.02.01 Right of Way		44	5/25/27	7/26/27																7/26/27, 80089926.02.01 Right of Way
80089926.02.01.02 Easements		44	5/25/27	7/26/27																7/26/27, 80089926.02.01.02 Easements
PRE1000	Identify Required Takes & Easements	22	5/25/27	6/24/27	Not Started		RE	Jennifer Valente												Identify Required Takes & Easements
PRE1010	Obtain Required Takes & Easements	22	6/25/27	7/26/27	Not Started		RE	Jennifer Valente												Obtain Required Takes & Easements
Hyde Park BESS - Battery Energy Storage System Solution - Traditional				00: EMA - Remaining (MLJ)					1/27/26 14:38											
Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMENT	Resources	2026	2027	2028	2029								
	DE1700	Conduct Initial Risk Management & SCLL Review (Project Initiator & PM)	4	12/16/27	12/21/27	Not Started		ENG.SYS.SYSP	Mariam A. Chatha, A											Conduct Initial Risk Management & SCLL
	DE2000	Project Constructability Review	5	12/22/27	12/28/27	Not Started		CONS	Joseph F. Didomenic											Project Constructability Review
	DE1600	Create Planning Grade Estimate (+/-25% Full Funding)	10	12/29/27	1/11/28	Not Started	EST-FUL	ENG.BQA.EST	Justin R. Nidemo											Create Planning Grade Estimate (+/-25%
	PAFF2000	Submit Full Funding PAF for Internal Review	10	1/12/28	1/26/28	Not Started		ENG.SYS.SYSP	Mariam A. Chatha, A											Submit Full Funding PAF for Internal Re
	80089926.01.02.05.02 EPAC - FF		5	1/26/28	2/2/28															2/2/28, 80089926.01.02.05.02 EPAC
	PAFF2200	EPAC Full Funding Meeting	0		1/26/28	Not Started	EPAC-MD	ENG.SYS.SYSP												EPAC Full Funding Meeting
	PAFF2210	Incorporate EPAC Full Funding PAF Review Comments	5	1/27/28	2/2/28	Not Started		ENG.SYS.SYSP	Mariam A. Chatha, A											Incorporate EPAC Full Funding PAF Re
	FF1010	EPAC Full Funding Approval	0		2/2/28	Not Started	EPAC-F	ENG.SYS.SYSP												EPAC Full Funding Approval
	80089926.01.02.05.03 Power Plan - FF		15	2/3/28	2/24/28															2/24/28, 80089926.01.02.05.03 Pow
	FF1000	Full Funding - PowerPlan Approval	15	2/3/28	2/24/28	Not Started	PPA-FF	PM	Adam S. Kelley, Mari											Full Funding - PowerPlan Approval
	80089926.01.03 Civil Conceptual & Preliminary Engineering Design		75	2/8/27	5/24/27															5/24/27, 80089926.01.03 Civil Conceptual & Preliminary
	80089926.01.03.02 Civil Preliminary Design		75	2/8/27	5/24/27															5/24/27, 80089926.01.03.02 Civil Preliminary Design
	80089926.01.03.02.01 Civil Preliminary Design -Foundations & Structures		75	2/8/27	5/24/27															5/24/27, 80089926.01.03.02.01 Civil Preliminary Design
	DE2150	Engineer Develop Civil Substation Preliminary Design–Foundations & Structures	60	2/8/27	5/3/27	Not Started		ENG.ST.SS.CIVL	James D. Curtis											Engineer Develop Civil Substation Preliminary Design–Fou
	DE2160	ES Engineering Review Civil Substation Preliminary Design - Foundation & Structures	15	5/4/27	5/24/27	Not Started	ES-PREL	ENG.ST.SS.CIVL	James D. Curtis											ES Engineering Review Civil Substation Preliminary Desi
	80089926.01.03.02.1 Site Design - Preliminary Design - Stormwater, Site Plan, Topo		75	2/8/27	5/24/27															5/24/27, 80089926.01.03.02.1 Site Design - Preliminary
	DE2220	Engineer Develop Civil Substation Preliminary Design - Stormwater, Site Plan, Topo	60	2/8/27	5/3/27	Not Started		ENG.ST.SS.CIVL	James D. Curtis											Engineer Develop Civil Substation Preliminary Design- Stc
	DE2230	ES Engineering Review Civil Substation Preliminary Design - Stormwater, Site Plan, Topo	15	5/4/27	5/24/27	Not Started	ES-PREL	ENG.ST.SS.CIVL	James D. Curtis											ES Engineering Review Civil Substation Preliminary Desi

Hyde Park BESS - Battery Energy Storage System Solution - Traditional				00: EMA - Remaining (MLJ)						1/27/26 14:38												
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMENT	Resources												
											2026 JFMA MJJJASND 2027 JFMA MJJJASND 2028 JFA JJJASND 2029 JFA JJJA											
80089926.04.02.01.02 ISO 1.3.9				493	3/16/26	2/24/28					2/24/28, 80089926.04.02.01.02: ISO											
80089926.04.02.01.02.1 System Impact Study				110	3/16/26	8/17/26					8/17/26, 80089926.04.02.01.02.1 System Impact Study											
ISO1240		Perform System Impact Study (SIS) Short Circuit Study		100	3/16/26	8/3/26	Not Started		ENG.SYS.SYSP	Amsa Mangga	Perform System Impact Study (SIS) Short Circuit Study											
ISO1250		Dynamic Study - Distribution Syst Planning		50	6/9/26	8/17/26	Not Started		ENG.SYS.SYSP	Mariam A. Chatha	Dynamic Study - Distribution Syst Planning											
80089926.04.02.01.02.2 Soln Pkg & Estimate				63	8/4/26	11/2/26					11/2/26, 80089926.04.02.01.02.2 Soln Pkg & Estimate											
ISO1130		Solution Pkg revised, -25/ +25 Estimate Complete - Distribution Syst Planning		0		11/2/26	Not Started		ENG.SYS.SYSP		Solution Pkg revised, -25/ +25 Estimate Complete - Distribution Syst Planning											
ISO1120		Revise Solution Pkg, -25/ +25 Estimate - Distribution Syst Planning		63	8/4/26	11/2/26	Not Started		ENG.SYS.SYSP	Mariam A. Chatha	Revise Solution Pkg, -25/ +25 Estimate - Distribution Syst Planning											
80089926.04.02.01.02.4 Reliability Committee				51	11/3/26	1/19/27					1/19/27, 80089926.04.02.01.02.4 Reliability Committee											
ISO1160		Draft the PPA Form - (Proposed Plan Application)		15	11/3/26	11/24/26	Not Started		ENG.SYS.SYSP	Melanie Bennett	Draft the PPA Form - (Proposed Plan Application)											
ISO1170		Submit PPA Form to ISO		0		11/24/26	Not Started		ENG.SYS.SYSP		Submit PPA Form to ISO											
ISO1180		Reliabilty Committee (RC) Review/ Recommendation		21	11/25/26	12/24/26	Not Started		ENG.SYS.SYSP	Melanie Bennett	Reliability Committee (RC) Review/ Recommendation											
ISO1190		Final Report/ Presentation		15	12/28/26	1/19/27	Not Started		ENG.SYS.SYSP	Melanie Bennett	Final Report/ Presentation											
80089926.04.02.01.02.5 P&C NPCC Directory 4				51	12/14/27	2/24/28					2/24/28, 80089926.04.02.01.02.5 P											
NPCC1090		Approve Application Diagram, One Lines, BOM, Floor Plan (Preliminary P&C IFC)		1	12/14/27	12/14/27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	Approve Application Diagram, One Lines, t											
NPCC1000		Compile Support Drawings Package - NPCC Directory 4 Submittal		5	12/15/27	12/21/27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	Compile Support Drawings Package - NP											
NPCC1010		Complete D4 Form (PSRF) - NPCC Directory 4 Submittal		2	12/22/27	12/23/27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	Complete D4 Form (PSRF) - NPCC Direc											
NPCC1020		Complete Cover Letter- NPCC Directory 4 Submittal		3	12/24/27	12/28/27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	Complete Cover Letter- NPCC Directory .											
NPCC1030		Develop + Provide Power Point Presentation - NPCC Directory 4 Submittal		5	12/29/27	1/4/28	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	Develop + Provide Power Point Presenta											
NPCC1040		Reivew & Comment - NPCC Directory 4 Submittal		3	1/5/28	1/7/28	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	Reivew & Comment - NPCC Directory 4											
NPCC1050		Submit NPCC Directory 4 Submittal		1	1/10/28	1/10/28	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	Submit NPCC Directory 4 Submittal											
NPCC1060		Receive Approval (30 Day Lead Time) - NPCC Directory 4 Submittal		30	1/11/28	2/23/28	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	Receive Approval (30 Day Lead Time											
NPCC1080		Begin P/C Construction		1	2/24/28	2/24/28	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	Begin P/C Construction											
80089926.04.04 Local Approvals And Permits				485	3/16/26	2/11/28					2/11/28, 80089926.04.04 Local Appr											
80089926.04.04.01 Permits				125	5/4/27	10/28/27					10/28/27, 80089926.04.04.01 Permits											
BD1000		Permits		125	5/4/27	10/28/27	Not Started		PM	Adam S. Kelley	Permits											
80089926.04.04.09 Special Use Article 80 or Conditional Use Permit				142	1/4/27	7/23/27					7/23/27, 80089926.04.04.09 Special Use Article 80											
BD1040		Local Siting ReviewArticle 80E Application		142	1/4/27	7/23/27	Not Started		SITING	Leah Zambemardi	Local Siting ReviewArticle 80E Application											
BD1050		BPDA Article 80E Approved		0		7/23/27	Not Started		SITING		BPDA Article 80E Approved											
80089926.04.04.11 Zoning / Planning Board				485	3/16/26	2/11/28					2/11/28, 80089926.04.04.11 Zoning /											
FCN2170		Vendor Prepare Preliminary Visual Renderings, Landscape & Traffic Management Plans		40	3/16/26	5/8/26	Not Started		SITING	Generic Progett Sitin	Vendor Prepare Preliminary Visual Renderings, Landscape & Traffic Management P											
FCN2200		Siting Fire Safety Plan		40	3/16/26	5/8/26	Not Started		SITING	Generic Progett Sitin	Siting Fire Safety Plan											
FCN2210		Siting - Site Layout (Battery System Design)		40	3/16/26	5/8/26	Not Started		SITING	Generic Progett Sitin	Siting - Site Layout (Battery System Design)											
FCN2180		ES Review and Approve Visual Renderings, Landscape & Traffic Management Plans		10	5/11/26	5/22/26	Not Started		SITING	Leah Zambemardi	ES Review and Approve Visual Renderings, Landscape & Traffic Management Pla											
BD1002		Prepare Package for Inspectional Services Department (ISD)		45	9/28/26	12/2/26	Not Started		SITING	Leah Zambemardi	Prepare Package for Inspectional Services Department (ISD)											
BD1020		Submit / File - Package for Inspectional Services Department (ISD)		0		12/2/26	Not Started		SITING		Submit / File - Package for Inspectional Services Department (ISD)											
BD1030		ISD Issues Zoning Refusal		20	12/3/26	12/31/26	Not Started		SITING	Leah Zambemardi	ISD Issues Zoning Refusal											
BD1010		Boston Zoning Board of Appeal (ZBA) Review - Conditional Use Permit		120	7/26/27	1/13/28	Not Started		SITING	Leah Zambemardi	Boston Zoning Board of Appeal (ZBA) R											
BD1060		ZBA Approve - Conditional Use Permit		0		1/13/28	Not Started		SITING		ZBA Approve - Conditional Use Permit											
BD1070		Boston Zoning Board of Appeal (ZBA) Appeal Period - Conditional Use Permit		20	1/14/28	2/11/28	Not Started		SITING	Leah Zambemardi	Boston Zoning Board of Appeal (ZBA)											
80089926.04.04.13 Municipal Notification				180	12/3/26	8/18/27					8/18/27, 80089926.04.04.13 Municipal Notification											
BCN2250		Obtain Distribution - Street Opening Permit - 9 month prior to Construction		180	12/3/26	8/18/27	Not Started		CONTR	Construction Resoun	Obtain Distribution - Street Opening Permit - 9 mon											
80089926.04.04.14 Sound Study				60	3/16/26	6/8/26					6/8/26, 80089926.04.04.14 Sound Study											
SS1010		Project Scoping		10	3/16/26	3/27/26	Not Started				Project Scoping											
SS1020		Regulatory Review		10	3/30/26	4/10/26	Not Started				Regulatory Review											
SS1030		Baseline Noise Monitoring		15	4/13/26	5/1/26	Not Started				Baseline Noise Monitoring											
SS1040		Noise Modeling and Prediction		10	5/4/26	5/15/26	Not Started				Noise Modeling and Prediction											
SS1050		Impact Assessment		5	5/18/26	5/22/26	Not Started				Impact Assessment											
SS1060		Reporting & Recommendations		10	5/26/26	6/8/26	Not Started				Reporting & Recommendations											
80089926.05 ENGINEERING DETAILED / IFC DESIGN				633	5/18/26	11/15/28					11/15/28, 8008992											
ENG1000		Engineering Detailed / IFC Design - WBS Summary		633	5/18/26	11/15/28	Not Started		ENG.ST.SS.ELEC		Engineering Detail											
80089926.05.01 Civil Engineering				279	10/7/26	11/16/27					11/16/27, 80089926.05.01 Civil Engineering											

Actual Work

Remaining Work

Critical Remaining Work

Milestone

Summary

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TASK filters: No Level of Effort, NOT CLOSED.

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Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMENT	Resources	2026202720282029											
									JFMAMJJJASNDJFMAMJJJASNDJF											

Hyde Park BESS - Battery Energy Storage System Solution - Traditional				00: EMA - Remaining (MLJ)						1/27/26 14:38											
Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMENT	Resources													
									2026202720282029												
									JFMAMJJJASNDJFMAMJJJASNDJF A JJJASNDJF A JJJ A												
A2930	Assign Lead Commissioning Engineer	2	5/15/28	5/16/28	Not Started		ENG.ST.SUBTEC	Carlos E. Ortiz, Jame	Assign Lead Commissioning Er												
A2950	Develop Test and Commissioning Plan - Balance of Plant, no Battery System Commissioning	55	8/8/28	10/25/28	Not Started	V-COMPI	TEST	LEAD COMMISSION	Develop Test and C												
A3300	STE assist compiling & review Vendor's Acceptance Test Procedures	15	10/26/28	11/15/28	Not Started	ES-COM	ENG.ST.SUBTEC	Carlos E. Ortiz	STE assist compilin												
80089926.05.11 Standards		598	5/18/26	9/26/28					9/26/28, 80089926.04												
80089926.05.11.1 Construction		140	5/25/27	12/13/27					12/13/27, 80089926.05.11.1 Construction												
STD1010	Develop Installation / Construction Standard	80	5/25/27	9/15/27	Not Started		SG	John J. Porazinski	Develop Installation / Construction Standard												
STD1020	Construction Standards Issued for Comments	20	9/16/27	10/14/27	Not Started		SG	John J. Porazinski	Construction Standards Issued for Comments												
STD1030	Construction Standards Review & Comments	20	10/15/27	11/12/27	Not Started		SG	John J. Porazinski	Construction Standards Review & Comment												
STD1040	Construction Standards Issue for Approval	10	11/15/27	11/29/27	Not Started		SG	John J. Porazinski	Construction Standards Issue for Approval												
STD1050	Construction Standards Sign-off	10	11/30/27	12/13/27	Not Started		SG	John J. Porazinski	Construction Standards Sign-off												
80089926.05.11.2 Procurement		240	7/14/26	6/25/27					6/25/27, 80089926.05.11.2 Procurement												
STD1060	Develop Installation / Procurement Standard	180	7/14/26	4/1/27	Not Started		SG	John J. Porazinski	Develop Installation / Procurement Standard												
STD1100	Procurement Standards Issued for Comments	20	4/2/27	4/29/27	Not Started		SG	John J. Porazinski	Procurement Standards Issued for Comments												
STD1070	Procurement Standards Review & Comments	20	4/30/27	5/27/27	Not Started		SG	John J. Porazinski	Procurement Standards Review & Comments												
STD1080	Procurement Standards Issue for Approval	10	5/28/27	6/11/27	Not Started		SG	John J. Porazinski	Procurement Standards Issue for Approval												
STD1090	Procurement Standards Sign-off	10	6/14/27	6/25/27	Not Started		SG	John J. Porazinski	Procurement Standards Sign-off												
80089926.05.11.3 Commissioning		150	2/25/28	9/26/28					9/26/28, 80089926.04												
STD1110	Develop Installation / Commissioning Standard	120	2/25/28	8/14/28	Not Started		SG	John J. Porazinski	Develop Installation / Cor												
STD1150	Commissioning Standards Issued for Comments	10	8/15/28	8/28/28	Not Started		SG	John J. Porazinski	Commissioning Standar												
STD1120	Commissioning Standards Review & Comments	10	8/29/28	9/12/28	Not Started		SG	John J. Porazinski	Commissioning Standa												
STD1130	Commissioning Standards Issue for Approval	5	9/13/28	9/19/28	Not Started		SG	John J. Porazinski	Commissioning Stand												
STD1140	Commissioning Standards Sign-off	5	9/20/28	9/26/28	Not Started		SG	John J. Porazinski	Commissioning Stand												
80089926.05.11.4 Work Methods		140	5/4/27	11/19/27					11/19/27, 80089926.05.11.4 Work Methods												
STD1160	Develop Installation / Work Methods Standard	80	5/4/27	8/24/27	Not Started		SG	John J. Porazinski	Develop Installation / Work Methods Standard												
STD1200	Work Methods Standards Issued for Comments	20	8/25/27	9/22/27	Not Started		SG	John J. Porazinski	Work Methods Standards Issued for Comments												
STD1170	Work Methods Standards Review & Comments	20	9/23/27	10/21/27	Not Started		SG	John J. Porazinski	Work Methods Standards Review & Commen												
STD1180	Work Methods Standards Issue for Approval	10	10/22/27	11/4/27	Not Started		SG	John J. Porazinski	Work Methods Standards Issue for Approval												
STD1190	Work Methods Standards Sign-off	10	11/5/27	11/19/27	Not Started		SG	John J. Porazinski	Work Methods Standards Sign-off												
80089926.05.11.5 Fire Protection		165	5/18/26	1/12/27					1/12/27, 80089926.05.11.5 Fire Protection												
STD1210	Develop Installation / Fire Protection Standard	120	5/18/26	11/4/26	Not Started		SG	John J. Porazinski	Develop Installation / Fire Protection Standard												
STD1250	Fire Protection Standards Issued for Comments	10	11/5/26	11/19/26	Not Started		SG	John J. Porazinski	Fire Protection Standards Issued for Comments												
STD1220	Fire Protection Standards Review & Comments	20	11/20/26	12/18/26	Not Started		SG	John J. Porazinski	Fire Protection Standards Review & Comments												
STD1230	Fire Protection Standards Issue for Approval	10	12/21/26	1/5/27	Not Started		SG	John J. Porazinski	Fire Protection Standards Issue for Approval												
STD1240	Fire Protection Standards Sign-off	5	1/6/27	1/12/27	Not Started		SG	John J. Porazinski	Fire Protection Standards Sign-off												
80089926.06 PROCUREMENT LONG LEAD MATERIAL		628	11/25/24 A	7/18/28					7/18/28, 80089926.06 PR												
PLLM1000	Procurement Long Lead Material - WBS Summary	628	11/25/24 A	7/18/28	In Progress		MAE.PURCH		Procurement Long Lead M												
80089926.06.01 Civil Long Lead Material		121	5/25/27	11/15/27					11/15/27, 80089926.06.01 Civil Long Lead												
CMAT1020	Civil (BG) Material DR Complete	0		5/25/27	Not Started		ENG.ST.SS.CIVL		Civil (BG) Material DR Complete												
CMAT1000	Prepare Civil Material DR	1	5/25/27	5/25/27	Not Started	DR-M	ENG.ST.SS.CIVL	James D. Curtis	Prepare Civil Material DR												
CMAT1010	Civil Material Fab & Delivery	120	5/26/27	11/15/27	Not Started		PURC	Purchasing Departm	Civil Material Fab & Delivery												
80089926.06.02 Electrical Long Lead Material		581	5/1/25 A	5/10/28					5/10/28, 80089926.06.02 Elect												
80089926.06.02.2 Material Procurement - Battery System - Proc Item #1		479	12/16/25 A	12/16/27					12/16/27, 80089926.06.02.2 Material Pro												
EMAT1280	BATTERY SYSTEM - ES Selects Contractor & Issue DR	5	12/16/25 A	1/30/26	In Progress	DR-M	PURC	Purchasing Departm	BATTERY SYSTEM - ES Selects Contractor & Issue DR												
EMAT1600	BATTERY SYSTEM - Additional Bid Clarification from Nomad	0	1/12/26 A	2/2/26	In Progress				BATTERY SYSTEM - Additional Bid Clarification from Nomad												
EMAT1560	BATTERY SYSTEM - DR Approved/Contracts Sent to Supplier for Review/Signature	16	2/2/26	2/24/26	Not Started		PURC	Purchasing Departm	BATTERY SYSTEM - DR Approved/Contracts Sent to Supplier for Review/Signature												
EMAT1100	BATTERY SYSTEM - ES Awards P.O, LNTP, Vendor Design	6	3/6/26	3/13/26	Not Started		PURC	Purchasing Departm	BATTERY SYSTEM - ES Awards P.O, LNTP, Vendor Design												
EMAT1180	BATTERY SYSTEM - Material Vendor Submits Design Drawings - Material Procurement	45	3/16/26	5/15/26	Not Started		PURC	Purchasing Departm	BATTERY SYSTEM - Material Vendor Submits Design Drawings - Material Procurer												
EMAT1190	BATTERY SYSTEM - ES Reviews Design Drawings - Material Procurement	20	5/18/26	6/15/26	Not Started		ENG.ST.SS.ELEC	Patrick J. Fitzgerald, ,	BATTERY SYSTEM - ES Reviews Design Drawings - Material Procurement												
EMAT1200	BATTERY SYSTEM - ES Approves Shop Drawings - Material Procurement	20	6/16/26	7/13/26	Not Started		ENG.ST.SS.ELEC	Patrick J. Fitzgerald, ,	BATTERY SYSTEM - ES Approves Shop Drawings - Material Procurement												
EMAT1210	BATTERY SYSTEM - Fab / Manufacture - 15 months	300	7/14/26	9/20/27	Not Started		PURC	Purchasing Departm	BATTERY SYSTEM - Fab / Manufacture - 15 m												
EMAT1230	BATTERY SYSTEM - Delivery	60	9/21/27	12/16/27	Not Started		PURC	Purchasing Departm	BATTERY SYSTEM - Delivery												
Actual Work Remaining Work Critical Remaining Work Milestone Summary				Page 7 of 12					TASK filters: No Level of Effort, NOT CLOSED.					© Oracle Corporation							



**Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review**

Project Title: Hyde Park Battery Energy Storage System	☐ Prior Constructability Review Performed?
	☐ Program Related Site-Specific Review?
Project Numbers(s): 21334	Project Constructability Phase: ☒ Conceptual Engineering Design ☐ Preliminary Engineering Design ☐ Detailed Engineering Design
Date(s) of Constructability Review: 10/31/2023	
07/20/2023	
Project Scope Statement: Installation of a 10MW/20MWh utility-scale battery storage system (BESS) along Cummings Highway near Hyde Park Substation #496 in Hyde Park, MA. This constructability review is being based off an initial project walkdown and tabletop review and is being completed prior to Conceptual Engineering for the BESS development. More encompassing reviews will be completed as the project progresses through engineering stages.	

Department Review/Walkdown Attendees*:

Required [X]	Attended [X]	Department	Name	Required [X]	Attended [X]	Department	Name
☐	☒	Project Management	A.Kelley	☐	☒	S/S Engineering	P. Fitzgerald
☐	☐	Asset Management		☐	☒	P&C Engineering	E. Revi
☐	☒	Constr. Management	R. Taves, J. Didomenico	☐	☒	Capital Projects Eng.	J. Zicko, L. Williams
☐	☐	S/S Technical / LCE		☐	☒	Telecom	B. Khoury
☐	☐	S/S Operations**		☐	☒	Environmental	C. Newhall
☐	☐	Field Engineering		☐	☐	Real Estate	
☐	☐	Line Engineering		☐	☐	ROW T&D Rights & Survey	
☐	☐	Vegetation Management		☐	☐	Survey Engineering	
☐	☒	Siting	Leah Zambernardi	☐	☐	Cost Estimating	
☐	☐	Safety		☐	☒	Project Services	E. Snedeker
☐	☐	SCADA		☐	☐	Material Procurement	

* Sign-off in the table above for the Detailed Engineering Design Constructability Review indicates that all Action Items/Concerns from prior Constructability Reviews have been mitigated/resolved as documented in this form. Not every department is required to attend walkdowns for every project but should be consulted for project impact considerations.

** Internal Eversource Substation Operations must be included in the review even if external construction resources are planned

[Project Manager] certifies that this review was performed to completeness, that all applicable departments were informed of the project and have had opportunity to provide input to this review:



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

6/5/24

Name / Signature

Date

Documents available for use prior to Constructability Review					
[X]	Document	[X]	Document	[X]	Document
☒	Initial Scope -Birdseye site plan -General Arrangement	☐	Preliminary Drawings -Civil Site Drawing -Electrical Drawing -Section Drawings	☐	Prints in hand for marking up -Panel Front/Read View -Control Enclosure Layout -Floor / Wall Plans
☐	Detailed Scope Document	☒	Proposed Schedule	☒	Construction Sequence/Timeline
☐	Previous Notes / Comments / Observations	☐	Major Material / Equipment List	☐	Replace / Reuse of equipment

Action Items and Resolutions:

Item No.	Const. Review Action Item/Concern	Design Resolution/Mitigation:
1		
2		
3		
4		
5		

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
A	Engineering & Asset Management		



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
1	Scope of Work - Is scope adequately developed and sufficiently detailed for intended deliverable? - Any complimentary projects or other adjacent obsolete equipment for inclusion? - Any overlapping projects that could impact work/outages? - Consider Physical Installation, P&C Scope, Civil Scope, Engineering studies, Design standard adherence - Any remote station impacts? - Engineering Estimates for Design and Technical Support? (EE and EOC) Total Project	☐	Scope of work has been completed for both internal Eversource engineering and external Battery Energy Storage System vendor. Next steps following partial funding will be to progress engineering deliverables up to Preliminary Engineering where many of the items in the section will be addressed.
2	Materials Identified/Ordered -Any concerns with BOM/Major Materials? Review BOM lead times and project need dates -Specialty Major Materials: Identify Specialty Equipment & Materials potentials - temporary/replacement - long lead items? -Any Materials needed to facilitate construction access?	☐	Major material lead times for batteries, grounding transformers, and inverters are estimated to be at 12 months +/- . This duration has been built into the project schedule including standard procurement practices based on funding schedule.
3	Control House - Review Unmapped equipment interferences, Control/Battery Enclosure Layout, HVAC System (if equipment added), space for new or modified equipment, AC / DC station breaker availability and service loading, Panel Space, Expansion, one-line and/or riser type panelboard drawings, other.	☐	Indicative pricing effort has provided us with multiple equipment layouts for the site. Further engineering will be required to finalize layout, but the site is able to accommodate the installation of the equipment in multiple locations.
4	Ratings & Dimensions for Major Equipment. Any concerns upon field review? Equipment Rating One Line (and/or equipment thermal ratings tables) - Review for any errors in field	☒	No site concerns. As engineering progresses, we must ensure that the new access road can handle the equipment.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
5	Wall, Fence, Gate, and Foundation Considerations -Any permanent / temporary alterations or relocations required? -Pre-Project Sound Measurements Needed? Sound Walls? -Take measurements for Foundation quantities, dimensions and concrete volumes -CIP Review/Ballistic needs. Any Physical Security Requirement Consideration? -Assess condition of existing yard equipment foundations and viability for reuse (if proposed) -Review station expansion areas (if needed) for structures, old foundations, etc. -Any Steel Structural Plan Considerations? -Any impacts to construction plan?	☐	Site is currently fenced, and the future Battery Energy Storage System and associated equipment will also be fenced. Sound studies will be completed further along in the engineering process to ensure system adheres to dB requirements.
6	Below Grade / Enclosure Raceway - If possible, evaluate existing raceway and review spacing requirements. Review for any subsurface investigation needs or potential below grade obstructions. GPR/test pits needed? Geotechnical Investigation - is soil boring or resistivity investigation needed?	☐	See item 7 below.
7	Underground Obstructions -Drawing review concerns? -Is preconstruction or test excavation needed? -Is reuse of underground control cables / Conduits /replace Non-Shielded Cable feasible (if applicable) -Are spare conduits available? Are they useable? -Review if any areas will need to be plated for increased load bearing for equipment access (i.e., large Xfmr, etc.)	☐	Underground obstructions and conditions to be determined through required soil borings. It is understood that the site is a former landfill for construction debris and efforts will be made during the design process to limit excavation when possible.
8	Grounding Plan (Ground Grid) - Will footprint impact layout or other equipment? Identify locations to improve grounding. Impacts to foundations and ground grid plans. Observations on wetlands or ability to perform tests	☐	This is a new location, so a new ground grid will be installed in the location of the BESS. Coordination with the foundation plan and other equipment will be done to ensure the proper ground grid size. No site conditions pose any challenges for the ground grid.



Constructability Review Form: M6-PM-2012 - Eversource

Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
9	Nomenclature / Switching One-Line Diagram -Any obvious deviations from diagrams provided?	☒	No
10	BPS Separation Requirements - Is this BPS Substation? Y/N (part of PAF) - Changes to pre-existing BPS standards?	☒	No
11	IEC 61850 Substation Automation and Communication Requirements - Any Impacts/Retrofit needed?	☐	No
12	Any Upgrades to ES Standards - Is retrofit needed to bring up to ES standards? Can that be done?	☐	Capital Projects Engineering has written technical specifications used in the Battery Energy Storage System Request for Proposal. These technical specifications are being reviewed and cross referenced with a draft document that the Eversource Standards group has drafted.
B	Construction		
1	Any safety hazards specific to the facility? i.e., Low Bus, Deterioration, etc.	☐	No
2	Prior known Site-Specific Conditions -Common or Obvious station Issues -Missing scope items that could impact costs?	☐	Unknown subsurface obstructions are being carried as a risk until soil boring is completed.
3	Identify/Consider Required Depts for Construction? -Who is constructor: Internal vs External Resources -S/S Ops, GO, Test, Field Engineering, etc. -Specialty Contractors Required? -Required Oversight Needed - Estimated level of CR and LCE support needed	☐	Outside vendor will complete Construction of Battery Energy Storage System. Civil work for underground distribution circuit extension to be completed by outside vendor while cable pulling and termination to be completed by internal resources.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
4	Site Plan, Access, and Temporary Construction Facilities: -Confirm maintenance access for proposed new or replaced equipment -Verify clear footprint for yard equipment / Property Plan -General access to site, can areas outside substation be utilized? -Substation Entry Plan (driveway and drive path modifications) - Identification of Unique Requirements, i.e. (access bridges, station service, etc.) -Site Development Plan (grading and drainage) with Cut/Fill -Existing Rock Conditions? Potential Impacts (rock quantity)? -Any police details needed for deliveries, restricted hours, and/or need for extended hour approvals -Any Sanitary Civil Requirements? (Septic tanks, etc.) -Review turning radius for delivery trucks (if critical) -Any Substation Access and site/yard access Limitations? -Office Trailer Needs: Facilities, Utilities, Restroom, Container, Dumpster Space -Weather impacts to Site (Snow Placement) -Identify Material Laydown and Security -Where will new vs removed equipment go? -Construction permit for trailers outside substation fence?	☐	Layout requirements have accommodated for both maintenance clearances between equipment as well as around equipment and fence perimeter. Existing access road will be improved and utilized to access location. Further site and grading plans to be developed as engineering progresses as well as a pre-construction plan that will address the other items.
5	Unique Equipment Considerations: Gas insulated substation, Gas insulated line, Gas insulated bus, Pass Breaker, Statcom, Sync Condenser	☒	This is Eversource's second BESS installation, and this should still be considered a piece of unique/specialty equipment to be installed.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
6	Outage & Energization Plan and Construction Sequence: <i>Review electrical working clearances for equipment during construction, Detailed look at protection zone changes, Locked sequence or random outages, Outage durations minimized. Considerations:</i> -Approach Distances – Outage Requirements to maintain electrical clearances throughout proposed installation -Distribution Clearances -Clearance Holder Assignments -Drawings vs Field Conditions -Outage boundaries and interconnections -Phased approach needs for additional mobilizations -Verify outage sequence supports the planned construction sequence and timeline. (Lead Commissioning Engineer "LCE", Substation Operations, and Outage Planner input) -Any design alterations or considerations to Eliminate SCLL Conditions, Reduce Recall Time, Reduce Outage durations, Special Conditions -Additional outage requirements: Railroad Catenaries, Fiber outage, SCADA outage, Relay permissions -Seasonal or weather conditions impacts	☐	Will be completed further along in the engineering process.
7	Equipment Testing & Acceptance -Grounding plans / Resources? Will we need grounds hung? -Hi-pot testing and where will the equipment be set up?	☐	Will be completed further along in the engineering process.
8	Need for mobile generator and where connected (if needed)	☒	Mobile generators not required for construction. If construction is delayed then spot generation could be considered.
9	Need for mobile battery and where connected (if needed)	☒	Mobile batteries not required for construction. Mobile batteries could be considered as a partial solution to assist with schedule compression mitigation.
10	OT work required or Notable Permit Conditions (special hours)?	☐	TBD
11	Coordination of boundaries/seams (Identification of point of demarcation for	☐	Will be completed further along in the engineering process.



Constructability Review Form: **M6-PM-2012 - Eversource**
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
	connection to existing equipment and/or for coordination with other municipalities)		
C	Environmental		
1	Soil Pre-Characterization or Grab Sampling Completed/Planned	☐	Soil borings and characterization planned.
2	Desktop investigation and EAD Review -Required Permits -Stormwater Considerations -Dewatering requirements -Wetland and buffer review	☐	EAD completed. Below are known permitting requirements at this time: -DEP Waste Management Program Post-Closure Use Permit for authorization to construct on closed landfill facility. -If soil is to be disposed offsite from Eversource property, soil samples will need to be taken to characterize the soil and a soil management plan will need to be developed to get approval from a disposal facility or landfill. - Requires annual Tier II reporting to MassDEP for batteries. - If project is expected to disturb ≥1 acre of land, then will require submittal of eNOI for coverage under the EPA 2022 Construction General Permit, incl. development of a SWPPP and associated inspections are required.
3	Risk or presence of PCB's/Asbestos/Lead/SF6	☐	Possible risk. Additional communication with DEP required along with review of landfill closure documents.
4	Transformer Spill Containment Recommended?	☐	Required spill containment will be determined with further engineering.
5	Expanding Footprint / New Buildings or future Expansion Plans? (Enviro Impact)	☐	Equipment will be ground mounted. A building will not be constructed.
6	Environmental oversight considerations? (Inspection type/Interval)	☐	Oversight will be required but length and frequency will be determined at permit application and approval stages.
7	Environmental mitigations - Any matting or seasonal needs? Review D&M Plan with CR.	☒	Not anticipated at this time.
D	Siting, Project Services, and ROW T&D Rights & Survey		
1	Any regulatory concerns or approvals needed? - Any concerns noted that could be mitigated by scope/design alterations? - Likely to require visual screening? What considerations (feasibility or location)? -Zoning Relief and Permits required?	☐	Views of BESS from abutting neighborhood and concerns for losing open land could be mitigated by scope/design alterations. Visual screening will likely be required, siting BESS closer to shopping plaza & away from neighborhoods. Screening/buffering likely required for abutting residential neighborhood. Zoning Relief and Permits are required. City of Boston Article 80 & Zoning Variance.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
2	Other siting considerations?	☐	Neighborhood is concerned about losing open land. Also interested in green energy and energy savings solutions.
3	Neighbor/Abutter impacts: - Any noise, visual impacts, or sensitive abutters - Any changes to lighting or additional lighting needed?	☐	Visual impact of equipment to be reviewed with City of Boston and community and accommodations to be made during engineering phase for any required screening of faux façade.
4	Review for roadway, gate access, trailers, equip height, noise, weekend/night work, or any potential public disturbance related to work	☐	Increased traffic will be seen during construction. Post construction, minimal traffic can be expected to the location for routine maintenance purposes.
5	Installation within Eversource Rights - Is Rights and Restrictions Review required?	☐	Eversource is owner of property
6	Expanding Footprint / New Buildings / Fence line or future Expansion Plans?	☐	Location of Battery Energy Storage System to take into consideration future development of the parcel for other electrical needs.
7	Proposed Equipment on same property? - Any Encroachments	☐	No encroachments anticipated
F	Protection & Controls		
1	Panel Graphic additions and/or Modifications	☐	Not applicable
2	Space on existing panels for new equipment? (front & rear of panel fronts)	☐	Not applicable
3	Panel fronts conform to drawings	☐	Not applicable
4	Access to bring new panels into control enclosure	☐	Not applicable
5	Nameplate list and location of nameplates	☐	Will be provided as engineering progresses
6	Verifies that relays and meters that will be an interconnected project work are as shown on the drawings.	☐	As engineering progresses drawings will be reviewed to ensure proper relaying and autotransfer scheme is achieved to allow the BESS to be supplying energy
7	Temporary protection, relays, control cable needed? How long will they be required?	☐	None anticipated at this time.
8	Will relay setting changes be needed? Any additional outage requirements for changes?	☐	At this stage of the engineering, it is not a requirement to do any additional relay setting changes as this will be interconnecting onto an existing distribution circuit. As the engineering progresses, the impact of the BESS interconnection on existing protection schemes will need to be assessed to determine the need for any remote setting changes.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
G	Telecom Engineering		
1	Review telecom equipment in existing control enclosure	☐	Not Applicable
2	Review path to get telecom circuits into the station and paths within the control house. Determine whether new underground duct or overhead tray work is needed.	☒	Fiber exists at the station and ADSS is available at 427 Cummins Highway if required. Wired or wireless options will be reviewed during the engineering phases.
3	Review current or proposed Telecom equipment power connections and validate with P&C any need for system 1 and system 2 separation.	☐	No need for system separation
4	Review current Telecom fiber diversity and validate with P&C any fiber diversity requirements.	☐	
5	Review and list any remote station impacts/updates	☐	Not Applicable
6	Confirm whether correct interface equipment is available (i.e.) for fiber termination or multiplexor termination	☐	This will be a new installation, and equipment will be specified to be installed
7	Confirm whether existing communications cabinets are enough to support additional telecommunications equipment installed	☐	New installation, will require new cabinets
8	Review space for any required additional communications cabinets which may be required to support additional telecommunications equipment installed	☐	New installation, will require new cabinets
9	Review third party Telecommunications at substation and confirm whether additional capacity or Ground Potential Rise protection is required	☐	

Photos

(Ex. Equipment control cabinet, Breakers, MODs, CCVTs, PTs, Transformers)

BESS High Level Scope

- Mid-term alternative before long-term solution is implemented
- Located off Cummings Hwy in Hyde Park

	Quantity	Total
Battery Size	1 installation	10 MW / 20 MWh
Inverter Size	6 x 2.5 MVA inverters 2 x 7.5 MVA sets	15 MVA



☐ **Manhole Installation**

- (3) concrete manholes with inside dimension of 6'x10'x8'

☐ **Conduit Installation**

- (1,270') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

☐ **Primary Underground Cable Removal**

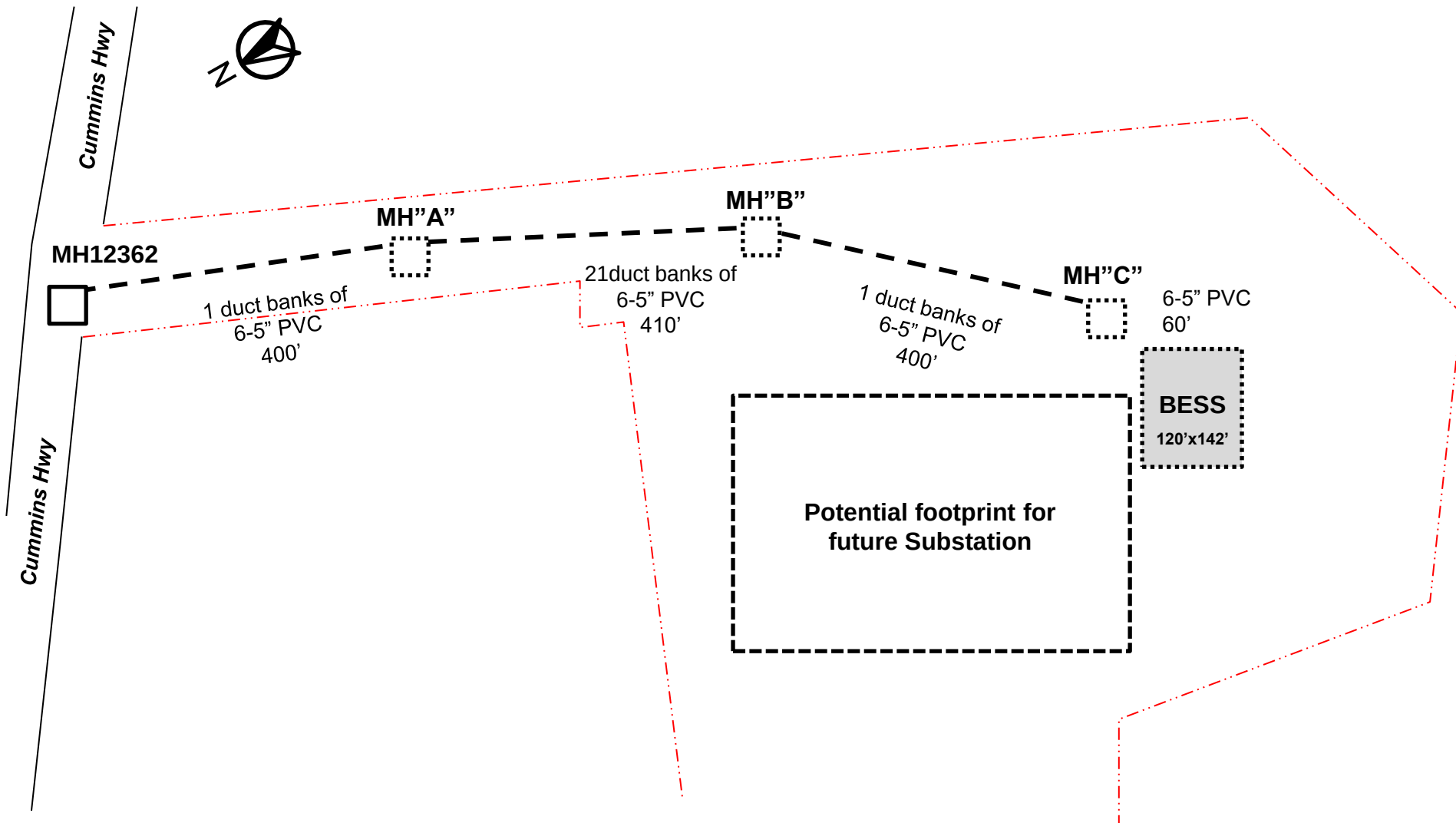
- (13,065') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (37 sections)

☐ **Primary Underground Cable Installation**

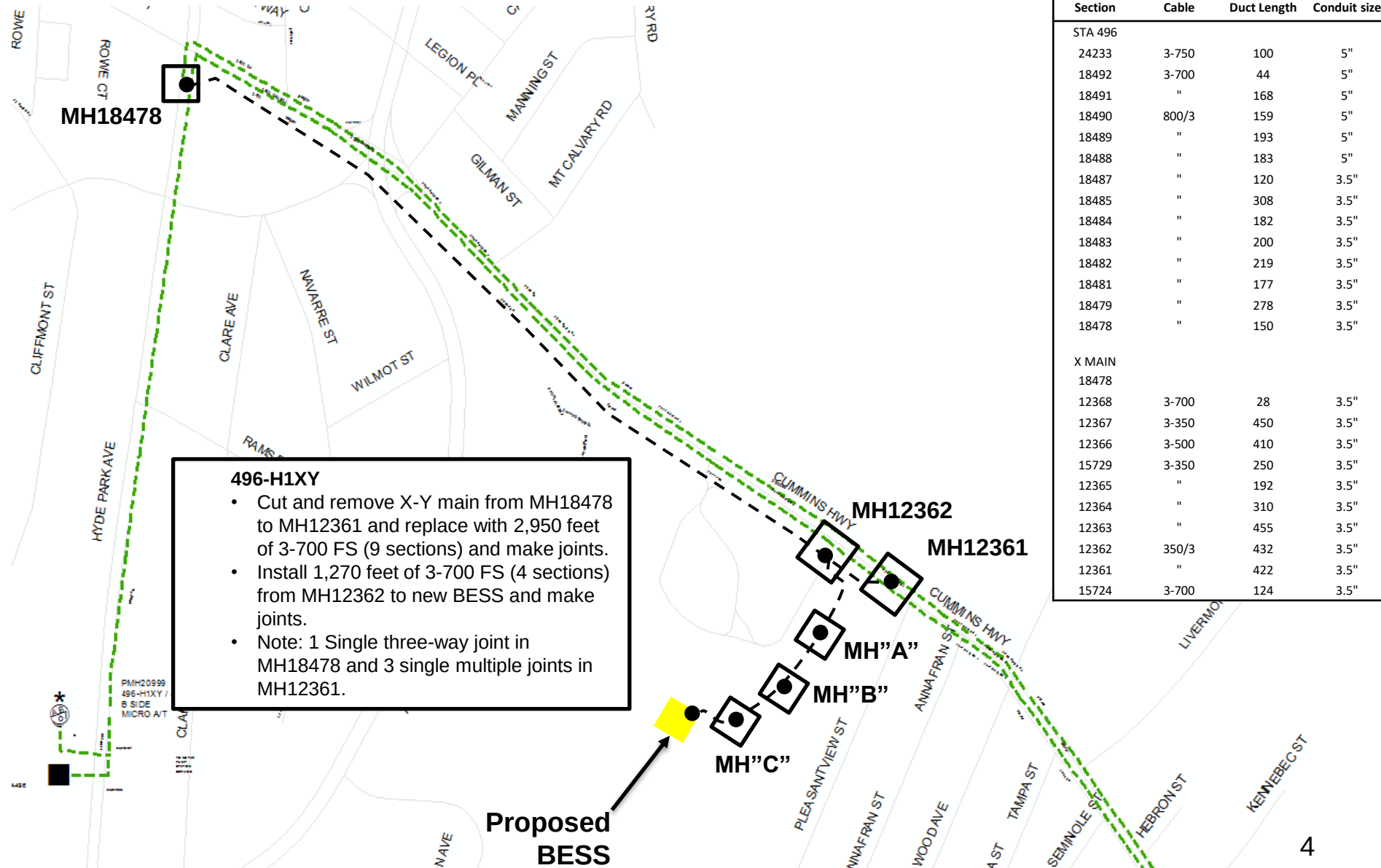
- (16,395') of 3-700 FS 15kV primary cable (47 sections)

Total Cost Estimate = \$4,459,417.00

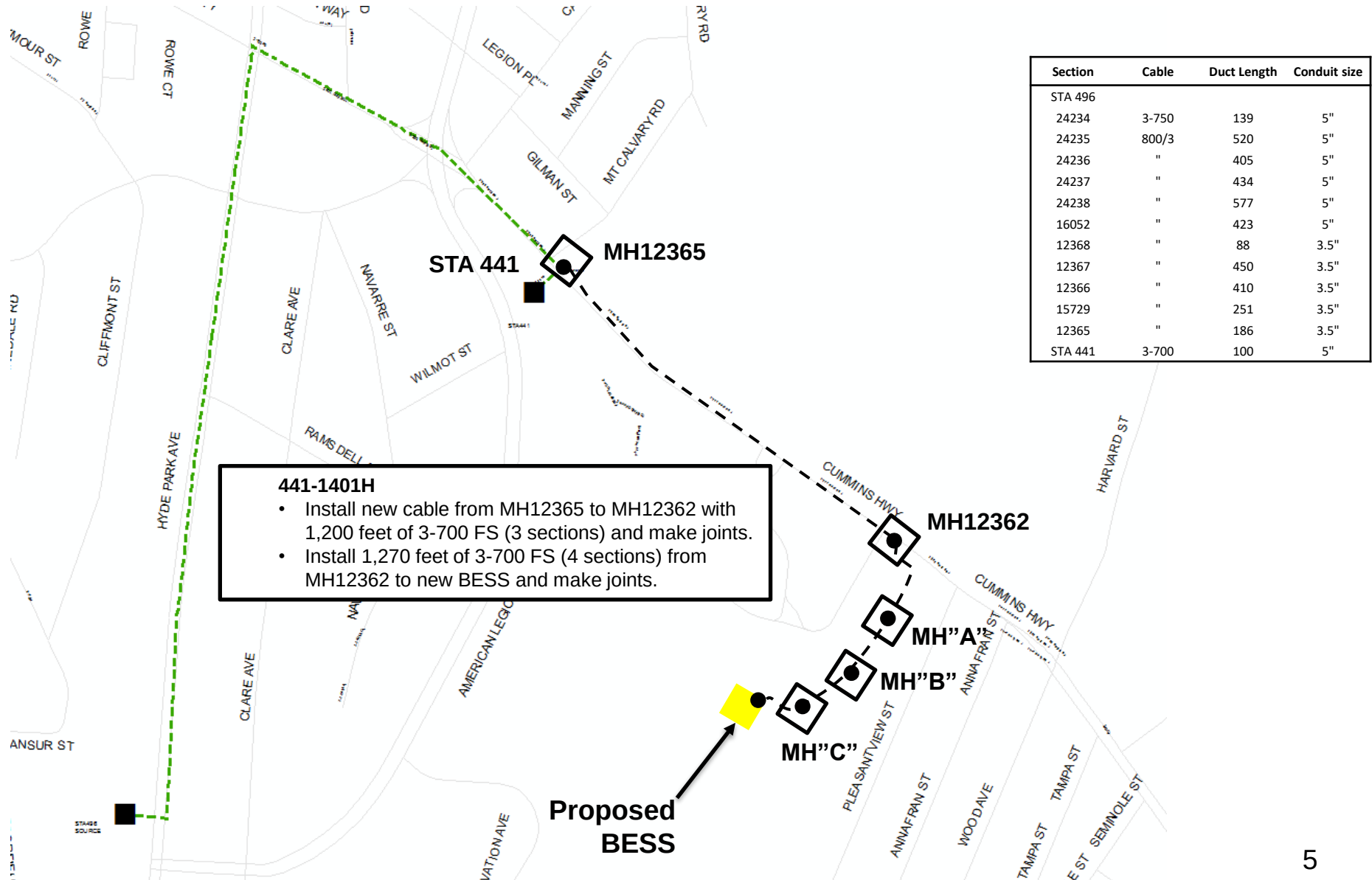
BESS Preliminary Design



BESS – 496-H1XY (Bus 1)



BESS – 441-1401H (Bus 2)

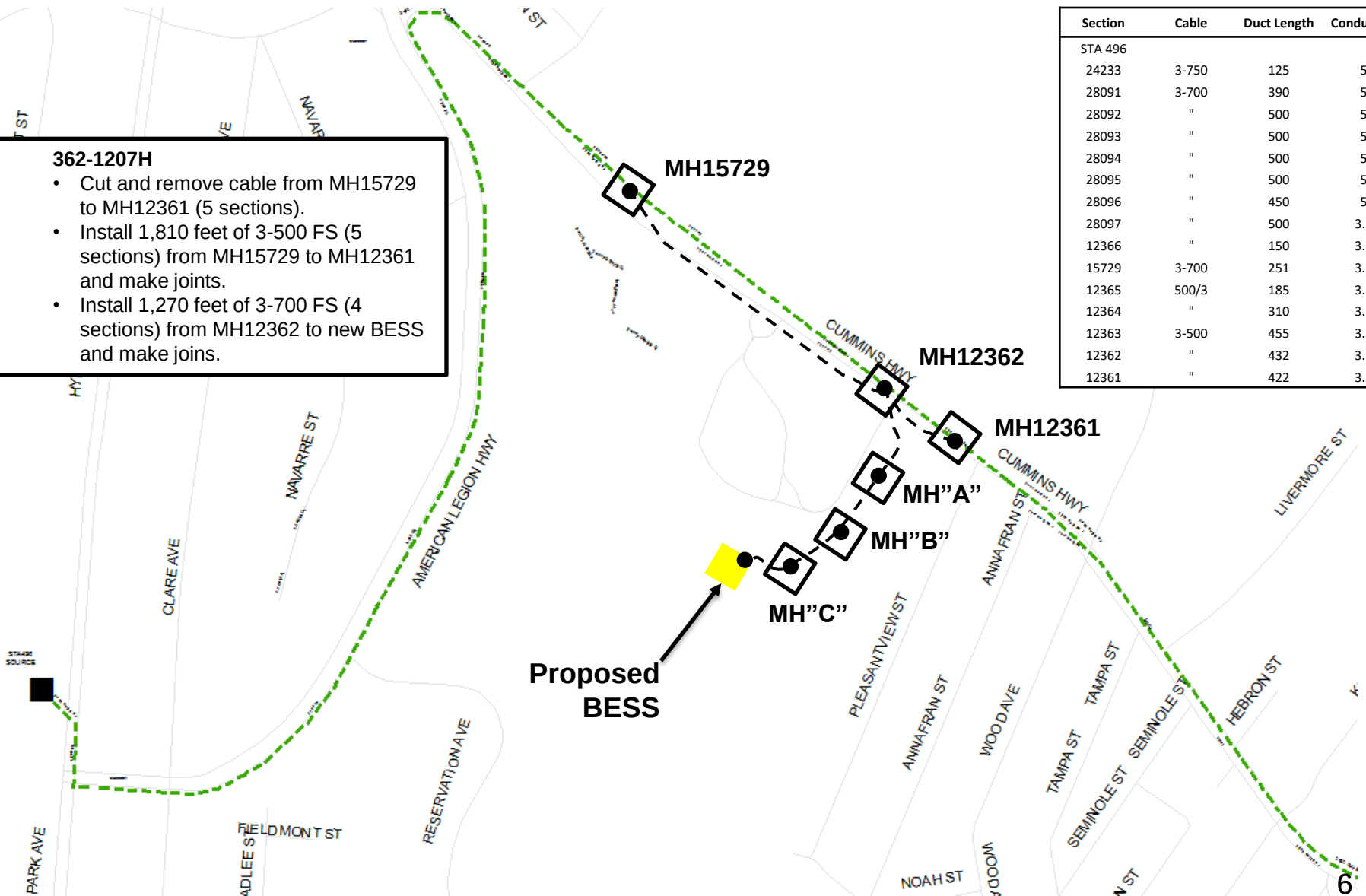


BESS – 362-1207H (Bus 3)

362-1207H

- Cut and remove cable from MH15729 to MH12361 (5 sections).
- Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.
- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Section	Cable	Duct Length	Conduit size
STA 496			
24233	3-750	125	5"
28091	3-700	390	5"
28092	"	500	5"
28093	"	500	5"
28094	"	500	5"
28095	"	500	5"
28096	"	450	5"
28097	"	500	3.5"
12366	"	150	3.5"
15729	3-700	251	3.5"
12365	500/3	185	3.5"
12364	"	310	3.5"
12363	3-500	455	3.5"
12362	"	432	3.5"
12361	"	422	3.5"

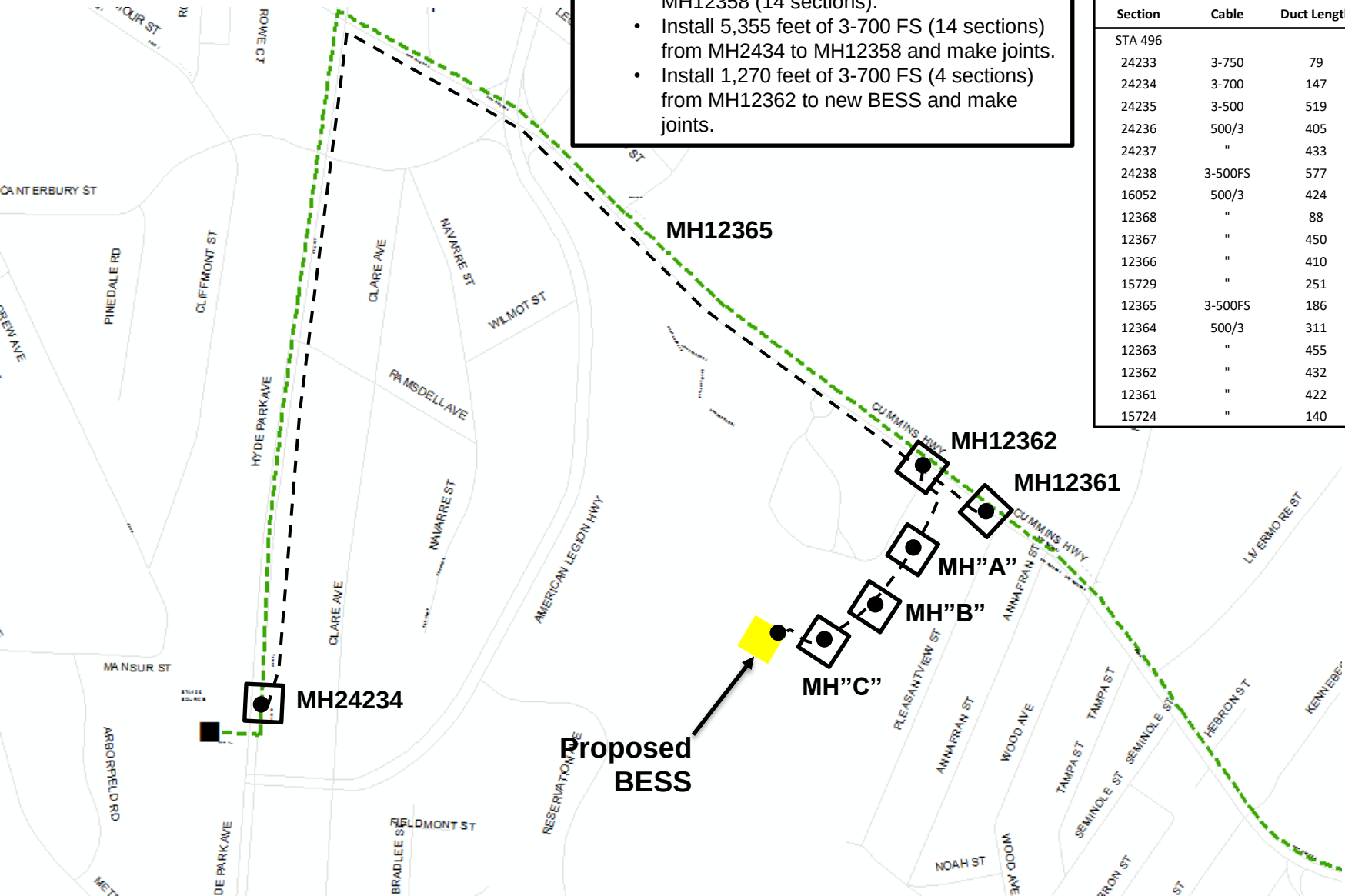


BESS – 362-1209H (Bus 4)

362-1209H

- Cut and remove cable from MH2434 to MH12358 (14 sections).
- Install 5,355 feet of 3-700 FS (14 sections) from MH2434 to MH12358 and make joints.
- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Section	Cable	Duct Length	Conduit size
STA 496			
24233	3-750	79	5"
24234	3-700	147	5"
24235	3-500	519	5"
24236	500/3	405	5"
24237	"	433	5"
24238	3-500FS	577	5"
16052	500/3	424	5"
12368	"	88	3.5"
12367	"	450	3.5"
12366	"	410	3.5"
15729	"	251	3.5"
12365	3-500FS	186	3.5"
12364	500/3	311	3.5"
12363	"	455	3.5"
12362	"	432	3.5"
12361	"	422	3.5"
15724	"	140	3.5"



BESS Preliminary Scope of Distribution Work

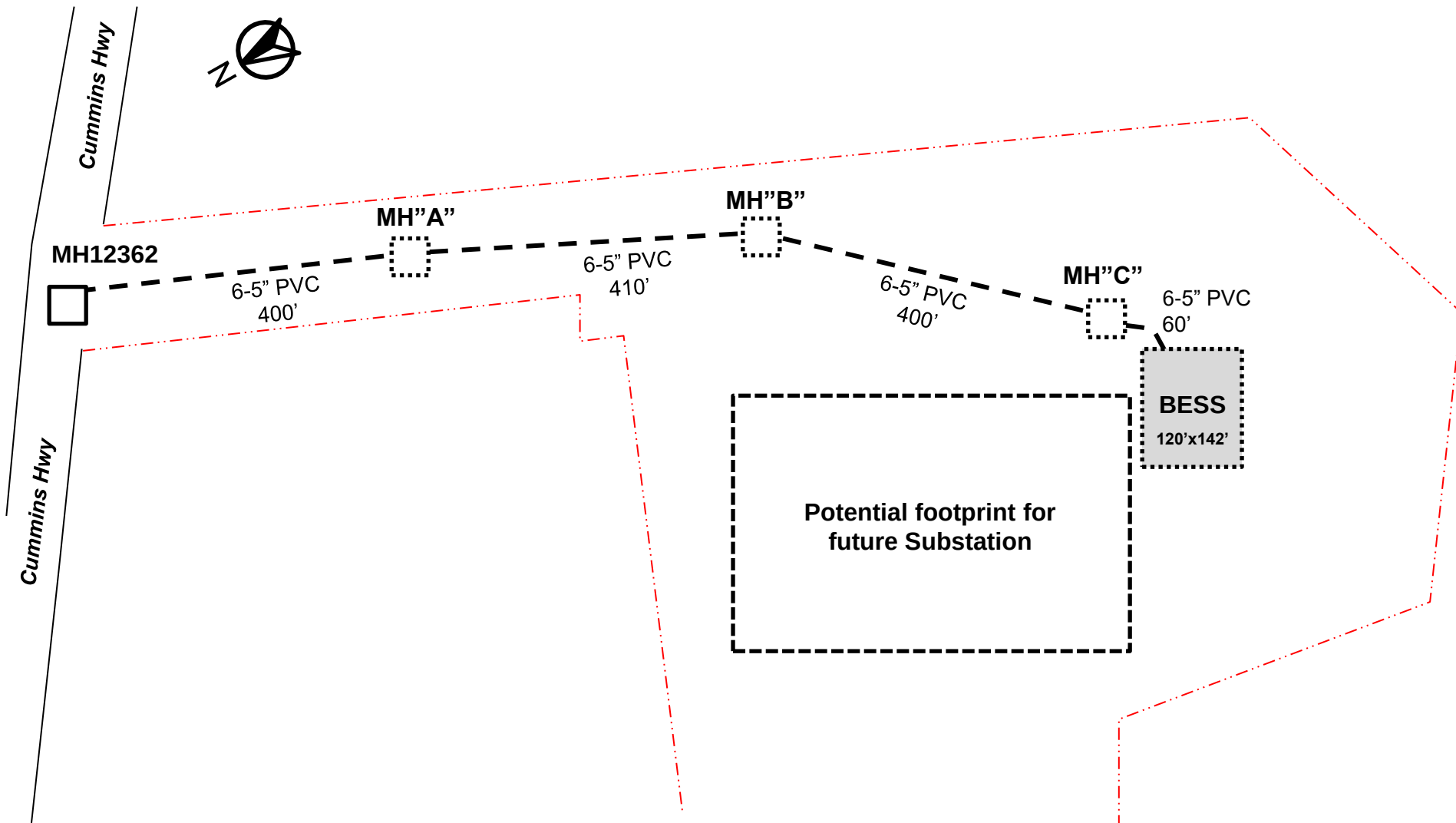
	Conduit	Manholes	Cable Install	Cable Removal	Straight Joint	Single 3-way	Single Multiple
-	1,270'	3	-	-	-	-	-
496-H7	-	-	1,270'	-	9	-	3
441-1401H	-	-	2,670'	186'	21	1	3
362-1207H	-	-	3,080'	1,810'	24	-	3
362-1209H	-	-	2,920'	1,650'	18	1	3
Total	1,270'	3	9,940'	3,646'	72	2	12

Notes:

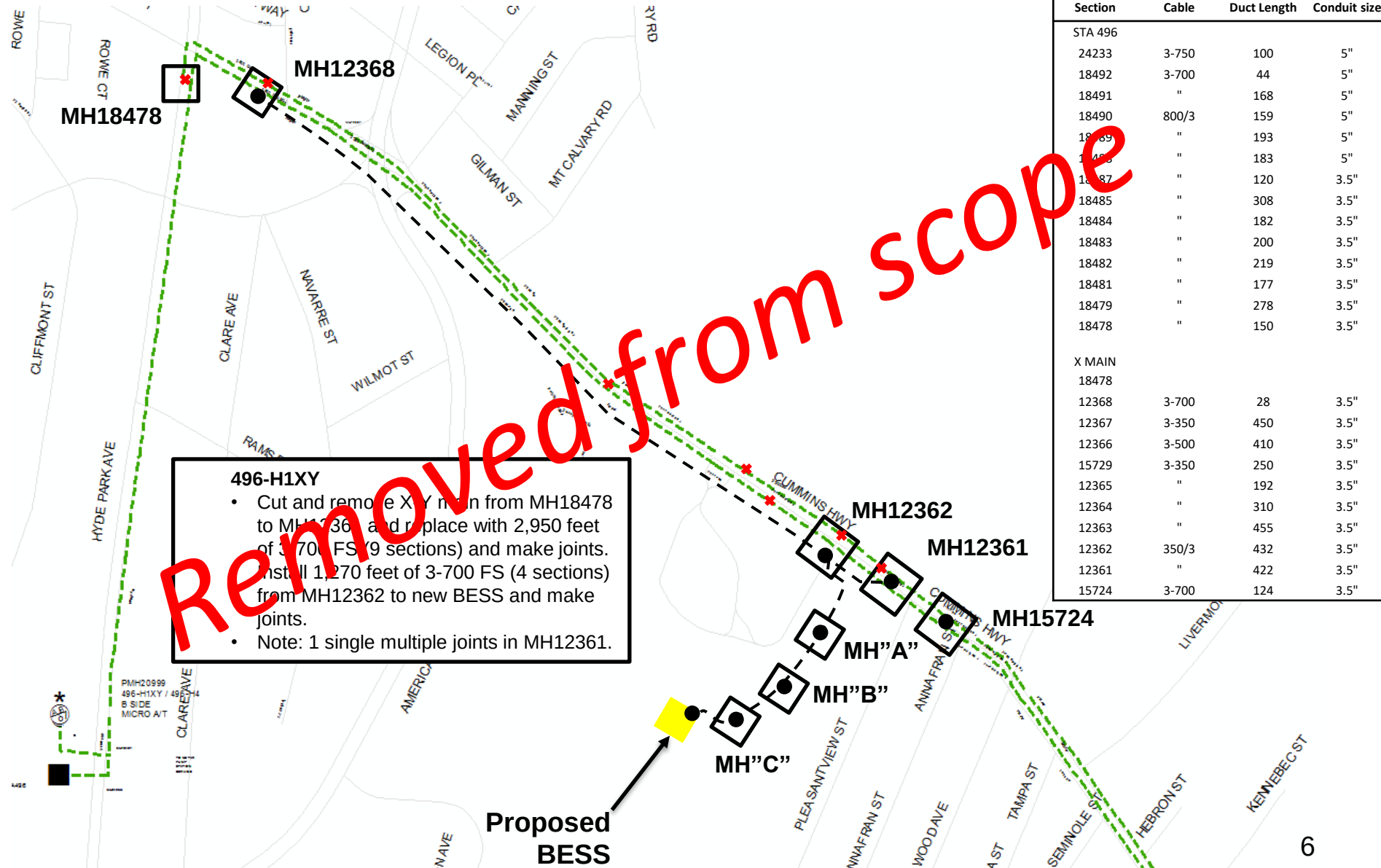
Conduit to be installed will be at least 6-5" PVC type "EB" encased in concrete for primary cable.

Manhole size will have inside dimensions of at least: 6' x 10' 8'.

BESS Preliminary Design



BESS – 496-H1XY (Bus 1)

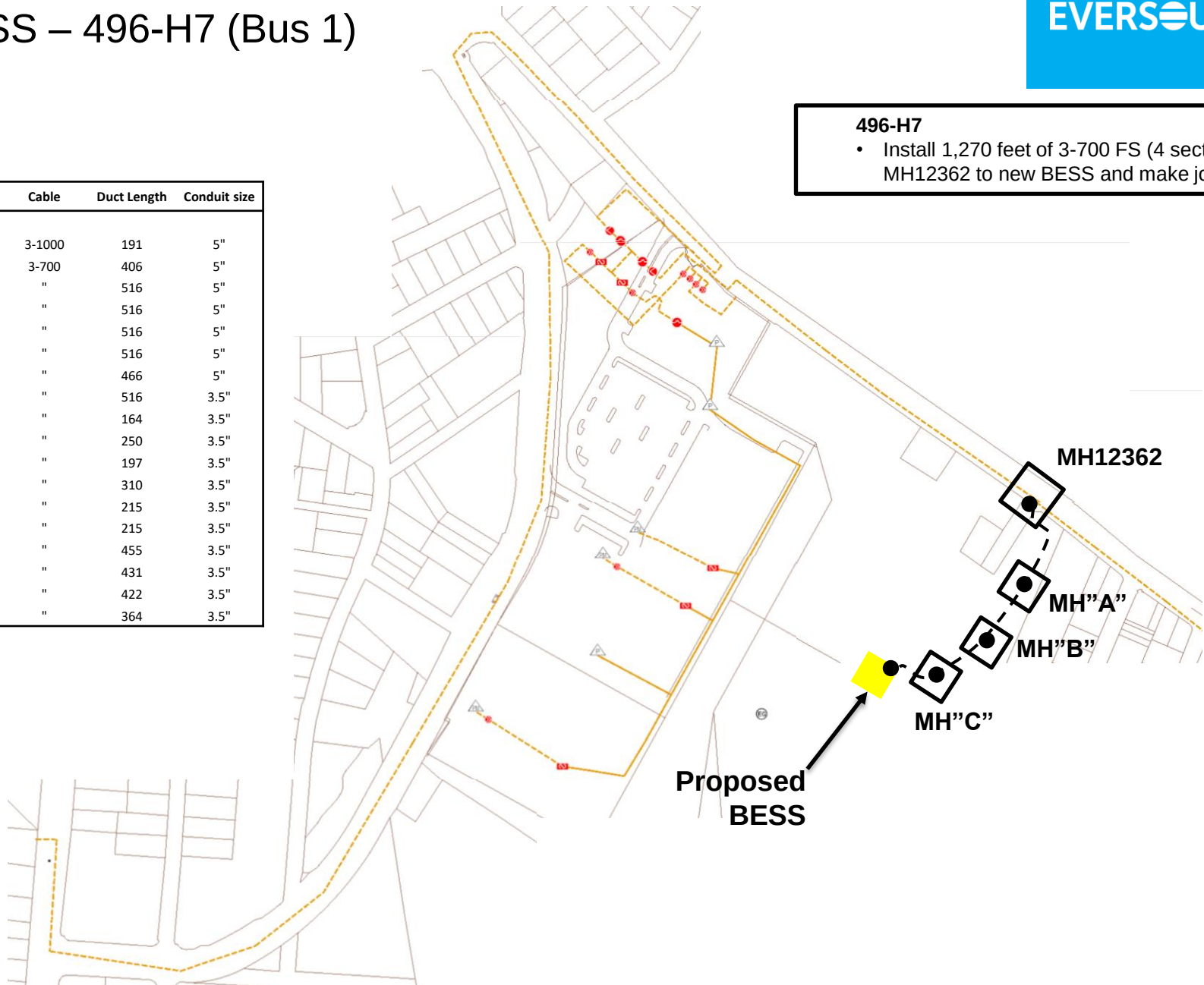


BESS – 496-H7 (Bus 1)

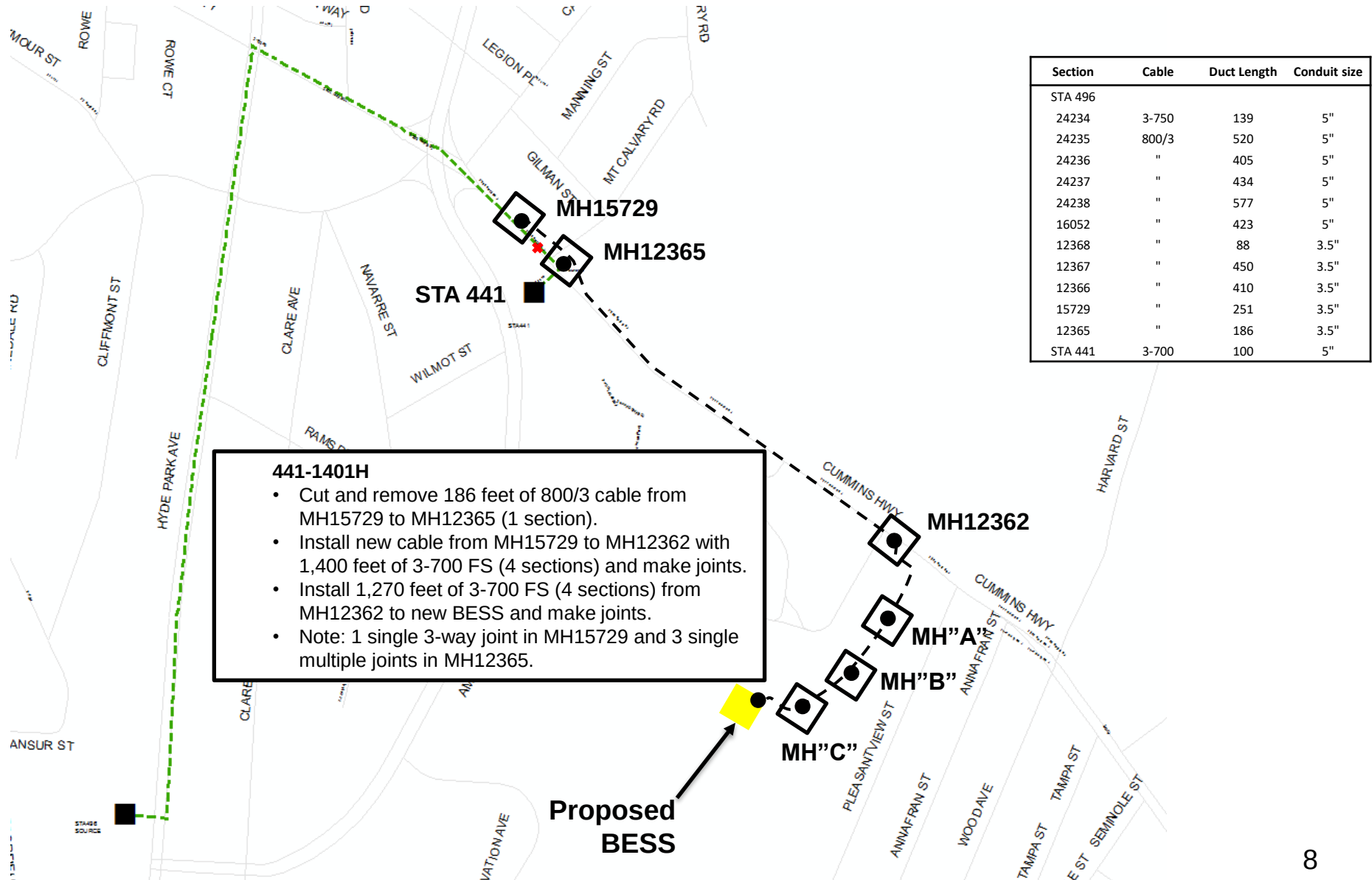
Section	Cable	Duct Length	Conduit size
STA 496			
24234	3-1000	191	5"
28091	3-700	406	5"
28092	"	516	5"
28093	"	516	5"
28094	"	516	5"
28095	"	516	5"
28096	"	466	5"
28097	"	516	3.5"
12366	"	164	3.5"
15729	"	250	3.5"
12365	"	197	3.5"
12364	"	310	3.5"
22934	"	215	3.5"
12364	"	215	3.5"
12363	"	455	3.5"
12362	"	431	3.5"
12361	"	422	3.5"
12360	"	364	3.5"

496-H7

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.



BESS – 441-1401H (Bus 2)



BESS – 362-1207H (Bus 3)

362-1207H

- Cut and remove cable from MH15729 to MH12361 (5 sections).
- Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.
- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Section	Cable	Duct Length	Conduit size
STA 496			
24233	3-750	125	5"
28091	3-700	390	5"
28092	"	500	5"
28093	"	500	5"
28094	"	500	5"
28095	"	500	5"
28096	"	450	5"
28097	"	500	3.5"
12366	"	150	3.5"
15729	3-700	251	3.5"
12365	500/3	185	3.5"
12364	"	310	3.5"
12363	3-500	455	3.5"
12362	"	432	3.5"
12361	"	422	3.5"

**Proposed
BESS**

MH15729

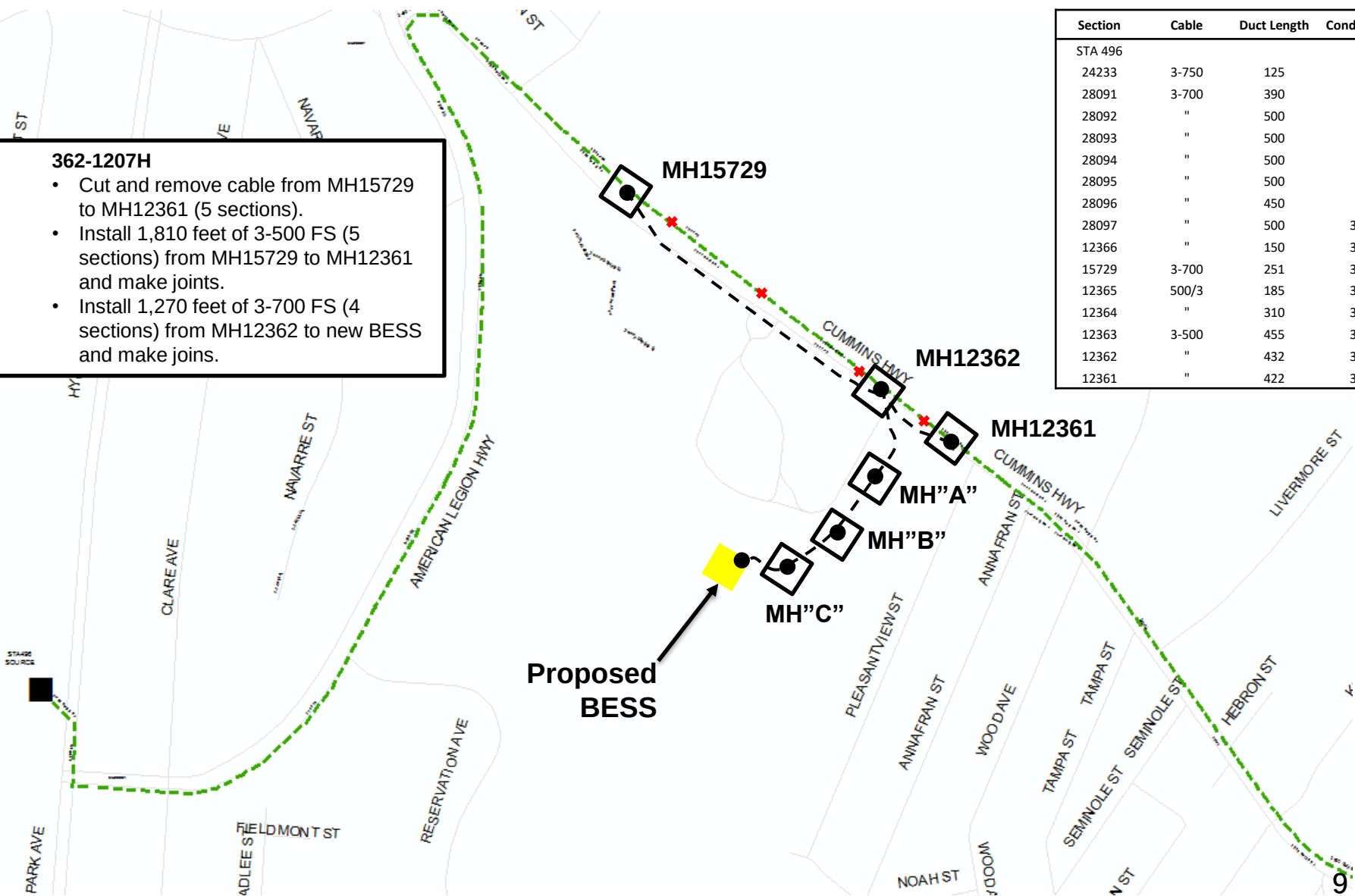
MH12362

MH12361

MH"A"

MH"B"

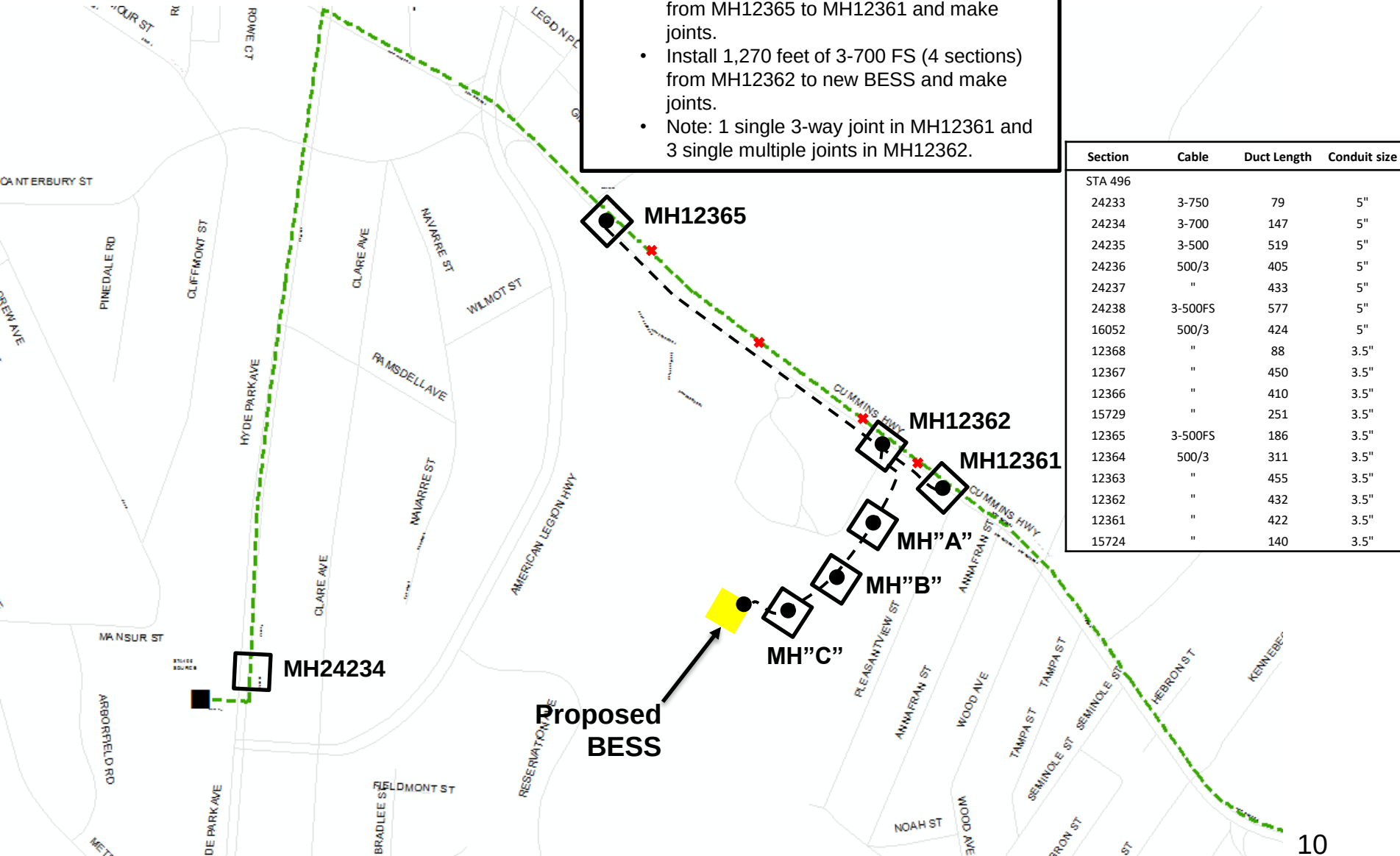
MH"C"



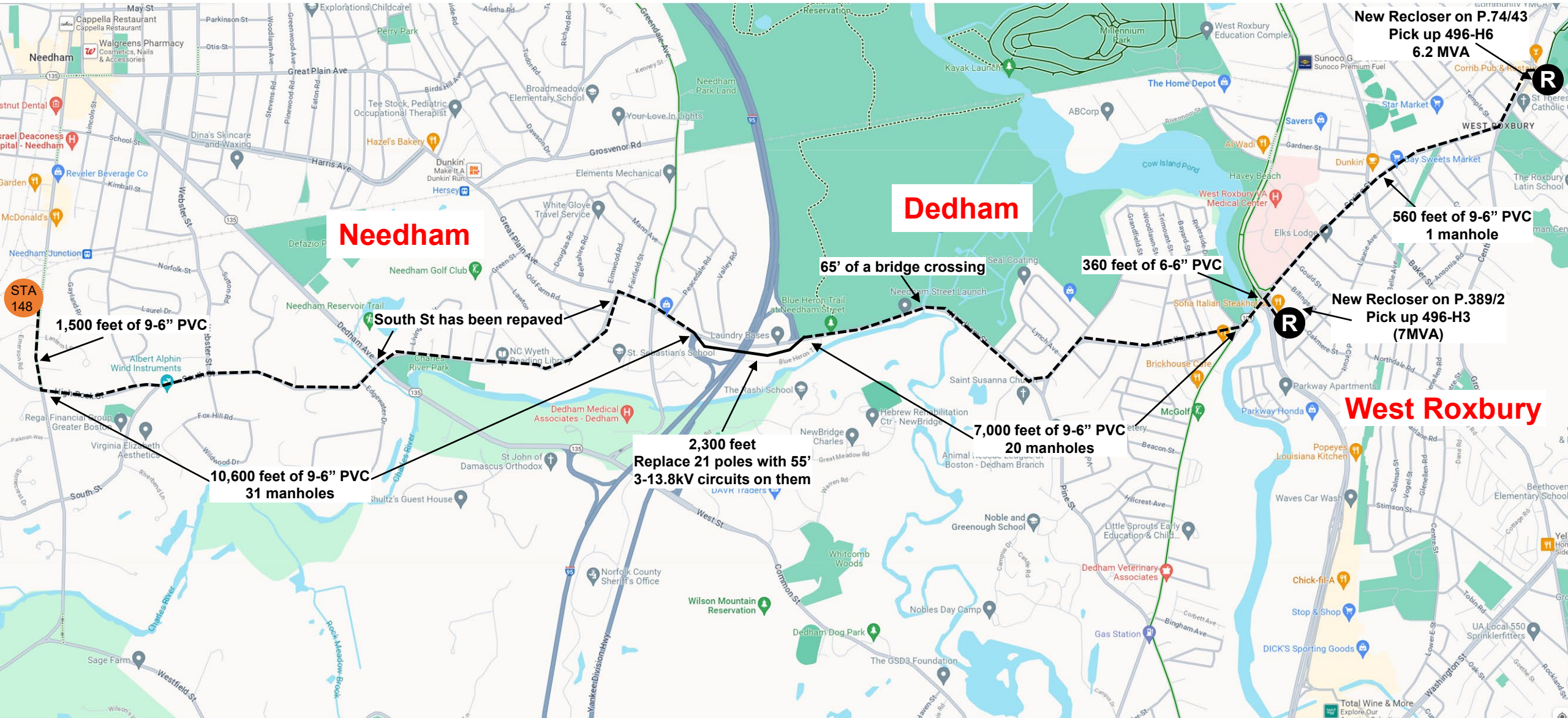
BESS – 362-1209H (Bus 4)

362-1209H

- Cut and remove cable from MH12365 to MH12361 (4 sections).
- Install 1,650 feet of 3-700 FS (4 sections) from MH12365 to MH12361 and make joints.
- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.
- Note: 1 single 3-way joint in MH12361 and 3 single multiple joints in MH12362.



Conduit and Route from STA 148 to 496-H3 & 496-H6



Scope of Work to relieve STA 496 via STA 148.

19,600 feet of 9-6" PVC conduit

52 manholes

360 feet of 6-6" PVC conduit

350 feet of 2-5" PVC conduit

21 (55 foot poles)

To Relieve 496-H3 (7 MVA)

19,900 feet of 3-700FS

2,450 feet of 477 ACSR

2 New Reclosers

2 New Underslung

2 new 45 foot poles

To Relieve 496-H6 (6.2 MVA)

25,900 feet of 3-700FS

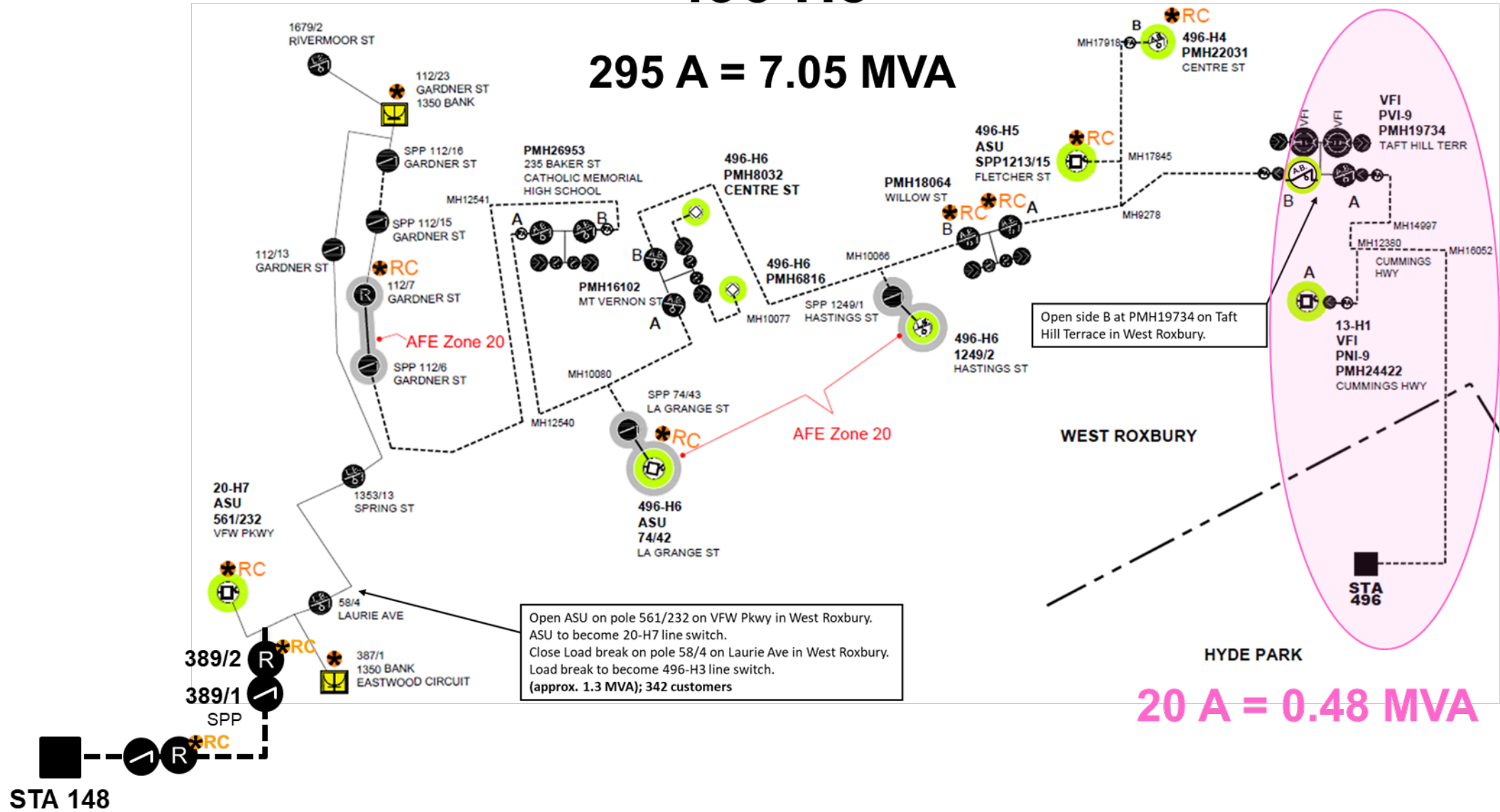
2,300 feet of 477 ACSR

2 New Reclosers

2 new Underslung

496-H3

295 A = 7.05 MVA



[illegible]

260 A = 6.21 MVA

20 A = 0.48 MVA

Open recloser on pole 1373/81 on Washington Street in West Roxbury.

WEST ROXBURY
DEDHAM



Hyde Park Substation Battery Energy Storage System (BESS)

**Distribution System Impact Study: Steady State and
Dynamic Analysis**

Report

System Planning MA

Prepared By: Mariam Chatha and Souresh Mukherjee

Revision History

Date	Rev.	Description
	1	Report

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Executive Summary

This report includes the dynamic analysis performed for the Battery Energy Storage System (BESS) being interconnected at four 13.8 kV Hyde Park distribution feeders at Cummins Highway in Boston on Eversource owned property. The four feeders are 496-H1XY, 441-1401, 362-1209H and 362-1207H supplied from Hyde Park Substation. There will be a 2.5MW/5MWH battery on each of the four feeders making the project consist of a total of 10 MW/20MWH Battery Storage. The Hyde Park BESS will be operated in parallel with Eversource's electrical grid. The BESS shall be comprised of a system of battery racks, inverters, control system, pad mounted transformers, and switchgear. The BESS will be located and interconnected to Eversource's existing 13.8 kV distribution infrastructure adjacent to the BESS facility.

The batteries would be dispatched with a peak load shave functionality. They will charge during the light load and discharge during the peak load in order to lower the substation peak. The substation peak exceeds the firm capacity of the station and is forecasted to keep rising due to step loads and EV charging load. The Battery can ride through multi-day peak load events. BESS can be connected to Baker St#110 side in the future through 25kV or 4kV feeders to support Baker Substation if needed, which will require underground cable construction.

Below is the preliminary distribution scope of work for the BESS interconnection to the four feeders.

Manhole Installation

-(3) concrete manholes with inside dimension of 6'x10'x8'

Conduit Installation

-(1,210') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

-(1,210') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes Note: this second duct bank is required only if the substation is constructed

-(60') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

Primary Underground Cable Removal

-(13,065') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (37 sections)

Primary Underground Cable Installation

-(16,395') of 3-700 FS 15kV primary cable (47 sections)

Below is the scope of work for connecting the BESS to feeder 496-H1XY.

- Cut and remove X-Y main from MH18478 to MH12361 and replace with 2,950 feet of 3-700 FS (9 sections) and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

- Note: 1 Single three-way joint in MH18478 and 3 single multiple joints in MH12361.

Below is the scope of work for connecting the BESS to feeder 441-1401H.

- Install new cable from MH12365 to MH12362 with 1,200 feet of 3-700 FS (3 sections) and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Below is the scope of work for connecting the BESS to feeder 362-1207H.

- Cut and remove cable from MH15729 to MH12361 (5 sections).

- Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Below is the scope of work for connecting the BESS to feeder 362-1209H.

- Cut and remove cable from MH2434 to MH12358 (14 sections).

- Install 5,355 feet of 3-700 FS (14 sections) from MH2434 to MH12358 and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

To study the impact of the BESS on the Hyde Park distribution system, a transient dynamic analysis of the preferred alternative was performed in the PSCAD (Power Systems Computer Aided Design) tool. The four Hyde Park 13.8 kV distribution feeders along with the proposed BESS systems were modeled¹ in PSCAD by using circuit load and impedance data collected from steady-state Synergi models. The dynamic Risk of Islanding (ROI) analysis, and the Transient Overvoltage (TOV) analysis were done to verify compliance with IEEE 1547-2018 Standard.

The ROI analysis determined the BESS inverter's response to unintended island conditions. According to the IEEE standard, for an unintended islanding condition, the distributed energy resource (which in this case is the BESS) shall detect the island, cease to energize the island, and trip within 2 seconds of the formation of the island.

¹ The BESS systems on the distribution circuits were modelled with a generic inverter model, not specific to any one manufacturer. After a vendor for the BESS is selected, the dynamic study will have to be re-run with the manufacturer specific inverter model.

When interconnecting distributed energy resource there is a concern that during step changes in load (such as tripping of an upstream device), the proposed inverters may cause transient over voltages in excess of 1.2 per unit, which can potentially cause damage to the customer's equipment and/or utility equipment. The TOV analysis was conducted to verify if the expected transient overvoltage conditions caused by the BESS inverters during step changes in load were within standard limits.

The ROI and the TOV analyses were performed on the normal (N-0) configuration and several contingency (N-1) configuration scenarios. The N-1 operational scenarios included one of the four feeder outages, a transformer outage, and a bus outage. The BESS system passed the ROI test for all scenarios tested. The BESS system also does not contribute to any TOV conditions for any of the tested scenarios.

Risk of Islanding (ROI) Screening

Based on the suggested guidelines for anti-islanding screening from Sandia National Laboratories and Eversource's distributed energy resources (DER) screening for ROI assessment (shown in Figure 1), additional requirements of ROI are necessary. Further study was performed to determine if the BESS inverters are reacting appropriately and not cause violations.

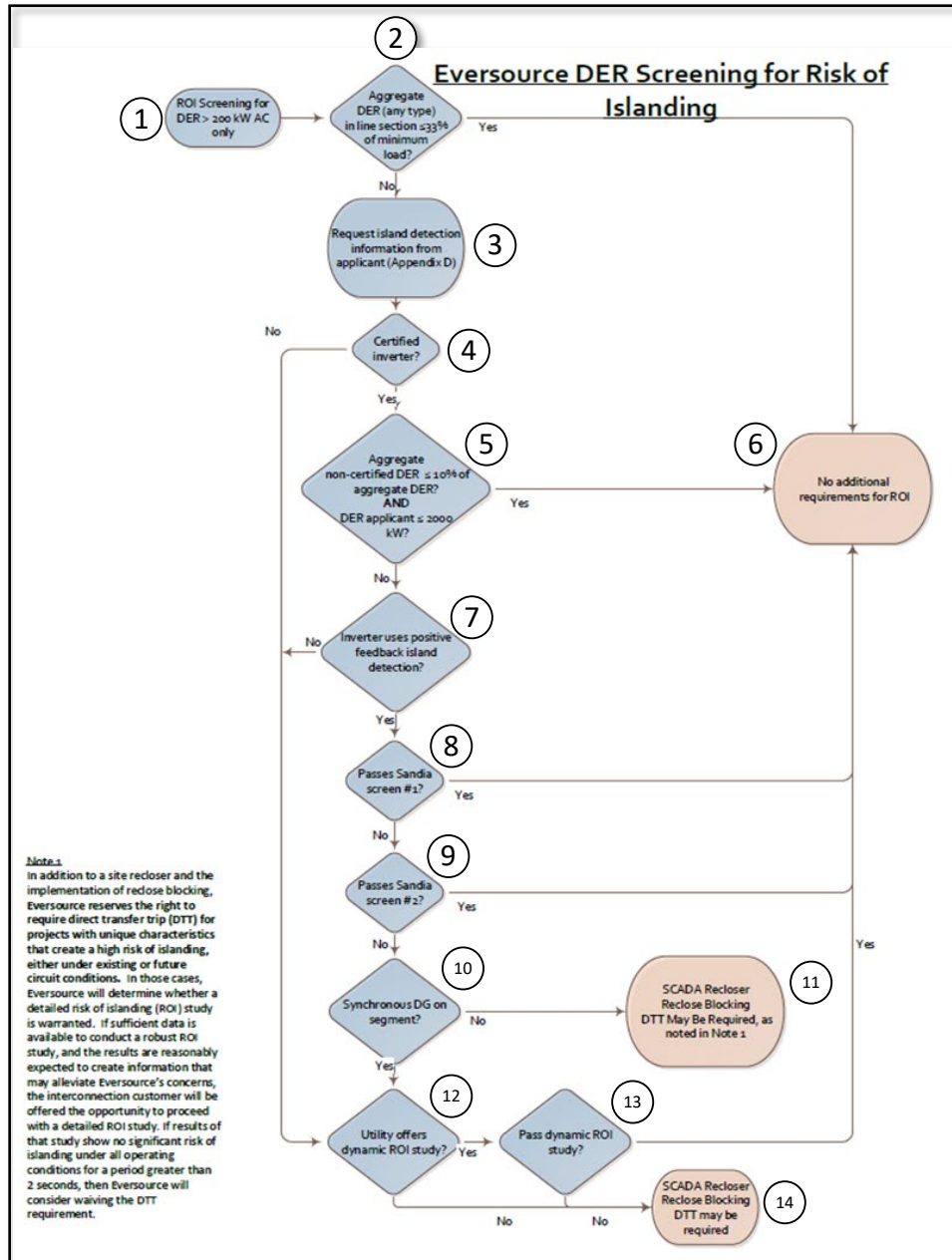


FIGURE 1. RISK OF ISLANDING (ROI) SCREENING PROCESS

During the dynamic study, in PSCAD, a detailed risk of islanding analysis was performed. The BESS system passed the ROI test for all four feeder interconnections.

Feeder	DER Interconnection	ROI Test
441-1401H	2.5 MW BESS	PASS
496-H1XY	2.5 MW BESS	PASS
362-1207H	2.5 MW BESS	PASS
362-1209H	2.5 MW BESS	PASS

TABLE 1: HYDE PARK BESS DYNAMIC STUDY – ROI RESULTS

PSCAD analysis shows that the BESS inverter trips within 2 seconds of unintentional island formation.

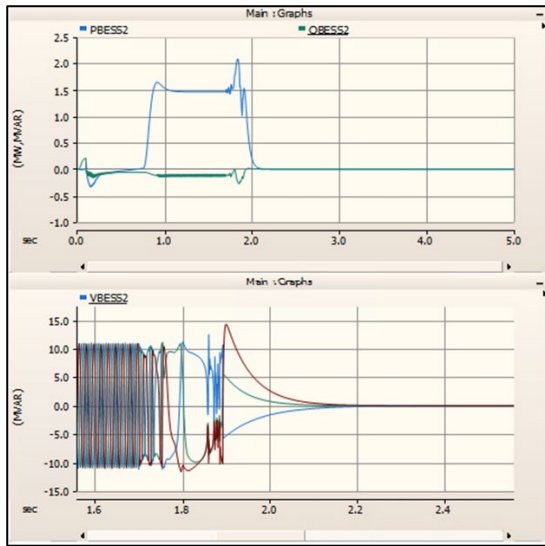


FIGURE 2: ROI TEST FOR FEEDER 441-1401H

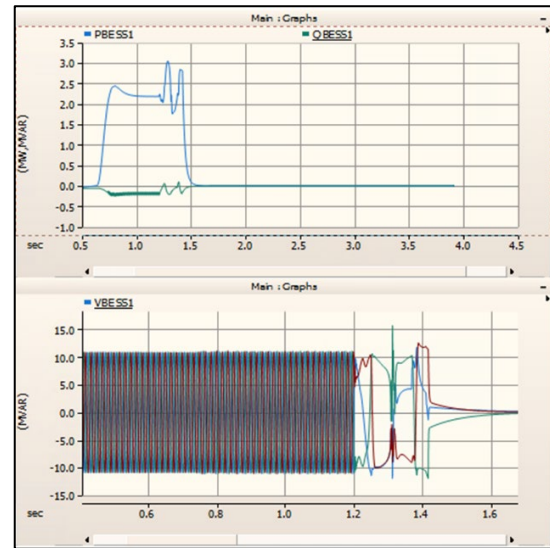


FIGURE 3: ROI TEST FOR FEEDER 496-H1XY

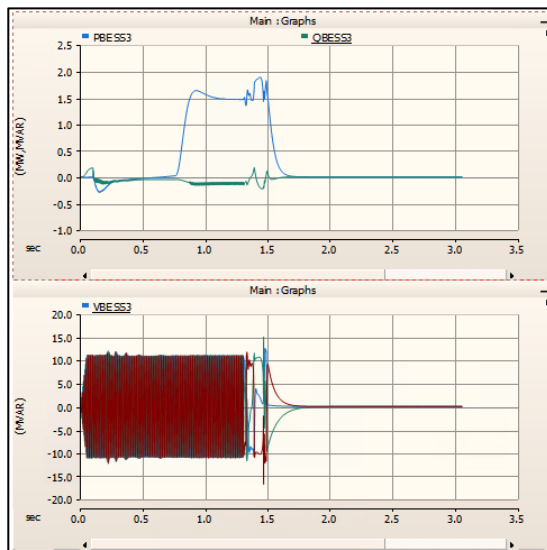


FIGURE 4: ROI TEST FOR FEEDER 362-1207H

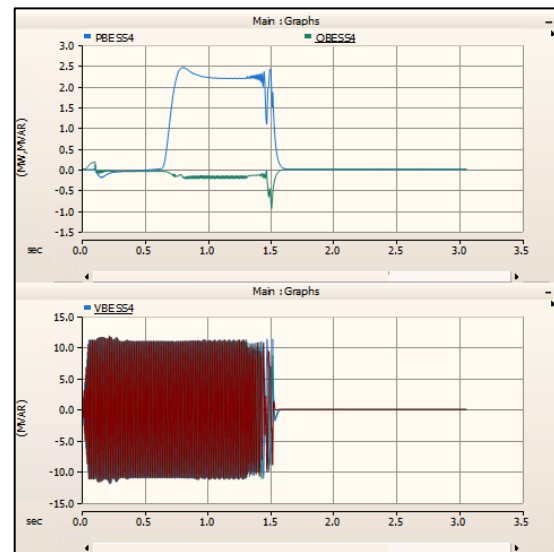


FIGURE 5: ROI TEST FOR FEEDER 362-1209H

Transient Overvoltage (TOV) Screening

If aggregate DER is greater than or equal to 115% of minimum gross load of the circuit, then there is a risk of Transient Overvoltage (TOV) at the feeder level and/or substation level. The same PSCAD dynamic study required for Risk of Islanding is further necessary to evaluate the TOV concerns. For dynamic transient overvoltage study, circuit load and impedance data are collected from steady-state Synergi models and used to build dynamic feeder models. The BESS inverters were modelled using generic PSCAD models. All other generators on the distribution circuits are modelled with a generic inverter model, not specific to any one manufacturer. The TOV analysis includes both Load Rejection Overvoltage (LROV) and Ground Fault Overvoltage (GFOV) studies.

Load Rejection Overvoltage (LROV) occurs when an upstream circuit breaker or recloser opens, which can cause an overvoltage on the isolated circuit due to stored energy in line and transformer inductances. LROV is modelled in PSCAD by operating all generation at 100% output in a minimum load case and opening the feeder circuit breaker. The voltage at the Point of Interconnection (POI) is measured.

Ground Fault Overvoltage (GFOV) is simulated by using the same minimum load, maximum generation scenario as the TOV scenario. A single-line-to-ground fault is initiated at the high side of the BESS interconnection transformer which persists until the upstream feeder breaker interrupts the fault after 10 cycles (0.16667 seconds). This is a conservative simulation and models a failure of protection selectivity.

Both LROV and GFOV durations must not exceed the following limits specified in IEEE 1547.

Transient Overvoltage Magnitude (per unit)	Maximum Allowable Duration (ms)
1.2	166
1.3	16
1.4	3
1.7	1.6
2.0	0

TABLE 2: IEEE 1547 STANDARD

Both, the LROV and GFOV analyses were performed on the normal (N-0) configuration and several contingency (N-1) configuration scenarios which included any one element out of service (OOS). The N-1 operational scenarios included:

- i. Transformer #1 OOS
- ii. Bus #1 OOS
- iii. Bus #3 OOS

The normal configuration of the proposed BESS at Hyde Park station is shown on the one-line diagram in figure 6. The N-1 contingency scenarios and corresponding breaker operations are shown in table 3.

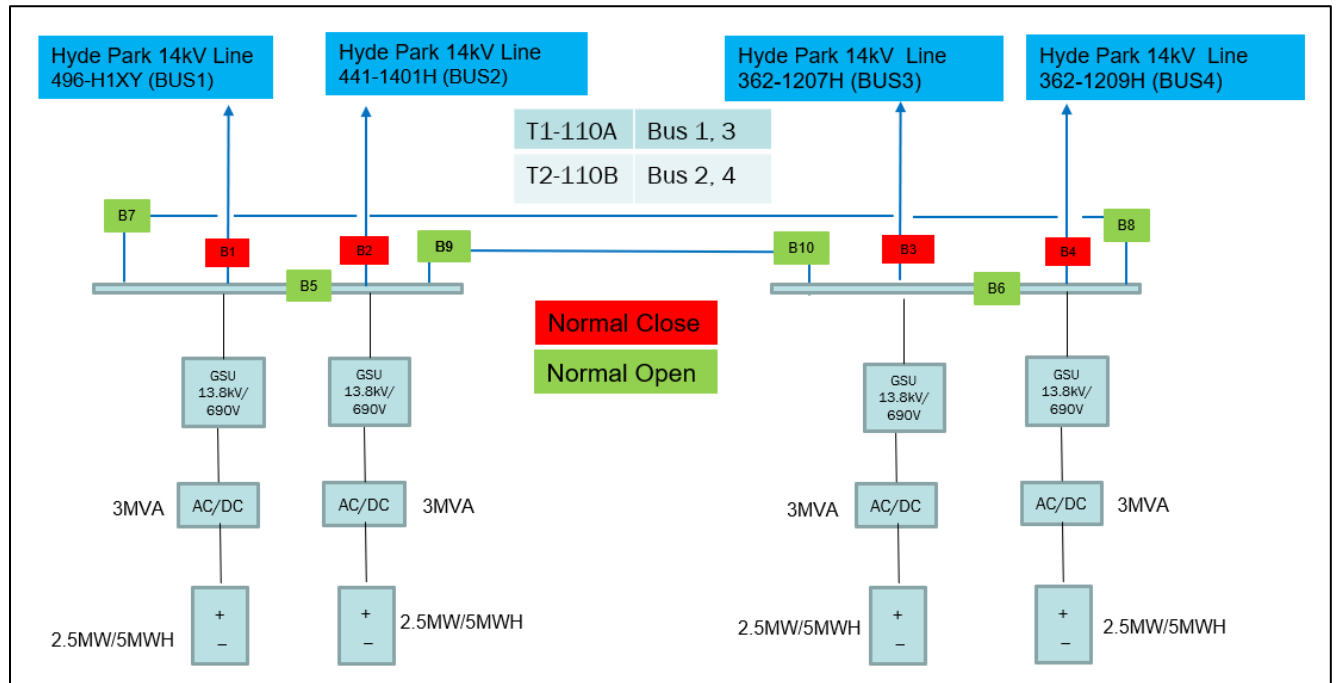


FIGURE 6: HYDE PARK PROPOSED BESS ONE-LINE DIAGRAM

Scenarios	Close	Open
Normal	B1, B2, B3, B4	B5, B6, B7, B8, B9, B10
T1- 110A OOS	B2, B4, B5, B6	B1, B3, B7, B8, B9, B10
Bus #1 OOS	B2, B3, B4, B5	B1, B6, B7, B8, B9, B10
Bus #3 OOS	B1, B2, B4, B6	B3, B5, B7, B8, B9, B10

TABLE 3: LIST OF TEST SCENARIOS

Results at Project (BESS) POI under Normal Configuration (N-0):

Load Rejection Overvoltage (LROV) Results – Feeder 441-1401H BESS Interconnection

BESS does not contribute to load rejection overvoltage on feeder 441-1401H.

Pre-project				
Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0
1.3 pu	16	0	0	0
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Post-project				
Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0.56	0
1.3 pu	16	0	0	0
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 441-1401H BESS Interconnection

BESS does not contribute to ground fault overvoltage on feeder 441-1401H.

Pre-project				
Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0
1.3 pu	16	0	0	0
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Post-project				
Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0
1.3 pu	16	0	0	0
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results – Feeder 496-H1XY BESS Interconnection

BESS does not contribute to load rejection overvoltage on feeder 496-H1XY.

Pre-project				
Feeder 496-H1XY				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0
1.3 pu	16	0	0	0
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Post-project				
Feeder 496-H1XY				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	2.10	1.80	0
1.3 pu	16	1.66	1.36	0
1.4 pu	3	1.26	0.98	0
1.7 pu	1.6	0.22	0	0
2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 496-H1XY BESS Interconnection

BESS does not contribute to ground fault overvoltage on feeder 496-H1XY.

Pre-project				
Feeder 496-H1XY				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0
1.3 pu	16	0	0	0
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Post-project				
Feeder 496-H1XY				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	24.54
1.3 pu	16	0	0	0.52
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results –Feeder 362-1207H Interconnection

BESS does not contribute to load rejection overvoltage on feeder 362-1207H.

Pre-project					Post-project				
Feeder 362-1207H					Feeder 362-1207H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.20	0	0.02
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0.02
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results –Feeder 362-1207H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1207H.

Pre-project					Post-project				
Feeder 362-1207H					Feeder 362-1207H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	36.76	0	0
1.3 pu	16	0	0	0	1.3 pu	16	1.42	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to load rejection overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	25.84	0
1.3 pu	16	0	0	0	1.3 pu	16	0	7.88	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Results at Project (BESS) POI under several Contingency Configurations (N-1):

(i) Transformer #1 Out of Service:

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	0.02
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.04	4.42	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

(ii) Bus #1 Out of Service:

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0.08	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

(iii) Bus #3 Out of Service:

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.02	4.20	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Steady State Analysis Summary Results

Power Flow Study

The below graphs show that the four (4) 2.5MW/5 MWh battery units can reduce the peak load to approximately 95% of the Load Carrying Capacity of the Hyde Park Substation in year 2032 or approximately 148 MVA. The BESS would be interconnected to four (4) 14kV distribution feeders, 496-H1XY, 441-1401, 362-1209H and 362-1207H. One 2.5MW/5 MWh battery will be connected to each of the four feeders.

The project is designed to meet the 10 year forecast, where the highest load being is in 2032. Year 2032 is chosen as the year with the highest load forecast of 155 MVA for the Hyde Park Substation without any batteries installed. The batteries act as a load and charge from 04:00 AM to 09:00 AM and discharge from 17:00 PM to 21:00 PM to decrease the Substation peak load as shown below.

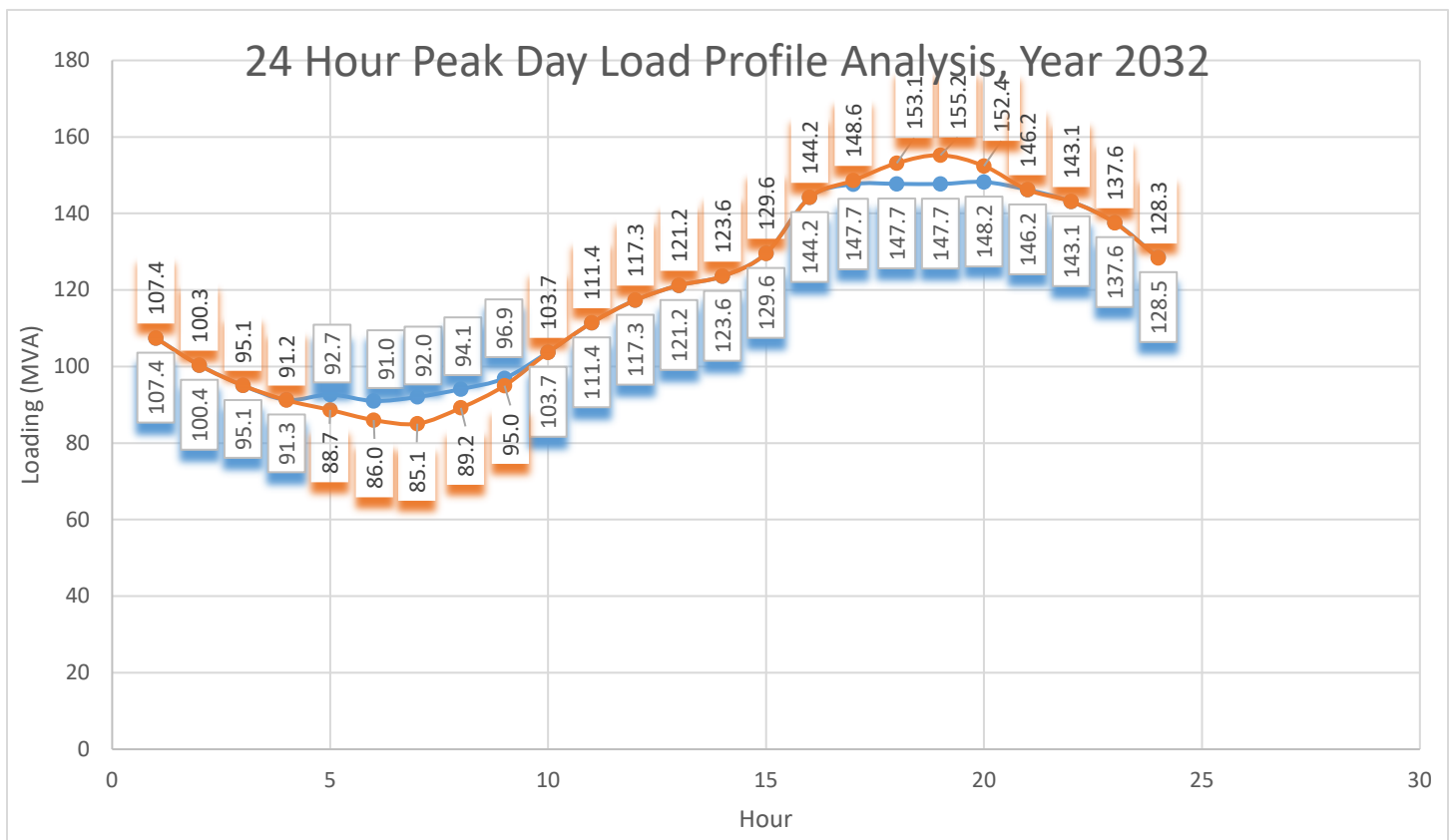


Figure 7

The orange line represents the Hyde Park Substation load without any batteries installed and the blue line represents the Substation load with the four batteries installed.

The below graph shows the resultant load profile for the substation during the peak load day under contingency assuming 13.17 MVA of distribution load transfers and the battery is operational. This decrease load to 141 MVA is due to a load transfer of 13.17 MVA which occurs during N-1 condition through distribution feeder ties and the decrease to 133.7 at 19:00 is due to the BESS.

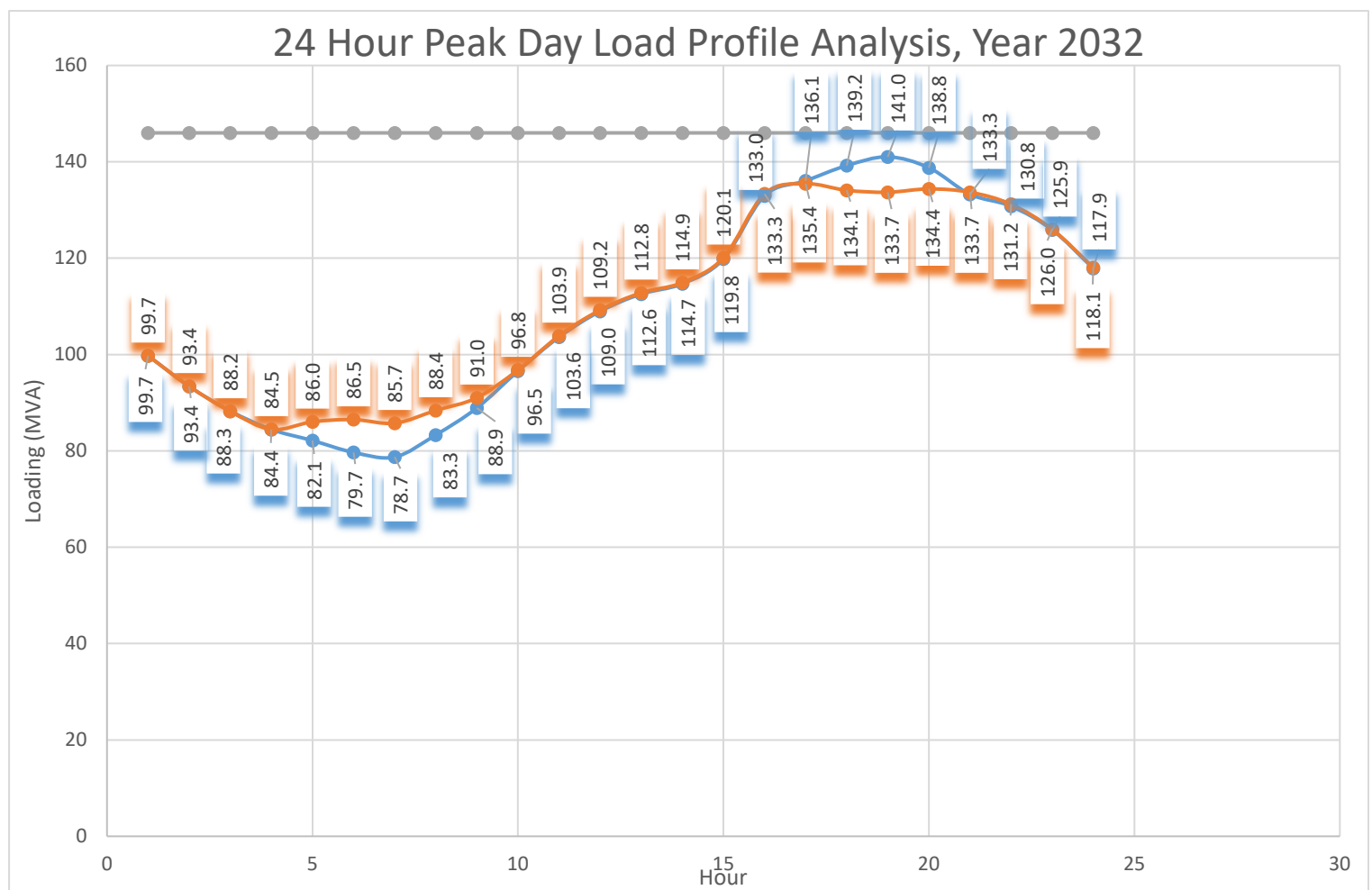


Figure 8

Thermal Rating Analysis

Below graph shows the 2032 24 hour peak day loading on each of the four feeders that the BESS is interconnected to and assuming the BESS is operational. It shows the 24 hour peak load

day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 to 09:00 and discharge from 17:00 to 21:00 to decrease the Substation peak load.

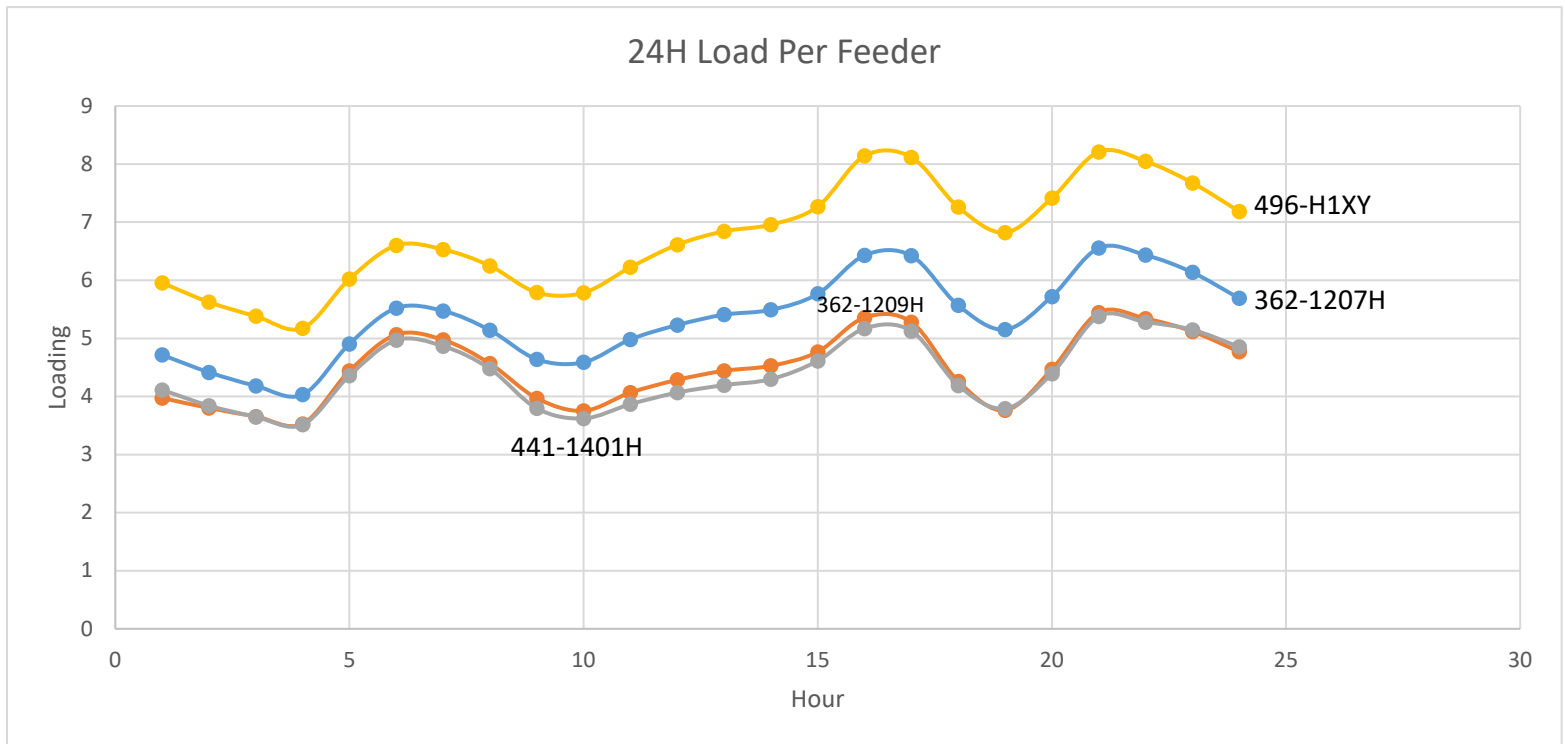


Figure 9

As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 4 AM to 9 AM and acting as a load.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 AM	4.9	4.4	4.4	6.0
6:00 AM	5.5	5.1	5.0	6.6
7:00 AM	5.5	5.0	4.9	6.5
8:00 AM	5.1	4.6	4.5	6.3
9:00 AM	4.6	4.0	3.8	5.8
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

The exact loading for the year 2032 cannot be predicted at this stage. However, based on the above loading data, there is a flexibility to charge the BESS on three of the above feeders instead of four if need be. Based on the above loading, feeder 441-1401 has the lowest amount of load and would be able to accommodate another 2.5 MW of BESS while feeder 496-H1XY may not. System Planning will be monitoring and modifying the battery schedule based on the EV charging profile and its load on the Hyde Park distribution circuits.

Below is a graph showing circuit loading for each of the four feeders under a Hyde Park Substation transformer contingency with the BESS connected to each of the four feeders. It shows the 24 hour peak load day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 to 09:00 and discharge from 17:00 to 21:00 to decrease the Substation peak load.

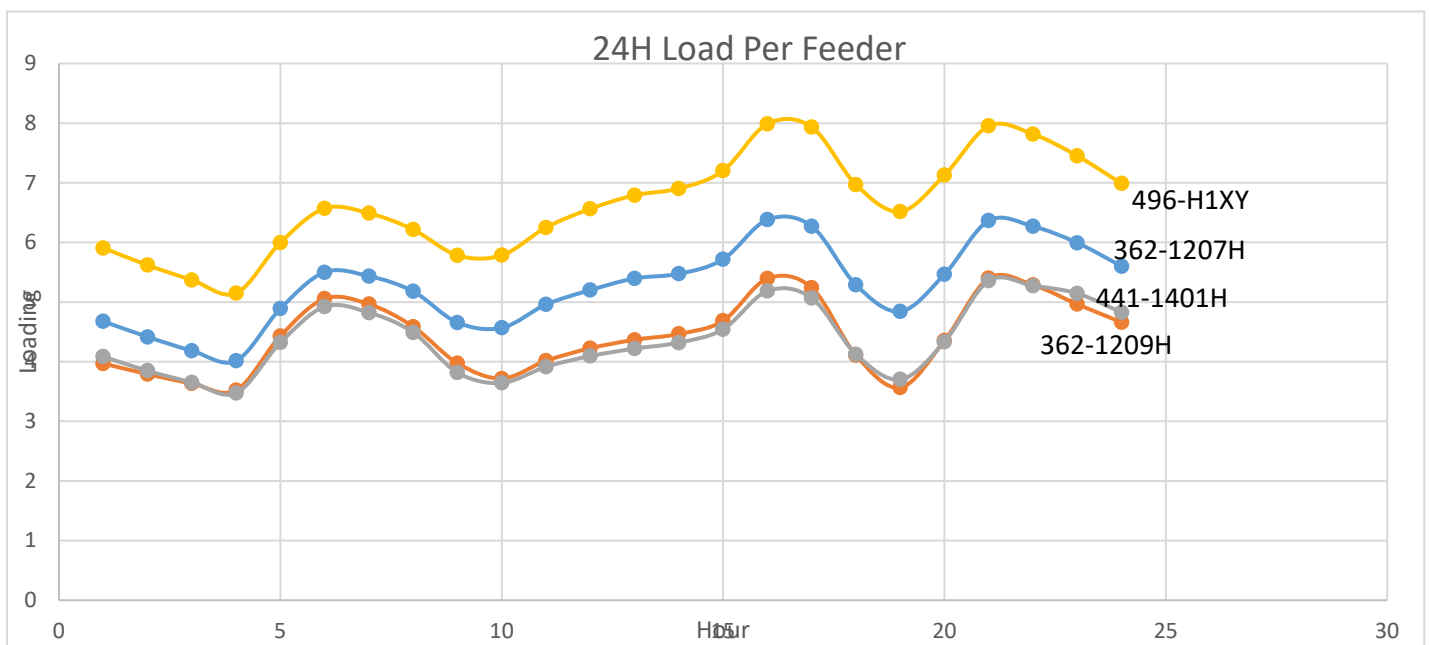


Figure 10

As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 4 AM to 9 AM and acting as a load.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 AM	4.89	4.43	4.32	5.996

6:00 AM	5.50	5.06	4.92	6.57
7:00 AM	5.43	4.96	4.82	6.49
8:00 AM	5.18	4.59	4.49	6.21
9:00 AM	4.66	3.97	3.82	5.783
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

The BESS also helps to bring down the peak load on these 14 kV feeders. The Hyde Park circuit 496-H1XY historically runs 80-90% of its summer normal ampacity in the summer. The circuit 362-1207 supplied historically runs 60-80% of its summer normal in the summer months. The circuit 362-1209 historically runs 60-90% of its summer normal in the summer months.

Reverse Power Flow Evaluation

Due to the small size of the battery, there is no reverse power flow into the Hyde Park substation transformers. To evaluate the reverse power flow, the worst case scenario was analyzed where two batteries with a total size of 5MW/10MWH were connected to one feeder and the BESS was operating at a 100% output. This is assuming there is a contingency on one of the four 14 kV feeders and there would need to be two 2.5MW/5MWH BESS on one feeder. It is seen that the load on the transformer supplying that feeder had just decreased by 5 MW due to the battery. Therefore, there is no reverse power flow because the battery would just decrease the load and is not large enough to cause reverse power flow into the electric grid.

For example, the below tables show the loading on the transformer supplying feeder 496-H1XY before and after 5MW/10MWH BESS was connected to it and operating at 100%output.

Transformer: STA_496_110A_2

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Transformer	View analysis results					
LTC / LDC	Result	A/AB	B/BC	C/CA	Avg	Mx/T
Settings	▼ General					
Tertiary	Ft Into	495578	465396	485737	482237	495578
Net Protector	Ft Away	---	---	---	---	0
Rates	Nom kV	ABC	ABCN	---	---	13.80
Info	▼ Taps and LTC					
Results	Vrr	---	121.7	---	121.7	---
	Amps Out	1599	1611	1615	1608	1615
	Tap Pos	11	11	11	11	11
	▼ Volts and Amps					
	Cumul Volt Drop	0.0	0.0	0.0	0.0	0.0
	Volts Out	121.8	121.7	121.7	121.8	121.8
	Amps Into	206	206	207	206	207
	Pct pf	92.0	92.0	92.0	92.0	---
	Imbalance	V,I,In	0.0%	0.3%	0 A	---
	▼ Power Flow					
	MVA Into	13.70	13.70	13.70	13.70	41.10
	MW Into	12.60	12.60	12.60	12.60	37.80
	Mvar Into	5.38	5.38	5.38	5.38	16.14
	MW Demand	12.46	12.55	12.58	12.53	37.59
	Mvar Demand	3.46	3.47	3.47	3.47	10.40
	Nom kvar Into	D,S,LC	0	0	0	---
	Nom kW Into	D,S,LC	0	0	0	---

Power Flow Without BESS

Transformer: STA_496_110A_2



Transformer	View analysis results
LTC / LDC	
Settings	
Tertiary	
Net Protector	
Rates	
Info	
Results	

Result	A/AB	B/BC	C/CA	Avg	Mx/T
General					
Ft Into	495578	465396	485737	482237	495578
Ft Away	---	---	---	---	0
Nom kV	ABC	ABCN	---	---	13.80
Taps and LTC					
Vrr	---	121.8	---	121.8	---
Amps Out	1412	1424	1427	1421	1427
Tap Pos	11	11	11	11	11
Volts and Amps					
Cumul Volt Drop	0.0	0.0	0.0	0.0	0.0
Volts Out	121.8	121.8	121.8	121.8	121.8
Amps Into	182	182	183	182	183
Pct pf	90.4	90.4	90.4	90.4	---
Imbalance	V,I,In	0.0%	0.3%	0 A	---
Power Flow					
MVA Into	12.11	12.11	12.11	12.11	36.32
MW Into	10.94	10.94	10.94	10.94	32.83
Mvar Into	5.18	5.18	5.18	5.18	15.55
MW Demand	10.81	10.91	10.94	10.89	32.66
Mvar Demand	3.69	3.69	3.69	3.69	11.07
Nom kvar Into	D,S,LC	0	0	0	---
Nom kW Into	D,S,LC	0	0	0	---

Power Flow With BESS

As shown, the load on the transformer supplying that feeder had just decreased by 5 MW due to the battery and there is no reverse power flow.

Flicker

The flicker values comply with the Eversource's allowable limit of 2.0% at the distribution circuits. The voltage difference when the BESS is off and when the BESS is operating at 100% output was analyzed and was below 2% during normal and contingency scenarios. The worst case scenario where there are two BESS with a total size of 5 MW/10MWH operating at 100% output on one feeder was also analyzed. The voltage difference still remained below 2%. This voltage difference was analyzed at the peak load of 155 MVA.

Transformer LTC Impact

The tap position range from -16 to 16 and the voltage is maintained at 122 Volts at the 14 kV buses. When the BESS is connected, the voltage increases slightly due to the decrease in load. Therefore, the BESS may cause the LTC to move to a slightly lower tap position to maintain the voltage at 122 volts. Overall, due to the small size of the battery, it would not have much impact on the transformer LTC tap positions.

Short Circuit

Due to the small size of the battery, it would not have much impact on the short circuit current.